Geofinance: Following Sovereign Capital Home*

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Abstract

Sovereign wealth funds (SWFs) invested around $1.28 trillion in cross-border private markets between 2002 and 2022. I develop a geofinance framework in which a low-cost financier offers capital to induce real activity toward its home country. The model predicts that such direction is strongest for recipients with costly external finance and that the resulting bilateral links crowd in third-party capital. Consistent with these predictions, deal-level evidence shows that a one-standard-deviation increase in a firm's external financing cost raises the probability of expansion toward the SWF home country relative to the co-investor's home country by 3.7 percentage points, about half the baseline rate. During the 2007–2008 financial crisis, banks with larger pre-crisis write-downs increase syndicated-loan deal counts disproportionately toward SWF home countries. At the aggregate level, FDI inflows and cross-border banking claims to SWF home countries rise by about 5–8 percent across the two designs. Sovereign capital does more than finance firms. It can reshape the geography of cross-border economic activity.

1 Introduction

In May 2025, Saudi Arabia's Public Investment Fund and Google Cloud announced a $10 billion joint investment to build an artificial intelligence hub in Saudi Arabia.[1] Finance flowed one way while economic activity followed the other. Deals of this kind are no longer unusual. Sovereign

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[1] Google Cloud and PIF Advance AI Hub in Saudi Arabia

wealth funds (SWFs), which collectively manage over $12 trillion, deployed roughly $1.28 trillion in cross-border private-market investments over 2002–2022.[2] This raises the question of whether these investments simply supply funding or redirect the recipient's cross-border activity toward the investor's home country.

I develop a geofinance framework in which a financier with a lower cost of capital offers financing terms to induce cross-border activity toward its home country. The mechanism builds on a distinction between two ways states project economic power. States can exploit existing dependencies through trade and financial networks, or they can build new bilateral links through their own investments. Existing geoeconomic frameworks emphasize the first channel, in which states exercise leverage through trade and financial networks (Clayton, Maggiori and Schreger, 2025, 2026). This paper studies the second: directed financing creates bilateral links that lower frictions for outside investors. SWFs are the empirical focus, but the mechanism is not unique to sovereign capital. It applies whenever a financier is willing to trade off part of its financial return for directed real activity or another strategic objective.[3]

The mechanism operates through a financing wedge, the gap between the financier's funding cost and the recipient's outside financing cost. When this wedge is large, directed activity is more likely; when it is small, the effect is weak or absent. The model separates the financing terms from directed activity. Favorable financing terms compensate recipients for accepting costly directed activity, but the directed response depends on the recipient's outside option. Recipients with higher external-finance costs value the financing wedge more and are more willing to accept directed activity. Directed activity also accumulates into a stock of bilateral links. As this stock rises, bilateral frictions fall, making subsequent investment in the financier's home country more attractive to outside investors.

The model yields two broad predictions for sovereign capital. At the micro level, among SWF-backed recipients, directed expansion toward SWF home markets should be strongest when external financing costs are high. At the macro level, if directed activity reduces bilateral frictions, third-party capital should crowd in toward SWF countries. I test these predictions in four empirical settings. The first two, PitchBook co-investments and crisis-era bank recapitalizations, provide

[2]For scale, the $12 trillion figure is comparable to the $12.7 trillion in total loans and leases held by U.S. commercial banks (Federal Reserve H.8 release, March 2025). The $1.28 trillion global figure is the author's calculations from GSWF and SWFI transaction databases for 2002–2022 and is roughly 6% of cumulative global inward foreign direct investment (FDI). In the United States alone, SWF investments account for about 8% of cumulative inward FDI.

[3]SWFs are a natural application because funding from commodity revenues, foreign-exchange reserves, or fiscal surpluses lowers their shadow cost of capital. The mechanism also applies to other financiers, such as development finance institutions or strategic corporate investors, that accept lower financial rents in exchange for influence over real decisions.

micro evidence of directed allocation. The last two test the corresponding macro implications: the private-investment channel in bilateral foreign direct investment (FDI), and the bank-lending channel in cross-border banking claims.

I first test this financing-friction gradient in 882 PitchBook co-investment deals across 21 SWF countries. Within-deal comparisons show that SWF involvement alone does not predict directed allocation. The differential appears only among higher-discount firms: a one-standard-deviation increase in the discount raises the SWF–non-SWF expansion gap by 3.7 percentage points, relative to a 6.5% baseline probability of entering a non-SWF co-investor's home market. I then test the model's separation between financing terms and directed activity. Favorable financing terms can compensate recipients for accepting costly directed activity, but the directed response depends on the recipient's outside financing option. Consistent with this separation, post-money valuation to revenue, deal size to revenue, and up-round status have little explanatory power for the SWF–non-SWF directed-expansion differential once valuation discount is included. The relevant margin is recipient financing friction, not round-level pricing.

I next turn to a crisis setting that provides plausibly exogenous variation in banks' need for outside capital: recapitalizations during the 2007–2008 financial crisis. During the acute phase of the crisis, SWFs were a major source of external bank recapitalization.[4] Banks with larger write-downs, driven by differences in pre-crisis portfolio exposure, had greater need for outside capital. Using DealScan syndicated lending data, I find that banks with larger write-downs relative to pre-crisis common equity disproportionately redirect lending toward SWF countries. With the full set of fixed effects and bilateral controls, a one-standard-deviation increase in write-down severity is associated with roughly 20% higher deal counts among bank–country–quarter cells with positive lending.

I then test the crowd-in mechanism in two complementary macro settings. The question is whether the firm-level and bank-level reallocations documented above are followed by broader bilateral crowd-in in FDI and banking claims. I exploit the 2018 Foreign Investment Risk Review Modernization Act (FIRRMA), which tightened U.S. scrutiny of foreign government-linked investments (Zhang and Zhou, 2025). SWF countries with greater pre-2018 exposure to U.S. high-equity-stake deals were more affected by the shock and non-U.S. high-equity-stake SWF deal volume rose after 2018. If SWF-backed activity builds bilateral investment links that lower bilateral frictions, more third-party capital should flow from non-U.S. source countries to SWF countries. I measure pre-FIRRMA exposure using PitchBook records of U.S. high-equity-stake deals and use

[4] The International Monetary Fund's *Global Financial Stability Report* (April 2008). Public offerings and other institutional investors were the other major sources of funding.

World Bank bilateral FDI inflows as the outcome. Conditional on positive flows, the observed exposure levels in Singapore, China, and Australia imply roughly 5–8% higher post-FIRRMA FDI inflows relative to SWF destinations with zero exposure. This magnitude is comparable to benchmark estimates of the bilateral FDI effect of signing a bilateral investment treaty.[5]

Finally, I test whether the same crowd-in mechanism operates in banking using the Bank for International Settlements (BIS) Locational Banking Statistics. In a staggered-cohort design, I study SWF equity injections into banks and trace how lender-country banking claims on SWF home countries evolve after the first injection. Each additional \$1 billion of SWF equity is associated with a 5.8% increase in loans and placed deposits to SWF countries. The similar order of magnitudes across the FDI and banking settings suggests that SWF-backed relationships are followed by broader bilateral capital-flow responses, consistent with the model's crowd-in channel.

Sovereign capital transactions are large, strategic, and infrequent, so no single setting can fully isolate every margin of the mechanism. I therefore combine four complementary designs that vary in source of identification, outcome, and level of aggregation. The first two settings test directed allocation at the firm and bank level; the last two test whether those directed links are followed by broader bilateral responses in FDI and banking claims. Because the identifying threats differ across settings, the common pattern across designs is informative: sovereign capital is followed by real activity and capital flows toward SWF home countries, consistent with the model.

Related literature. This paper contributes to three literatures. First, it adds to work on geoeconomics by showing how state capital can create, not just exploit, cross-border dependencies. Clayton, Maggiori and Schreger (2025, 2026) provide theoretical frameworks in which states use existing economic relationships to project power, and Farrell and Newman (2019) formalize how control over network hubs enables coercion. These papers model geoeconomic *pressure*: the threat or exercise of power through existing economic dependencies. Empirically, Clayton et al. (2025) document how China sequences foreign investor entry into its bond market, and Kabir et al. (2025) show that state-directed trade finance through export credit agencies has large real effects on exports, especially for firms facing bilateral frictions. My paper formalizes a complementary channel, *geofinance*, in which state capital creates new bilateral ties rather than leveraging existing ones. The paper traces the mechanism from individual equity deals to aggregate bilateral capital flows.

Second, it adds to the literature on sovereign wealth funds and state capitalism by shifting attention from target-firm outcomes to the geography of post-investment activity. Existing work

[5] Egger and Merlo (2007) estimate that bilateral investment treaties raise bilateral FDI stocks by about 4.8% in the short run and 8.9% in the long run.

establishes that government-linked acquirers exhibit different behaviors than private corporate investors (Karolyi and Liao, 2017) and evaluates how SWF investments affect target firms' stock prices, governance, and operating performance (Kotter and Lel, 2011; Bernstein, Lerner and Schoar, 2013; Bortolotti, Fotak and Megginson, 2015). Recently, Zhang and Zhou (2025) show that SWF investments increase home-country exports, using FIRRMA as a source of identification. I study a different mechanism and a broader set of financial outcomes. The paper shows that directed activity is strongest when recipients face tighter financing constraints, and that it extends beyond trade to bilateral FDI and banking claims.

Third, it adds to international finance by showing SWF investment as a channel that creates bilateral investment links and lowers bilateral frictions for later investors. In both trade and international finance, bilateral frictions are central determinants of cross-border flows (Anderson and van Wincoop, 2003; Portes and Rey, 2005). In foreign direct investment, these frictions include institutional, informational, and control barriers: corruption and weak institutions depress FDI, while distance-like barriers shape bilateral ownership and investment patterns (Wei, 2000; Head and Ries, 2008; Daude and Fratzscher, 2008). A related strand emphasizes that bilateral frictions are not necessarily fixed. In Chaney (2014), firms trade more when they have larger networks because existing contacts reduce the search costs of finding new foreign buyers. My paper brings a similar dynamic logic to cross-border capital allocation. I show that SWF investments create bilateral links that lower relationship frictions between recipients and SWF home countries and crowd in third-party capital flows.

The remainder of the paper is organized as follows. Section 2 develops the geofinance model and derives the main empirical predictions. Sections 3 and 4 present the micro evidence from PitchBook and DealScan. Section 5 studies the macro crowd-in effects using FDI and BIS banking claims data. Section 6 concludes.

2 A Framework for Geofinance

I model geofinance as a contract in which a financier with low funding costs uses financing rents to induce costly activity toward its home market. The wedge between the financier's cost of capital and the recipient's outside financing cost is the source of that rent. The model has three implications. First, conditional on this contracting relationship, the financing-rent channel is stronger when recipients face high financing frictions. Second, the contract rate satisfies participation; directed activity is governed by the financier's network motive and the recipient's private marginal cost. Third, directed activity expands bilateral links, lowers relationship frictions for outside investors, and crowds in third-party capital. I refer to this mechanism as *geofinance*: financiers with a funding advantage exchange financing rents for non-pecuniary objectives.[6]

2.1 Setup

Let $\mathcal{I}$ be a finite set of potential recipient firms. Firm i faces an effective marginal financing cost $r_i > 0$ per unit of capital. I normalize the opportunity cost of internal equity to zero, so r_i captures the private external-finance wedge. In the bank application, this wedge reflects regulatory leverage constraints and balance-sheet distress; in the corporate application, it reflects costly outside financing.

Firm i also has ordinary home-market activity, pre-optimized at

$$A_i(r_i) \equiv \max_{k_{ih} \geq 0} \{F_i(k_{ih}) - r_i k_{ih}\}. \tag{1}$$

This value is additively separable from strategic-market deployment and drops out of the marginal conditions below. The firm chooses strategic-market activity $k_{is} \geq 0$, with return $G_i(k_{is})$. Strategic-market activity carries a proportional cost $\mu(Z)k_{is}$, where Z is the stock of bilateral links. Since $\mu'(Z) < 0$, accumulated links lower the cost of later strategic-market activity.[7] I assume the home-market problem defining $A_i(r_i)$ is well behaved; G_i is twice differentiable, strictly increasing, and strictly concave; $\mu(Z) > 0$ and $\mu'(Z) < 0$; and $U(Z)$ is increasing and weakly concave.

A strategic financier has cost of funds $\rho \geq 0$ and can offer capital at an effective financing rate

$$r_{is} < r_i, \tag{2}$$

conditional on the firm undertaking strategic-market activity. The recipient's financial rent from x_i units of strategic finance is $(r_i - r_{is})x_i$, while the financier's financial margin is $(r_{is} - \rho)x_i$. The baseline focuses on the interior regime in which the contract rate can adjust freely to satisfy participation.

I begin with the firm's allocation under purely commercial finance. Taking Z as given, the firm

[6] The framework draws on the geoeconomic setting of Clayton, Maggiori and Schreger (2025, 2026). They formalize geoeconomic power through trade networks and threats. My framework treats the reduction of financial friction as the source of power. I integrate the principal-multi-agent contracting structure of Segal (1999), the public-liquidity logic of Holmström and Tirole (1998), and the "large trader" coordination mechanism of Corsetti et al. (2004).

[7] The model treats $\mu(Z)$ as a reduced-form bilateral relationship friction. It can reflect search, information, contracting, or coordination costs that decline as bilateral activity accumulates. In Section 5, I show evidence that, in the sovereign-capital setting, this friction operates primarily through relationship channels.

solves

$$\max_{k_{is}\geq 0}\left\{A_i(r_i)+G_i(k_{is})-\mu(Z)k_{is}-r_ik_{is}\right\}. \tag{3}$$

For an interior solution, the first-order condition for strategic-market activity is

$$G_i'(k_{is}^{\text{out}})=\mu(Z)+r_i. \tag{4}$$

Strategic-market activity is limited by two frictions: a common bilateral friction $\mu(Z)$, the cost of operating in an unfamiliar market, and a firm-specific financing friction r_i, the private cost of external finance. Holding Z fixed, firms with higher financing costs choose less strategic-market activity.

2.2 The contract and the firm's response

The strategic financier offers $\{x_i, r_{is}, k_{is}\}$: capital $x_i \geq 0$ at rate r_{is}, conditional on strategic-market activity k_{is}. I interpret k_{is} as verifiable through project finance, ring-fencing, covenants, regulatory commitments, or other observable deployment restrictions. After accepting, the firm does not choose k_{is} freely. The amount x_i delivers the financing rent and need not be proportional to k_{is}.

The firm's payoff under the strategic contract decomposes into an operating component and a financial rent:

$$\Pi_i(x_i, r_{is}, k_{is}, Z)=\underbrace{\text{Fund}_i(k_{is}, Z)}_{\text{operating value given } k_{is}}+\underbrace{(r_i-r_{is})\,x_i}_{\text{financial rent}}, \tag{5}$$

where

$$\text{Fund}_i(k_{is}, Z)\equiv A_i(r_i)+G_i(k_{is})-\mu(Z)k_{is}-r_ik_{is}. \tag{6}$$

The rent term isolates the wedge-based transfer that compensates the firm for accepting a level of strategic-market activity it would not otherwise choose. The firm accepts the contract whenever

$$\Pi_i(x_i, r_{is}, k_{is}, Z)\geq v_i(Z), \tag{7}$$

where the outside option under commercial finance is

$$v_i(Z)=\max_{k_{is}\geq 0}\left\{A_i(r_i)+G_i(k_{is})-\mu(Z)k_{is}-r_ik_{is}\right\}. \tag{8}$$

The implemented level of strategic-market activity can be represented by an implicit marginal wedge τ_{is}:

$$G_i'(k_{is}) = \mu(Z) + r_i - \tau_{is}. \tag{9}$$

This equation is not an additional primitive constraint. It defines the wedge required to implement a given k_{is} relative to the commercial benchmark in Equation (4). A positive τ_{is} lowers the effective marginal cost of strategic-market activity and therefore induces $k_{is} > k_{is}^{\text{out}}$ at a given Z.

Finally, let $C \subseteq \mathcal{I}$ denote the contracted set of strategic-finance recipients. I take this set as given in the benchmark contracting problem and characterize optimal contracts conditional on C. Firms outside C choose their commercial strategic-market activity after observing the network stock. The aggregate network stock is

$$Z = \sum_{i \in C} k_{is} + \sum_{j \notin C} k_{js}^*(Z), \tag{10}$$

where the second term captures crowd-in. Under the maintained assumptions, outside firms respond positively to a larger network stock:

$$\frac{\partial k_{js}^*}{\partial Z} = \frac{\mu'(Z)}{G_j''(k_{js}^*)} > 0. \tag{11}$$

For local stability of the network fixed point, I impose

$$0 \leq \sum_{j \notin C} \frac{\partial k_{js}^*}{\partial Z} < 1. \tag{12}$$

2.3 The financier's problem and main results

The financier acts as a Stackelberg leader conditional on a contracted recipient set C. It values aggregate network stock Z because its strategic payoff depends on the overall bilateral network it builds, rather than on any single contract in isolation. With strategic-objective weight $\theta > 0$, aggregate funding capacity $\bar{X}$, and convex implementation cost $\frac{c}{2}x_i^2$ with $c > 0$, it chooses contracts $\{x_i, r_{is}, k_{is}\}_{i \in C}$, and the resulting network stock Z to solve

$$
\begin{aligned}
\max_{\{x_i, r_{is}, k_{is}\}_{i\in C},\, Z} \quad & \sum_{i\in C} \left[(r_{is} - \rho)x_i - \tfrac{c}{2}x_i^2 \right] + \theta U(Z) \\
\text{subject to:} \quad & \Pi_i(x_i, r_{is}, k_{is}, Z) - v_i(Z) \geq 0, \qquad \forall i \in C, \quad [y_i] \\
& \bar{X} - \sum_{i\in C} x_i \geq 0, \qquad [\lambda] \\
& \sum_{i\in C} k_{is} + \sum_{j\notin C} k^*_{js}(Z) - Z = 0, \qquad [\Psi_Z] \\
& x_i \geq 0, \quad k_{is} \geq 0, \qquad \forall i \in C.
\end{aligned}
\tag{13}
$$

All inequality multipliers are nonnegative. The equality multiplier Ψ_Z is unrestricted; under the sign convention in Equation (13), a positive value is the financier's shadow value of expanding the bilateral network stock. The first-order conditions below characterize contracted recipients with $x_i > 0$, interior k_{is}, and an interior contract rate, conditional on the contracted set C.

The contract-rate first-order condition is

$$y_i = 1. \tag{14}$$

To see this, only two terms involve r_{is}: $(r_{is} - \rho)x_i + y_i(r_i - r_{is})x_i$. The first-order condition is $x_i(1 - y_i) = 0$, so for $x_i > 0$, $y_i = 1$ and participation binds. The rate sets surplus division in the interior pricing regime; it does not set funding volume or directed activity.

The remaining first-order conditions yield three main results, stated below as propositions. Appendix A provides the full derivations.

Proposition 1 (Financing Power and Induced Action)**.** *For any contracted recipient with an interior strategic-market choice, the optimal implementation wedge is*

$$\tau^*_{is} = \Psi_Z. \tag{15}$$

Therefore, holding Z fixed, the financier induces $k_{is} > k^{out}_{is}$ whenever $\Psi_Z > 0$.

Intuition. Financing rents compensate the firm for moving away from its commercial benchmark. The wedge Ψ_Z is the financier's shadow value of expanding the network stock. When the financier places positive shadow value on expanding the bilateral network, it uses cheap finance to support a directed allocation above the firm's commercial choice.

Proposition 2 (Financial friction and financing volume)**.** *For any contracted recipient with an* *interior financing choice, equilibrium financing volume satisfies*

$$x_i^* = \frac{r_i - \rho - \lambda}{c}. \tag{16}$$

Under the nonnegativity constraint on financing, active funding for a contracted recipient requires

$$r_i > \rho + \lambda. \tag{17}$$

Thus, conditional on participation feasibility and the chosen contracted set, financing volume loads on the recipient's effective outside financing cost r_i, the financier's own cost ρ, and aggregate funding scarcity λ; the contract rate r_{is} drops out of the marginal funding condition.

Intuition. Once $y_i = 1$, the contract rate cancels from the funding first-order condition.[8] The rate allocates surplus, while funding volume depends on the value of replacing outside finance. For a candidate contract, a higher r_i supports a larger financing volume and makes the funding margin easier to satisfy. This result does not imply that financing volume maps one-for-one into directed activity. The financing amount x_i provides the participation rent; the directed action k_{is} is pinned down by the network wedge Ψ_Z.[9]

Proposition 3 (Network multiplier and crowd-in)**.** *Strategic finance affects outside firms through the endogenous network stock Z. Under the stability condition in Equation (12), the shadow value of network expansion is*

$$\Psi_Z = \left[1 - \sum_{j \notin C} \frac{\mu'(Z)}{G_j''(k_{js}^*)}\right]^{-1} \left[\theta U'(Z) - \mu'(Z) \sum_{i \in C} (k_{is} - k_{is}^{out})\right]. \tag{18}$$

The multiplier term is positive and finite, and $\Psi_Z > 0$ whenever

$$\theta U'(Z) - \mu'(Z) \sum_{i \in C} (k_{is} - k_{is}^{out}) > 0. \tag{19}$$

Intuition. The first bracketed term is the crowd-in multiplier. It is finite by the stability condition. It

[8]The cancellation of r_{is} does not require literal risk neutrality by the recipient. It follows from transfer separability: the contract rate and funding volume enter the recipient's payoff through the financing rent $(r_i - r_{is})x_i$. The result can fail when the rate changes payoff risk, default incentives, limited-liability constraints, or the marginal cost of directed activity.

[9]The model does not solve the discrete choice of C; instead, it characterizes the contracting margin conditional on a recipient set. The separation result is an interior-pricing result. With transfer separability and unrestricted transfers, bargaining affects surplus division but not the implemented allocation. Additional binding feasibility constraints can alter the allocation condition, in which case the financier may scale back the action or drop the firm.

equals one when the network does not lower outside frictions ($\mu' = 0$) and exceeds one when a larger network stock lowers outside firms' bilateral frictions ($\mu' < 0$). The second bracket contains the financier's direct value of network expansion and the friction-reducing effect of a larger network. Crowd-in is stronger when $|\mu'(Z)|$ is large and outside firms respond more elastically (small $|G_j''|$).

Extensions. Appendix A studies two extensions. In a repeated relationship, the firm complies because deviating would end future access to cheap financing. In the binding-dynamic incentive compatibility, slack-participation regime, the dynamic extension gives $\tau_{is}^{\text{dyn}} = \beta\tau_{is}^{\text{static}} = \beta\Psi_Z$, where $\beta \in (0, 1)$ is the firm's discount factor. Relational enforcement therefore attenuates the static covenant-enforceable wedge, since more patient firms can sustain a larger fraction of that static wedge. The appendix also studies a first-best planner, clarifying which externalities the strategic financier does and does not internalize.

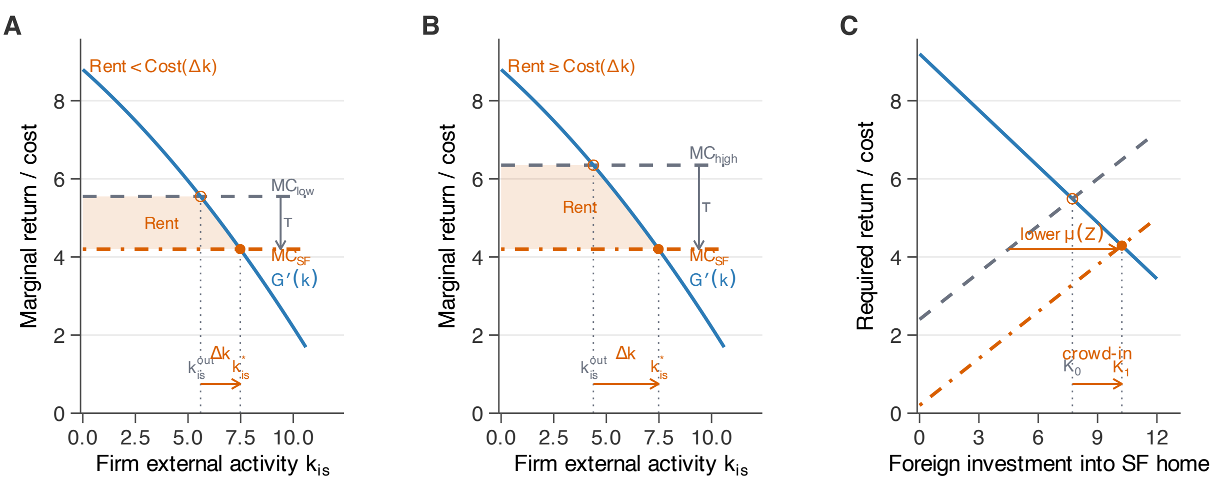

Figure 1: Geofinance: Firm Financing Wedge and Aggregate Spillover

Notes: In Panels A and B, the firm's marginal return on strategic-market activity is $G_i'(k_{is})$. The commercial benchmark is $\text{MC} = \mu(Z) + r_i$, and the strategic-finance contract lowers the effective marginal cost to $\text{MC}_{\text{SF}} = \mu(Z) + r_i - \tau_{is}^*$. In the baseline model without tied financing, $\tau_{is}^* = \Psi_Z$. Panel A illustrates a case in which the available financing rent is too small to satisfy participation at the desired directed activity. Panel B illustrates a case in which the contract implements a positive wedge $\tau_{is}^* > 0$, generating $k_{is}^* > k_{is}^{\text{out}}$. Panel C shows how the induced activity raises Z, lowers $\mu(Z)$, and expands third-party capital from K_0 to K_1.

2.4 From model to tests

Figure 1 summarizes the mechanism and its empirical implications. The main testable gradient is conditional: among recipients in a geofinance relationship, the SWF-directed expansion gap should be larger when outside finance is costly, because the financing rent is more valuable to the recipient.

In PitchBook, I test this within financing rounds by asking whether the SWF–non-SWF expansion differential is larger for targets raising capital at larger valuation discounts. The non-SWF co-investors provide the empirical benchmark for ordinary investor-linked expansion. In the bank setting, the analogous financing-friction proxy is pre-crisis write-down severity. Proposition 2 implies that pricing terms need not predict directed activity once financing frictions are accounted for. I test this in PitchBook by comparing valuation-discount proxies with round-level pricing and financing-package measures. Proposition 3 implies that directed activity lowers bilateral frictions for other investors and crowds in third-party capital. I test this aggregate implication using bilateral FDI inflows and BIS cross-border banking claims. These specifications are reduced-form counterparts to the model's comparative statics; they do not separately identify Ψ_Z, θ, or $\mu(Z)$.

3 Directed Expansion in Private-Market Co-Investments

This section tests the model's relative financing-friction prediction in private-market co-investments. The sample consists of PitchBook financing rounds in which at least one SWF and one non-SWF institutional investor invest in the same target. For each investor–deal pair, I measure whether the target later expands into the investor's home country. The design therefore compares expansion toward the SWF investor's home country with expansion toward non-SWF co-investor home countries within the same financing round.[10] SWF participation alone does not predict directed expansion. The SWF–non-SWF expansion differential widens only when the target raises equity at a larger valuation discount relative to sector–year peers.

3.1 Data and empirical design

Data. Using PitchBook, I construct a 1998–2025 sample of co-investment rounds in which at least one SWF and one non-SWF institutional investor participate in the same deal. Appendix B.1 documents the data construction and cleaning steps.

Sample. The unit of observation is the investor–deal pair. I impose three restrictions. First, I drop same-country pairs, where the target's headquarters country matches the investor's home country, to avoid classifying home-market activity as directed expansion. Second, I drop non-SWF investors located in the same country as the deal's SWF, because if the SWF induces entry into

[10] The within-deal investor-level design follows the within-borrower design of Khwaja and Mian (2008), applied here to investor–deal pairs.

its home market, those co-investors mechanically receive $Y = 1$ without independent evidence of redirection, which contaminates the control group.[11] Third, for companies with multiple rounds, I drop later-round rows where the same investor already appeared in an earlier round to avoid double-counting persistent investor-target relationships across rounds. The restrictions produce a broader cleaned co-investment panel of 10,101 investor–deal pairs across 2,476 deals and 2,417 companies, with SWF investors from 24 home countries. The main tests require a non-missing valuation discount, the financing-friction proxy introduced below. This valuation-discount sample contains 3,713 investor–deal pairs across 882 deals and 850 companies, with SWF investors from 21 home countries.[12]

Outcome variables. I identify expansion events using two identifier-based matching routes in PitchBook (Appendix B.1).[13] An expansion event is a subsequent PitchBook-recorded transaction by the target in the investor's home country, including acquisitions, joint ventures, buyouts, and venture investments. The extensive margin outcome is new market entry: $Y_{\text{ext}} = 1$ if the target expanded into the investor's home country within five years after the deal but had no prior expansion there in PitchBook's recorded history.[14] The other outcomes are the inverse hyperbolic sine of the total number of expansion events and their total recorded value (in $ millions) within the same five-year window.[15]

Key regressor: valuation discount. The model predicts that, conditional on SWF financing, the SWF-country expansion gap should be larger when the target's external finance is expensive. It does not require high-friction firms to expand more in absolute terms. For each deal d in sector s and year t, I construct a valuation discount as the negative of the deal's log post-money valuation

[11] Appendix Table 23 shows that this restriction is conservative. Re-adding the dropped rows leaves the SWF × Discount coefficient virtually unchanged, and dropping the full deal whenever the SWF's home country has more than one investor in the round strengthens the estimate.

[12] In the valuation-discount sample, the typical deal has one SWF and roughly three institutional co-investors: the mean is 4.21 investor rows per deal and the median is 3. 36% of deals pair a single SWF with a single non-SWF investor; 89% involve only one SWF.

[13] Matches use only PitchBook entity IDs and there is no fuzzy name matching. As a result, the measure may miss expansion activity conducted under alternate legal names, subsidiaries, or affiliates that are not linked to the focal firm in PitchBook. This creates potential attenuation from missed true expansions rather than false positives from incorrect matches.

[14] For later financing rounds whose five-year post-deal window extends beyond the latest observed PitchBook expansion data, outcomes are measured through the end of the observed expansion window. I do not condition on firm survival or continued PitchBook coverage; the issue is only calendar observability. Because the regressions include deal fixed effects, all investor-country rows within a financing round share the same available post-deal window.

[15] $\operatorname{asinh}(y) = \log\left(y + \sqrt{y^2 + 1}\right)$, which behaves like log at large y but is defined at $y = 0$. I use it here to keep zero-expansion observations in the sample.

residual relative to the sector×year mean[16]:

$$\text{ValDiscount}_d = -\left(\log(\text{PostVal}_d) - \overline{\log(\text{PostVal})}_{s(d),t(d)}\right), \tag{20}$$

where the benchmark is computed across all completed deals in sector s and year t from the full PitchBook deal universe.[17] Higher values indicate lower valuations relative to sector×year peers and therefore a higher cost of equity r_i. Because the measure is based on post-money valuation, it may also capture firm scale or round size, not just financing frictions.The robustness tests use alternative proxies (a pre-money-based valuation discount and the dilution ratio based on deal size over post-money valuation) and add revenue controls.[18]

Specification. I estimate

$$Y_{id} = \beta_1\,\text{SWF}_i + \beta_2\,(\text{SWF}_i \times \text{ValDiscount}_d) + \alpha_d + \alpha_{h(i)\times t(d)} + \varepsilon_{id}, \tag{21}$$

where Y_{id} is the post-deal expansion outcome for investor i in deal d. β_1 captures the within-deal SWF–non-SWF expansion differential at the mean discount. β_2 captures whether that differential is larger in higher-discount deals. Deal fixed effects α_d absorb firm- and round-specific shocks, while investor-country×deal-year fixed effects $\alpha_{h(i)\times t(d)}$ absorb time-varying differences in the attractiveness of investors' home markets.

3.2 Directed Expansion

Composition in SWF-backed rounds. Appendix Table 15 reports composition patterns for my main regression sample. In rounds that include an SWF and at least one institutional non-SWF investor, higher valuation discount and higher dilution are associated with a larger SWF fraction among the investors.[19] I treat this pattern as descriptive. It is not a direct test of the model's

[16] I use post-money rather than pre-money valuation because PitchBook reports post-money for roughly twice as many completed deals (21.2% versus 11.0% of the universe). The two values are correlated at over 95%.

[17] The measure is winsorized at the 1st and 99th percentiles and standardized to unit standard deviation.

[18] As an additional check, I also considered size, age, and round-history proxies for financing frictions. Hadlock and Pierce (2010) argue that firm size and age are among the most useful observable predictors of financial-constraint levels. In this sample, smaller-revenue and smaller-employee targets show the same standalone pattern as valuation discount, while firm age and down-round status are weaker and less precisely measured. When these proxies are horse-raced against valuation discount, they do not add independent explanatory power. Appendix Table 24 reports the related pre-money valuation and revenue-control checks.

[19] I use dilution, measured as deal size over post-money valuation, as an alternative proxy for financing frictions because firms facing tighter external-finance constraints typically sell a larger ownership stake to raise a given amount of capital.

financing volume x_i: the SWF fraction can vary with the number of retained co-investors in a round, and PitchBook rarely reports investor-level ticket amounts. The main test below therefore does not rely on this composition pattern. It asks whether, within the same financing round, the SWF–non-SWF directed-expansion differential is larger for higher-discount targets.

Main results. Column (1) reports the baseline within-deal SWF effect and shows no evidence that SWF-backed firms expand more into SWF markets on average. Column (2) adds the SWF×valuation-discount interaction with deal fixed effects. The interaction is positive, meaning that the SWF–non-SWF expansion differential is larger among targets priced at a deeper valuation discount. Column (3) adds controls for the target's pre-deal geographic orientation toward each investor country, and the coefficient changes little; the result is therefore not explained by firms already doing business in SWF countries before the financing round. Column (4) adds investor-home-country×deal-year fixed effects, absorbing time-varying differences in destination-country attractiveness. The coefficient rises to 0.037. A one-standard-deviation increase in valuation discount therefore raises the within-deal SWF–non-SWF differential in new-entry probability by 3.7 percentage points, slightly more than one-half of the non-SWF control mean of 6.5%.[20] For the aggregate outcomes, the corresponding coefficients are 0.103 for asinh(count) and 0.213 for asinh(value).[21] The value estimate is less precise because deal sizes are missing for about 54% of expansion events, but the sign and economic pattern are the same.

Total effect and the marginal firm. The interaction $\hat{\beta}_2$ shows how the within-deal SWF gap varies with valuation discount.[22] It is a differential slope: it does not say that high-discount firms expand more into SWF countries in absolute terms, but that the SWF–non-SWF expansion gap narrows and eventually turns positive as valuation discount rises. In column (4), the SWF indicator is -0.0322. At the mean valuation discount, an SWF-backed target is therefore less likely to expand into the SWF's home country than into a non-SWF co-investor's home country. This negative baseline is consistent with the fact that Singapore, the United Arab Emirates, and Kuwait are smaller and less accessible markets than the United States, the United Kingdom, or France. The

[20] Out of 100 non-SWF investor–deal pairs in the sample, the target expands into that non-SWF investor's home country within five years in roughly 6.5 cases. The low baseline reflects the sparsity of bilateral cross-border expansion. Non-SWF co-investors are concentrated in deep markets, while SWF investors are concentrated in smaller economies. For example, Singapore accounts for 20 new-entry events in the sample, compared with 121 for the United States.

[21] As a robustness check, I replace investor-home-country×deal-year fixed effects with investor-home-country×sector×deal-year fixed effects. The SWF×valuation-discount coefficients remain positive, significant and similar in magnitude.

[22] With deal fixed effects alone, this is the weighted covariance between the within-deal SWF–non-SWF expansion gap and the deal's financing-friction proxy.

Table 1: Valuation Discount, SWF Co-Investment, and Business Expansion

	New Entry				asinh(Count)	asinh(Value)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Valuation Discount (primary)						
SWF × Val. Discount		0.0219**	0.0260***	0.0371***	0.1028***	0.2131***
		(0.0087)	(0.0092)	(0.0117)	(0.0171)	(0.0522)
SWF	-0.0188	-0.0194**	-0.0226**	-0.0322	-0.0092	-0.1024
	(0.0207)	(0.0085)	(0.0088)	(0.0211)	(0.0382)	(0.0901)
Pre-acq. share			-0.0129**			
			(0.0062)			
Pre-acq. share × Val. Disc.			0.0128**			
			(0.0058)			
Panel B: Dilution (robustness)						
SWF × Dilution		0.0171**	0.0187**	0.0211**	0.0525***	0.1312***
		(0.0071)	(0.0074)	(0.0091)	(0.0142)	(0.0379)
Deal FE	✓	✓	✓	✓	✓	✓
Investor-country × Deal-year FE	✓	–	–	✓	✓	✓
R^2	0.647	0.540	0.545	0.650	0.783	0.730
SE clustered by company	✓	✓	✓	✓	✓	✓
Control mean	0.0649	0.0649	0.0649	0.0649	0.1897	0.3795
Observations (Panel A)	3,713	3,713	3,713	3,713	3,713	3,713
Observations (Panel B)	–	3,660	3,660	3,660	3,660	3,660

Note: Investor–deal level regressions. Column (1) regresses new entry on the SWF dummy under the most-saturated fixed effects (deal and investor-country×deal-year), providing the cleanest unconditional baseline. Columns (2)–(4) add the SWF×valuation discount interaction under progressively richer controls: deal FE only (2); deal FE plus pre-deal geographic orientation controls (3); deal FE plus investor-country×deal-year FE (4). Columns (5)–(6) report the most-saturated specification for asinh(count) and asinh(value). The new entry outcome equals one if the target had no prior expansion in the investor's home country before the deal but did expand within five years after the deal, or by the end of the observed PitchBook expansion window for later deals. The aggregate outcomes count and value are the total number of expansion events and the total recorded value of those events in the investor's home country over the same five-year window. For the asinh outcomes, the reported control mean is the mean of the transformed dependent variable for non-SWF co-investors. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

total within-deal SWF effect, $-0.0322+0.0371\times$ValDiscount, turns positive at roughly one standard deviation above the mean. This is the empirical counterpart to Proposition 1: the SWF-country gap turns positive only when the financing rent is large enough to offset the cost of entering the SWF market. [23]

Follow-on financing. The repeated-relationship extension in Appendix A gives another reason why directed expansion can be sustained after the initial financing round: a firm that values future SWF financing has an incentive to maintain the relationship. Appendix Table 16 examines this continuation margin directly. SWF investors are more likely than same-deal non-SWF co-investors to return to the same portfolio company within five years, even after controlling for each investor's prior follow-on propensity and same-home-country co-investors in the original round. I then ask whether this continuation margin explains the directed-expansion result. The new-entry coefficient on SWF×valuation discount is nearly unchanged after adding SWF-home and generic home-country-sector thickness measures. This suggests that first entry remains tied to the financing-friction proxy rather than to thick investor markets by themselves. For broader post-deal expansion, however, the count and value responses are larger in thicker generic home-country sectors, consistent with later expansion being stronger where follow-on financing opportunities are more abundant.[24]

Dynamic evidence. Figure 2 reports the dynamics for expansion count and volume. For each non-overlapping 12-month bin relative to the deal date, I re-estimate the same SWF×Valuation Discount interaction with deal FE and investor-country×deal-year FE, using a bin-specific outcome. The pre-period estimates are mostly small and imprecisely estimated, with no monotone run-up before the financing round. After the round, the count response turns positive and is strongest in the first two post-deal years. The value response is noisier, but it is positive in the first post-deal bins and is estimated most precisely at $\tau = +2$. The timing is consistent with directed expansion following SWF financing, rather than a smooth pre-existing trend.

[23] The crossover at ValDiscount $\approx$ 1 SD identifies the model's marginal firm, the threshold at which the rent transfer just compensates the cost of entering the SWF's market. The negative level combined with the positive interaction is also inconsistent with a pure selection story: if SWFs simply picked firms already planning to expand everywhere, the SWF level coefficient would be positive.

[24] I follow the relational-contracting logic in (Gil and Marion, 2013), which uses market opportunities to proxy the continuation value of a relationship. I implement this idea with an ex ante market-thickness measure: for each investor–deal row, I count prior five-year deals involving investors from the same home country and target firms in the same sector, take $\log(1+\text{count})$, and standardize the measure before estimation. The SWF-home measure counts SWF-backed deals; the generic measure counts non-SWF deep-pocket-investor deals.

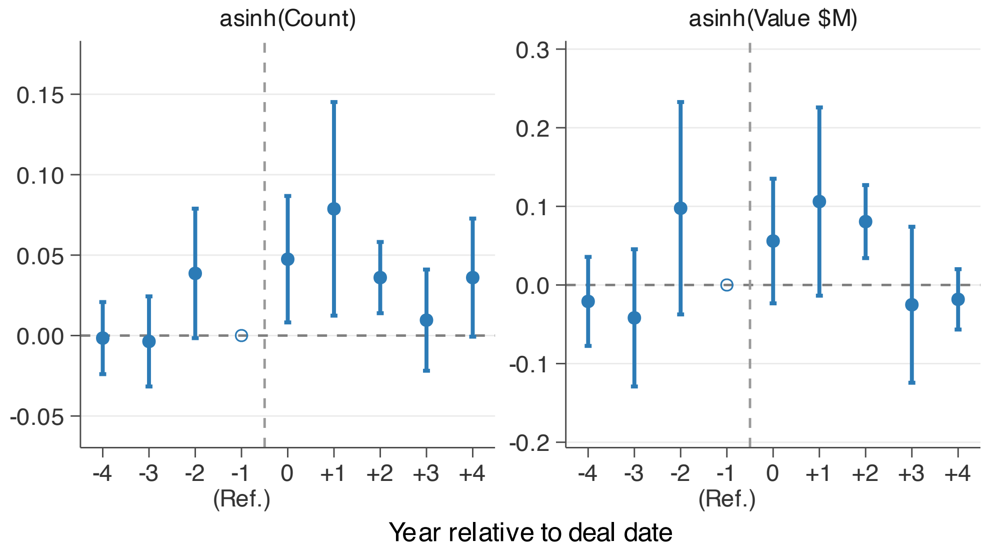

Figure 2: Dynamic Event Study: Expansion Count and Value

Notes: Each bin covers one non-overlapping 12-month window relative to the deal date. The underlying sample is the same investor–deal sample used in the baseline specification. All estimations include deal FE and investor-country×deal-year FE. Left: the asinh(expansion count) in the corresponding event-time bin. Right: the asinh(expansion value) (in $ millions) in the corresponding event-time bin. The value panel is less precise because PitchBook reports deal sizes for only 46% of expansion events.

3.3 Pricing Terms and Directed Expansion

Proposition 2 separates contract pricing from the directed response. Conditional on a recipient accepting the contract, pricing terms compensate participation; the directed activity depends on the recipient's outside financing cost. Because investors in the same round usually share the same headline valuation, I test this implication across SWF-backed rounds. Once valuation discount is included, round-level pricing and financing-package measures should have little independent power to predict directed expansion. I compare valuation discount, my proxy for the target's outside financing cost, with measures of round pricing and the size of the financing package relative to firm scale.

SWFs pay a financial premium. Figure 3 plots average log Price/Revenue (P/Revenue) for SWF and non-SWF deals within sector×year cells.[25] In roughly 66% of cells, SWF deals are priced higher.[26] Appendix Table 18 shows that the premium persists after adding sector×year fixed effects,

[25] Price/Revenue (P/R) denotes the deal's post-money valuation divided by the target firm's latest-twelve-month revenue.

[26] Using deal size relative to revenue as an alternative pricing proxy also shows that SWFs tend to participate in larger financing rounds.

round controls, and within-company fixed effects.

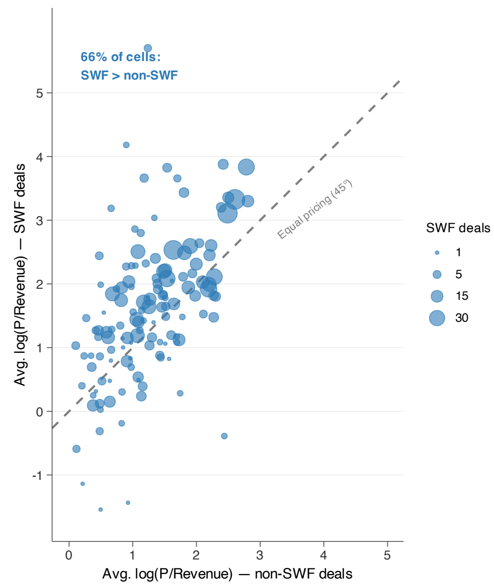

Figure 3: SWF Deals Command Higher Valuations Across Sector×Year Cells

Notes: Each dot represents one sector×year cell. The x-axis is the average log(Post-money valuation/Revenue) of non-SWF deals in that cell; the y-axis is the average of SWF deals in the same cell. Cells require at least one SWF deal and five non-SWF deals. Dot size is proportional to the number of SWF deals in the cell. Dots above the 45-degree line indicate that SWF deals in that sector-year were priced higher than non-SWF deals.

Round pricing does not predict directed expansion. Table 2 compares valuation discount with three round-level alternatives. Log(P/Revenue) and Up Round proxy for the price at which capital is provided. Log(DealSize/Revenue) captures the size of the financing package relative to firm scale. These measures are not investor-specific contract rates: PitchBook does not consistently report security-level prices, per-share terms or investor-level investments. The ideal test would compare the financing amount and contract prices offered by the same SWF across recipients. The available test is therefore more modest. It asks whether the SWF-directed expansion differential loads on observable round-level terms, or instead on the recipient's financing friction proxy.

The round-level alternatives are weak predictors of directed expansion. In the horse race, their interactions are small and statistically insignificant, while valuation discount remains positive and significant.[27] Across the three panels, the SWF–non-SWF directed-expansion differential loads on valuation discount rather than on round pricing or financing-package size.

[27] The coefficients are larger than those in the fully saturated main specification because Panels A and B condition on the smaller revenue-available subsample where the baseline valuation-discount effect is itself larger.

Table 2: Pricing Terms and Directed Expansion

	Proxy Only			Horse Race with Val. Discount		
	New Entry (1)	asinh(Cnt) (2)	asinh(Val) (3)	New Entry (4)	asinh(Cnt) (5)	asinh(Val) (6)
Panel A: log(P/Rev) as pricing proxy (r_{is}) — revenue subsample						
SWF × log(P/Rev)	-0.0175	-0.0002	0.0383	-0.0111	0.0154	0.0707
	(0.0153)	(0.0255)	(0.0622)	(0.0154)	(0.0243)	(0.0631)
SWF × Val. Discount				0.0664***	0.1575***	0.3238***
				(0.0213)	(0.0288)	(0.0904)
Observations	1,836	1,836	1,836	1,829	1,829	1,829
Panel B: log(DealSize/Rev) as financing quantity proxy — revenue subsample						
SWF × log(DealSize/Rev)	0.0134	0.0437	0.1322*	-0.0026	0.0081	0.0629
	(0.0186)	(0.0308)	(0.0752)	(0.0167)	(0.0281)	(0.0687)
SWF × Val. Discount				0.0736***	0.1629***	0.3178***
				(0.0209)	(0.0288)	(0.0891)
Observations	1,790	1,790	1,790	1,790	1,790	1,790
Panel C: Up Round as pricing proxy (r_{is}) — full sample						
SWF × Up Round	-0.0166	-0.0554	-0.1339	-0.0067	-0.0280	-0.0774
	(0.0269)	(0.0401)	(0.1207)	(0.0263)	(0.0386)	(0.1174)
SWF × Val. Discount				0.0367***	0.1012***	0.2086***
				(0.0116)	(0.0170)	(0.0514)
Observations	3,713	3,713	3,713	3,713	3,713	3,713
Deal + inv.-country × year FE	✓	✓	✓	✓	✓	✓

Note: Each panel horse-races a different pricing or financing-quantity proxy against valuation discount. Columns (1)–(3) include only the pricing proxy interacted with the SWF indicator; columns (4)–(6) add the SWF×valuation discount interaction. Panel A uses log(post-money valuation / revenue) as a deal-pricing proxy; Panel B uses log(deal size / revenue) as a financing-quantity proxy, measuring how much capital was raised relative to firm scale; Panel C uses an up-round indicator (1 if the round price exceeded the prior round). Panels A and B are estimated on the revenue subsample; Panel C uses the full sample. All specifications include deal and investor-country×deal-year fixed effects. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Redirection by financing friction. Figure 4 provides a descriptive complement to the separation regression. It conditions on deals with at least one post-deal expansion and asks where those expansion events occur. In Panel A, SWF countries account for 24% of post-deal expansion events in high-valuation-discount deals, compared with 10% in low-valuation-discount deals. In Panel B, there is no comparable difference by median P/Revenue: the SWF-country share is 10% on both sides of the median.[28]

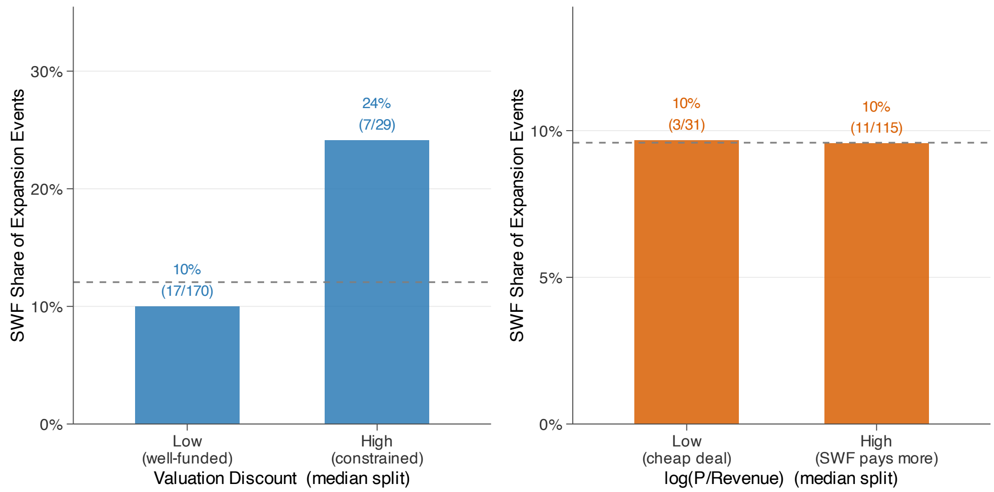

Figure 4: SWF-Directed Expansion: Redirection by Financing Frictions vs. Valuation Premium
Notes: Each bar reports the SWF share of post-deal expansion events among deals with at least one post-deal expansion. The numerator is the number of expansion events into the SWF's home country, and the denominator is the total number of expansion events in that bin. Panel A uses the full valuation-discount sample of 882 deals, of which 72 have any expansion, involving 199 investor–deal observations. It splits deals at the median valuation discount and shows that the SWF share of expansion events rises from 10% (17/170) below the median to 24% (7/29) above it. Panel B uses the revenue subsample of 399 deals, of which 47 have any expansion. It splits deals at the median log(P/Revenue) and shows nearly identical SWF shares below and above the median (10% (3/31) and 10% (11/115)). The dashed line marks the overall SWF share of expansion events in the corresponding panel.

3.4 Robustness

The robustness checks address three remaining selection and measurement concerns. First, firm-level selection could still matter if SWF-backed firms were already more likely to enter SWF markets despite the within-deal design. Second, measurement choices could affect the estimates: the valuation-discount proxy may partly capture firm scale or round size, and the asinh outcomes

[28] The Panel A pattern reflects a narrowing relative gap rather than an absolute increase in SWF-directed expansion. Absolute entry rates are lower in the high-valuation-discount bin for both SWF and non-SWF rows, but they fall much more sharply for non-SWF rows; Appendix Figure 9 reports the decomposition.

combine entry and scale responses. Third, the results could be driven by influential SWF home countries or by accidental assignment patterns within deals.

Matched counterfactual. The within-deal design absorbs deal-level heterogeneity, but it does not rule out the possibility that SWF-backed firms were already more likely to enter SWF countries even without sovereign participation. Appendix C.13 addresses this with a pair-level matched-counterfactual design. Treated and control observations are exactly matched by candidate SWF country, industry, broad headquarters region, and deal-year bin, with propensity-score overlap weights used within matched cells to balance remaining deal and firm characteristics. In the baseline, SWF-backed pairs expand into the SWF country at a 7.21% rate, compared with 3.23% for weighted control pairs. The overlap-weighted exact-cell regression estimates a 3.73 percentage-point increase (SE 1.17), slightly smaller than the raw weighted difference because exact-cell fixed effects absorb composition. Alternative weighting schemes and outcome windows remain positive, but the matched-counterfactual estimate is no longer statistically distinguishable from zero once Singapore-targeting pairs are excluded. This sensitivity is specific to the matched counterfactual; the within-deal leave-one-country-out estimator remains positive and significant when Singapore is omitted.

Valuation-discount robustness. Because valuation discount is constructed from post-money valuation, it may partly capture firm scale or round size rather than only financing frictions. Appendix Table 24 addresses this concern directly. In the revenue-available subsample, the post-money valuation-discount interaction remains positive and significant when horse-raced against SWF×log(revenue); the revenue interaction is small and statistically indistinguishable from zero. A parallel check using pre-money valuation, which removes the round-size component in post-money valuation, also predicts new entry and expansion count. The value coefficient is positive but imprecise. Neither firm scale nor the round-size component drives the directed-expansion result.

Conditional-positive outcomes. Because the inverse-hyperbolic-sine count and value outcomes pool zeros and positives, their coefficients combine new entry and scale effects.[29] Appendix Table 22 therefore reports log-on-positive specifications. On that subsample, the count and value coefficients are negative throughout, with some specifications estimated precisely. Among pairs that already expand, higher valuation discount is associated with smaller, not larger, SWF-relative transaction count and value.

[29] Chen and Roth (2024) show that the resulting average treatment effect has no clean percentage interpretation when the treatment also shifts the extensive margin.

Leave-one-out. Appendix Figure 8 drops each SWF country in turn and re-estimates the main new-entry specification. The coefficient remains positive across all 21 SWF countries and stays close to the full-sample baseline. Appendix Table 17 shows that the count and value results are stable to omitting individual sectors, although the new-entry result attenuates when consumer-facing targets are omitted. Appendix Tables 20 and 21 show that the result is stable to alternative winsorization and remains positive under 3-, 5-, and 7-year outcome windows. The window estimates rise modestly with the horizon, from 0.032 at three years to 0.041 at seven years.

Permutation inference. Appendix Table 19 reports two-sided permutation tests (1,000 draws). The model-free ratio statistic lies in the extreme tail of the placebo distribution ($p < 0.001$). For the regression-based estimates, the two-sided randomization-inference p-values are below 0.001 for new entry, expansion count, and expansion value. The observed pattern is therefore unlikely to be generated by randomly assigning SWF labels within deals.

The PitchBook evidence supports the micro-allocation prediction: SWF co-investments are followed by a relative tilt toward SWF home countries, compared with non-SWF co-investor home countries, but only when firms face stronger financing frictions. The design addresses firm-level selection through within-deal comparisons, matched counterfactuals, and permutation tests, but it does not make sovereign capital randomly assigned within financing rounds. I therefore interpret the estimates as evidence of directed expansion among co-investors in the same deal. I next turn to crisis-era bank recapitalizations, where pre-crisis write-down exposure generates cross-bank variation in demand for sovereign capital.

4 Directed Lending after Bank Recapitalizations During the Crisis

This section tests the financing-friction prediction in crisis-era bank recapitalizations. Using a bank–country–quarter DealScan panel of 18 large cross-border arrangers, I compare high- and low-write-down banks around 2007Q3. Banks with larger pre-crisis write-downs, which had greater demand for outside capital, increased syndicated-loan deal counts disproportionately toward SWF countries after the crisis cutoff.

4.1 Data and Empirical Design

Data. I assemble a bank–country–quarter panel of cross-border syndicated lending from Refinitiv DealScan for 2002–2015. Appendix B.2 details the panel construction, the exposure-allocation formula, and the loan-term variables. The sample contains 18 large global syndicated-loan arrangers, five of which received SWF equity during the 2007–2008 financial crisis and 13 of which did not.[30] For each bank–country–quarter cell I measure total loan volume and deal count. The realized panel contains 155 borrower destination countries, of which 14 are effective SWF-status destinations.

SWF equity injections. Five banks received equity injections from eight SWFs across seven SWF countries, for a total of ten events. I define 2007Q3–2008Q1 as the clean recapitalization window. Banks first reported large structured-credit write-downs in 2007Q3, and SWF equity injections began in 2007Q4. I end the treatment window in 2008Q1 to avoid contamination from later government recapitalization programs and late-crisis private placements.[31] This period also corresponds to the interval in which SWFs were a major source of external recapitalization, providing roughly 39% of all bank capital raised (International Monetary Fund, 2008).

Key regressors. I measure bank capital need with WriteDownRatio_b, defined as cumulative structured-credit write-downs in 2007Q3 and 2007Q4 divided by 2006 common equity.[32] Higher values indicate a larger crisis-era hit to bank capital. SWFStatus_c indicates whether country c was home to an active cross-border SWF as of 2006. This is an ex ante destination-level treatment: it classifies countries by the pre-crisis presence of sovereign investors, not by whether a particular bank ultimately received capital from that country. The baseline set contains 15 countries: the seven countries that injected equity into treated banks plus eight additional countries with SWFs that have documented pre-2007 cross-border private-market activity.[33] Post_t equals one in quarters

[30] The control sample is drawn from the 2011 Financial Stability Board's Global Systemically Important Bank (G-SIB) list with substantial cross-border lending activity in DealScan. Appendix Table 26 reports robustness to the broader 30-bank universe. In the realized DealScan lending panel, Wachovia does not appear as a separate parent after its late 2008 acquisition by Wells Fargo, so that robustness is estimated on 29 banks.

[31] The U.S. Troubled Asset Relief Program and the U.K. bank rescue scheme both began in 2008Q4. Once government equity became available, it becomes difficult to separate the lending response to SWF capital from the response to public recapitalization. Barclays is the main example: its late-2008 capital raise falls outside the clean window. I therefore keep Barclays in the control group rather than treat it as exposed, and the results are robust to dropping it.

[32] The Bloomberg write-down tracker compiles quarterly mark-to-market losses on subprime RMBS, CDOs, CMBS, leveraged loans, Alt-A securities, credit default swaps, and SIVs reported by financial institutions in their public filings.

[33] The seven injection countries are Singapore, the UAE, Kuwait, Qatar, Saudi Arabia, China, and South Korea. The eight additional SWF countries (FY06 AUM $\geq$ \$10bn) are Bahrain, Australia, Oman, Malaysia, New Zealand, Brunei, Libya, and Algeria. Norway (public-market-only mandate), Kazakhstan (no pre-2007 private-market activity), and the United States (home country of three treated banks) are excluded from the SWF-status destination definition. Libya

at or after 2007Q3.[34] Table 3 summarizes the injection events and the bank write-down ratios.

Specification and identification. SWF equity receipt is endogenous at the bank level, so I estimate a reduced-form design that interacts bank write-down severity, destination-country SWF status, and the crisis cutoff. Write-downs reflect pre-crisis portfolio exposure and mark-to-market losses rather than country-specific lending choices, so WriteDownRatio_b is plausibly exogenous to the cross-country allocation of lending. SWFStatus_c is defined ex ante using countries with active pre-crisis SWFs. The design is therefore an intent-to-treat specification: it uses broad SWF-destination exposure rather than only the ten realized bank–SWF-country injection pairs.[35] The estimating equation on the bank–country–quarter panel is

$$\begin{aligned} Y_{bct} &= \beta\left(\text{WriteDownRatio}_b \times \text{SWFStatus}_c \times \text{Post}_t\right) \\ &\quad + \sum_{\ell} \gamma_\ell \left(\text{WriteDownRatio}_b \times X^{\ell}_{bc} \times \text{Post}_t\right) + \alpha_{bc} + \alpha_{bt} + \alpha_{ct} + \varepsilon_{bct}. \end{aligned} \tag{22}$$

where Y_{bct} is new entry, log deal count on cells with positive lending, or log allocated loan volume on cells with positive loan volume, using the standard DealScan exposure-allocation rule (Sufi, 2007). Deal count is the cleaner intensive-margin outcome because bank participation is directly observed; loan volume is useful as a size measure but depends on the exposure-allocation rule. The log specifications condition on positive lending, so they describe intensive-margin changes among active bank–country–quarter cells; the new-entry outcome captures the extensive margin.[36]

β asks whether, after 2007Q3, high-write-down banks reallocate lending toward SWF destinations relative to low-write-down banks. $\mathbf{X}_{bc}$ is a vector of pair-level controls, including the pre-crisis lending share, FTA, common language, and political distance, each interacted with WriteDownRatio_b and Post_t. Bank×country fixed effects absorb time-invariant bilateral relation-

is retained in the ex ante SWF-country definition, but it has no realized syndicated-loan observations in the DealScan panel, so the estimating sample contains 14 effective SWF-status destination countries. Appendix Table 28 re-estimates the specification under PitchBook 21 SWF-country set; core results remain positive and significant.

[34] I use 2007Q3 as the post cutoff because large write-downs first became public in that quarter and recapitalization negotiations began before most SWF injections formally closed in 2007Q4 or 2008Q1. This dating captures the onset of the crisis shock without conditioning on the realized timing of bank–SWF matches. The main results are robust to using 2007Q4 or 2008Q1.

[35] In the estimation window, the baseline panel contains 1,729 bank–country pairs, including 225 pairs involving SWF-status destinations. The full-controls specification uses 1,586 bank–country pairs and retains all 225 SWF-status pairs.

[36] Appendix Table 27 reports zero-inclusive aggregate count and volume specifications using inverse-hyperbolic-sine outcomes. These specifications retain zero-lending cells and therefore combine entry and scale responses. The aggregate-count estimates remain positive and significant across specifications; aggregate-volume estimates are positive and significant before the full control bundle, and remain positive but less precise with full controls.

ships, bank×quarter fixed effects absorb bank-specific supply shocks, and country×quarter fixed effects absorb destination-specific demand shocks.[37] Appendix Figure 10 shows that WriteDownRatio is balanced on pre-crisis bank characteristics: none of the five standardized covariates is significant at the 5% level. The identification assumption is that, conditional on these fixed effects and controls, high-write-down banks would not otherwise have shifted lending disproportionately toward SWF countries after 2007Q3.

Table 3: SWF Equity Injections and Bank Write-Down Ratios

Panel A: Treatment Events

Bank	SWF	SWF ctry	Qtr	Inj. ($bn)	Term price	D-1 stock	Prem./ Disc.
Citi	ADIA	UAE	2007Q4	7.50	\$31.83–\$37.24	\$31.70	Prem.
Merrill Lynch	Temasek	Singapore	2007Q4	4.40	\$48.00	\$55.51	Disc.
Morgan Stanley	CIC	China	2007Q4	5.60	\$48.07–\$57.68	\$48.07	Mixed
UBS	GIC	Singapore	2007Q4	9.70	CHF 47.68	CHF 50.87	Disc.
UBS	Saudi	Saudi Arabia	2007Q4	1.75	CHF 47.68	CHF 50.87	Disc.
Citi	GIC	Singapore	2008Q1	6.88	\$31.62	\$29.06	Prem.
Citi	KIA	Kuwait	2008Q1	3.00	\$31.62	\$29.06	Prem.
Credit Suisse	QIA	Qatar	2008Q1	3.00	n.a.	CHF 47.31	n.a.
Merrill Lynch	KIA	Kuwait	2008Q1	2.00	\$52.40–\$61.31	\$55.97	Mixed
Merrill Lynch	KIC	South Korea	2008Q1	2.00	\$52.40–\$61.31	\$55.97	Mixed

Panel B: Write-down Ratio Construction

Bank	WD ratio	2006 Eq. ($bn)	Loss ($bn)
Merrill Lynch	0.703	39.00	27.40
UBS	0.435	44.62	19.40
Morgan Stanley	0.292	35.30	10.30
Credit Suisse	0.255	34.87	8.90
Citi	0.199	119.80	23.80

Note: Panel A: SWF recapitalization events with pricing benchmarks. Injection size is in USD bn; term price is the negotiated issue price on convertible or preferred securities; D-1 stock is the day-before common stock close; Prem./Disc. flags whether the term price was above or below the D-1 close. Panel B: write-down ratio used in the exposure term. Full summary statistics are in Appendix D.

4.2 Directed Lending

Bank write-downs and SWF investments. Table 4 shows that banks with larger crisis write-downs were more likely to receive SWF equity. In the 18-bank cross-section, a one-standard-deviation increase in WriteDownRatio (SD ≈ 0.18) is associated with about a 37 percentage point

[37] Without country×quarter fixed effects, the coefficient on SWFStatus×Post is negative, reflecting the post-crisis contraction in cross-border bank lending (Cetorelli and Goldberg, 2011; De Haas and Van Horen, 2013). The full specification absorbs this aggregate pullback and isolates the relative shift toward SWF countries among high-write-down banks.

increase in the probability of receiving an SWF injection.[38] This result supports using write-down severity as a bank-level proxy for demand for sovereign capital.

Table 4: Bank Write-Downs and SWF Investments

	18-bank baseline		30 G-SIB banks	
	(1) P(Injection)	(2) asinh(Total Amt.)	(3) P(Injection)	(4) asinh(Total Amt.)
WriteDownRatio	2.070*	5.775*	2.095**	5.830**
	(1.121)	(3.017)	(0.956)	(2.570)
Banks (total)	18	18	30	30
Treated banks	5	5	5	5
R^2	0.653	0.631	0.682	0.661
SE	HC3	HC3	HC3	HC3

Note: Cross-sectional bank-level OLS. Unit of observation is a bank. Columns (1)–(2): 18-bank baseline sample; columns (3)–(4): broader 30-bank sample of global systemically important banks from the Financial Stability Board's 2011 list. Outcome in columns (1) and (3): indicator for receiving any SWF equity injection (LPM). Outcome in columns (2) and (4): asinh(total SWF injection amount, USD bn) across the clean 2007Q4–2008Q1 treatment window. WriteDownRatio is in raw units (0–1 scale). Standard errors are heteroskedasticity-robust (HC3), used because bank-level clustering is uninformative in this small cross-section. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Main results. Table 5 reports the continuous DiD estimates on the $[-8, +5]$ quarter window around 2007Q3. High-write-down banks increase lending activity toward SWF countries after the crisis cutoff, relative to low-write-down banks. The response is strongest on the extensive and deal-frequency margins. Columns (2), (5), and (8) add WriteDownRatio×Pre-share×Post, where Pre-share is the bank–country lending share before the cutoff. This directly addresses the concern that the result simply reflects high-write-down banks expanding in destinations where they were already more active before the crisis. The new-entry and log-count estimates remain positive and significant after this adjustment. Columns (3), (6), and (9) add bilateral FTA, common-language, and political-distance controls, each interacted with WriteDownRatio and Post. In the full-control specifications, the new-entry coefficient is 0.029 and the log-count coefficient is 0.992. A one-standard-deviation increase in WriteDownRatio at 0.18 implies roughly 20% higher deal counts among positive-lending cells. As in PitchBook, the most stable response appears in whether and how often recipients direct activity toward SWF countries.

The loan-volume estimates point in the same direction but are less stable. The coefficient remains positive and significant after the pre-share adjustment, then falls from 1.149 to 0.734 and becomes statistically insignificant after adding the full geopolitical control bundle. The attenuation

[38] As in any linear probability model, fitted probabilities can exceed the unit interval at extreme write-down values in this small sample. The 0.18 standard deviation is computed across banks; the panel-weighted standard deviation is 0.154.

is driven mainly by the political-distance interaction: in the full specification, WriteDownRatio$_b$ × PoliticalDistance$_{bc}$×Post$_t$ is positive and significant. I therefore treat loan volume as complementary evidence. It suggests broader post-crisis reallocation toward politically distant sovereign-capital destinations, but not a precisely estimated incremental SWF-status volume effect.[39]

Table 5: Bank Write-Down Ratios, SWF Countries and Redirected Syndicated Loans

	New Entry			log(Count)			log(Volume)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
WD × SWFStatus × Post	0.037***	0.037***	0.029**	0.988***	0.890**	0.992**	1.306***	1.149**	0.734
	(0.009)	(0.010)	(0.011)	(0.307)	(0.328)	(0.374)	(0.431)	(0.401)	(0.588)
WD × Pre-share × Post	–	-0.025	-0.023	–	-1.475*	-1.317*	–	-2.161	-2.503**
		(0.015)	(0.053)		(0.752)	(0.735)		(1.559)	(1.033)
Add. controls (FTA, language, pol. dist.)	–	–	✓	–	–	✓	–	–	✓
Bank×Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Bank×Quarter FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country×Quarter FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
SE clustered by bank	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	24,206	24,206	22,196	7,356	7,356	6,707	6,862	6,862	6,262
R^2	0.302	0.302	0.311	0.820	0.820	0.824	0.802	0.802	0.810

Note: Bank–country–quarter panel within $[-8, +5]$ quarters of 2007Q3. Columns progress within each outcome group: (1)/(4)/(7) baseline; (2)/(5)/(8) adds WriteDownRatio$_b$ × Pre-share × Post, where Pre-share is the bank–country average lending share over 12 quarters before 2007Q3; (3)/(6)/(9) adds the full control bundle (Pre-share, FTA, political distance, common language). Outcome groups: new entry LPM (cols 1–3, equals 1 only in the first post-cutoff quarter with positive deals for pairs that had zero deals in all eight pre-period quarters), log(deal count) on cells with positive deal count (cols 4–6), and log(loan volume) on cells with positive allocated loan volume (cols 7–9). Three-way FE: bank×country, bank×quarter, country×quarter. SE clustered by bank. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dynamic evidence. Figure 5 reports the binned event study for log deal count, using the quarter immediately before the crisis cutoff as the omitted bin.[40] The pre-period coefficients are statistically indistinguishable from zero and do not show a systematic upward trend before the cutoff. After the cutoff, the coefficient becomes positive immediately: the first post-crisis bin is 0.724 and statistically significant. The later post-crisis bins remain positive, 1.036 and 0.912, but are estimated less precisely.

[39] In this sample, politically distant SWF-status destinations include U.S. banks' lending to Qatar, Kuwait, Saudi Arabia, the United Arab Emirates, and China.

[40] I focus the main dynamic figure on log deal count because the main table shows the most stable intensive-margin response on deal frequency, while the log-volume coefficient is positive but no longer statistically significant under the full control bundle. Appendix Figure 11 reports the corresponding full-controls event study for zero-inclusive aggregate count and volume outcomes; aggregate volume turns positive after the cutoff but is noisier. I aggregate individual quarters into two-quarter bins because the small number of treated banks (five) yields imprecise quarter-by-quarter estimates (Schmidheiny and Siegloch, 2023).

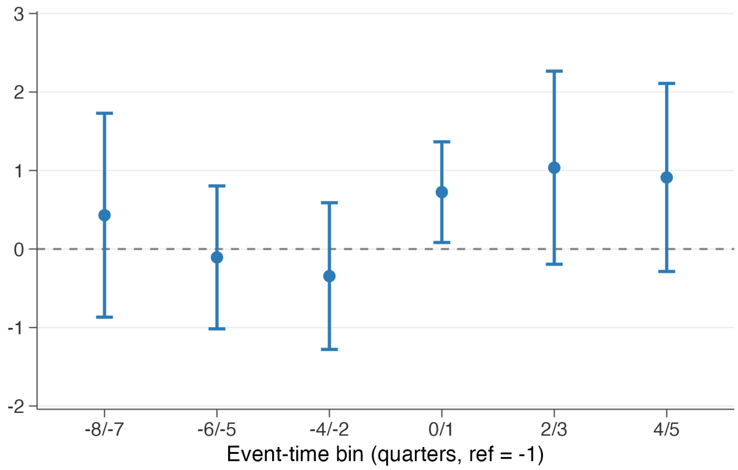

Figure 5: Binned Event Study: Deal Count

Notes: The figure shows $\text{WriteDownRatio}_b \times \text{SWFStatus}$ coefficients by binned event-time period relative to 2007Q3. Outcome: log(deal count), restricted to cells with positive deal count. Reference bin: $t = -1$. Error bars: 95% CIs; SE clustered by bank. Baseline specification; three-way FEs (Bank×Country, Bank×Quarter, Country×Quarter).

Leave-one-out. Appendix Figure 12 drops each SWF destination country in turn and re-estimates the full-controls log-count specification. The coefficient remains positive in every omission and remains statistically significant except when Australia is omitted. This exception is not central to the interpretation because Australia is part of the ex ante SWF-destination set but is not one of the realized sovereign capital providers in the crisis recapitalization window. The leave-one-out exercise therefore rules out a sign reversal or a result mechanically driven by a single capital-provider country.

Together, Sections 3 and 4 provide complementary evidence that sovereign capital is associated with directed activity when recipients face financing pressure. PitchBook offers broad cross-sectional coverage through within-deal variation, while DealScan offers cleaner identification from crisis-driven variation in bank capital need. The next question is whether these directed micro reallocations reduce bilateral frictions more broadly. Subsections 5.1 and 5.2 turn to that macro prediction by testing whether SWF-backed activity is followed by broader capital-flow responses: third-party FDI flows in FIRRMA and aggregate banking claims in BIS.

5 Bilateral Capital-Flow Responses

Proposition 3 predicts that directed activity lowers bilateral frictions for outside investors. I test this macro implication in two settings: bilateral FDI after the FIRRMA regulatory shock and cross-border banking claims after SWF bank equity injections.

5.1 FDI Responses to FIRRMA

This subsection tests the crowd-in prediction in bilateral FDI. I exploit FIRRMA, enacted in 2018, as a shock that raised the cost of U.S. investments for SWFs. The design asks whether SWF countries with greater pre-2018 exposure to this shock subsequently attracted more FDI from outside investors. The estimates are positive, consistent with SWF-backed activity strengthening bilateral investment links.

Institutional setting. FIRRMA, enacted in August 2018, expanded the authority of the Committee on Foreign Investment in the United States (CFIUS) to review and restrict foreign acquisitions in sensitive sectors. For SWFs, the relevant margin was large-equity transactions in the United States. Appendix Figure 14 documents the corresponding non-U.S. deal margin: non-U.S. high-dilution SWF deal counts rise after 2018, including among countries with positive pre-FIRRMA U.S. high-dilution exposure. I treat this figure as descriptive background rather than a first-stage regression, because the increase is not differential across exposure groups. The reduced-form test below asks whether SWF home countries with greater pre-FIRRMA U.S. exposure subsequently attract more third-party FDI.

Exposure construction. The main exposure measure is constructed using the PitchBook data in Section 3 as the 2014–2017 share of a country's SWF co-investment deals that both targeted the United States and involved above-median dilution:

$$\text{Exposed}_i^{\text{HD}} = \frac{\#\{\text{US-targeted, high-dilution deals by SWFs from } i\}_{2014\text{–}2017}}{\#\{\text{deals by SWFs from } i \text{ with non-missing dilution}\}_{2014\text{–}2017}}. \tag{23}$$

High-dilution deals are those with dilution above the median in the SWF co-investment sample. They are the natural margin for this test. They were the deals most likely to face tighter CFIUS scrutiny because they involved larger equity stakes, and Section 3 shows that directed allocation loads on this margin. I use the 2014–2017 window because SWF co-investment is thin and volatile before 2014. Under this measure, 9 of the 21 SWF home countries have positive exposure; the remaining SWF destinations form the zero-exposure comparison group. Appendix Figure 15 plots the cross-country distribution of the high-dilution FIRRMA exposure measure.

Specification and identification. I estimate whether SWF countries with greater pre-FIRRMA U.S. exposure subsequently attract more third-party FDI from non-U.S. source countries:

$$\text{asinh}(\text{FDI}_{jit}) = \beta\,(\text{Exposed}_i \times \text{Post}_t) + \mathbf{X}'_{jit}\gamma + \alpha_{jt} + \alpha_{ji} + \varepsilon_{jit}, \tag{24}$$

where j indexes source countries, i destination countries, and t years. The outcome is bilateral FDI inflows from the World Bank Harmonized Bilateral FDI Database.[41] The coefficient β compares post-2018 changes in FDI inflows toward more- and less-exposed destinations. Source×year fixed effects absorb shocks to each source country's outward FDI, pair fixed effects absorb time-invariant bilateral affinity, and $\mathbf{X}_{jit}$ adds standard gravity controls like those in the bank setting.[42] Appendix Figure 13 shows that exposure is not strongly related to destination GDP, GDP growth, pre-2018 FDI growth, or the number of source partners. It is higher for SWF destinations with greater pre-2018 U.S. trade exposure and lower pre-2018 FDI inflow levels, so the main table reports specifications that interact the post-FIRRMA period with both U.S. trade share and standardized pre-2018 FDI level. The identifying assumption is that, absent FIRRMA, more-exposed SWF countries would not have experienced systematically different post-2018 changes in FDI from the same source countries.

Main results. Table 6 reports the main specification. The high-dilution exposure coefficient is positive and stable across specifications in the main sample. With the gravity controls (column 2), the estimate is 0.692. Column (3) adds the two balance controls motivated by Appendix Figure 13: the destination's pre-2018 U.S. trade share interacted with post-FIRRMA and its standardized pre-2018 FDI inflow level interacted with post-FIRRMA. The high-dilution coefficient remains positive at 0.713, while both balance-control interactions are small and statistically insignificant. Column (4) restricts FDI source countries to the 49 non-U.S. countries that appear as targets of high-dilution SWF co-investment deals in PitchBook; the coefficient rises to 1.550. Columns (5)–(6) replace the baseline high-dilution exposure with two broader measures of pre-FIRRMA U.S. exposure: the U.S. share of all deal counts and the U.S. share of all deal values for each SWF country. The estimates remain positive and significant. I use the event study mainly for timing and the decomposition table to separate entry, continuation, and intensive-margin responses behind the asinh outcome.

Dynamic evidence. Figure 6 plots annual event-study coefficients from the balance-control specification in column (3) of Table 6. I use the figure as a timing check rather than as an estimate of the exact adjustment path, because annual FDI flows are volatile and the final post year, 2020, coincides with pandemic-era disruptions to global investment. The two leads closest to FIRRMA are close to zero and statistically indistinguishable from zero. The earliest lead is negative under

[41] Because bilateral FDI flows can be negative or zero, I use the inverse hyperbolic sine transformation.

[42] Controls include free trade agreement (FTA) membership, destination GDP level and growth, asinh(bilateral trade) and diplomatic distance measured by UN-voting ideal-point distance.

Table 6: FIRRMA Exposure and Third-Party FDI Inflows to SWF Home Countries

	High-Dilution Exposure				Count	Value
				Target Sources		
	(1)	(2)	(3)	(4)	(5)	(6)
SWF US Exposure × Post	0.550***	0.692***	0.713***	1.550***		
	(0.132)	(0.139)	(0.228)	(0.599)		
SWF US Exposure (count) × Post					0.431***	
					(0.162)	
SWF US Exposure (value) × Post						0.470***
						(0.146)
US Trade Share × Post			0.289	0.208	1.805	1.802*
			(1.399)	(3.345)	(1.131)	(1.081)
Pre-FIRRMA FDI Level × Post			0.039	0.107	-0.032	-0.008
			(0.084)	(0.172)	(0.067)	(0.069)
asinh(Bilateral Trade)		-0.708**	-0.729**	-1.316**	-0.731**	-0.717**
		(0.304)	(0.309)	(0.610)	(0.309)	(0.309)
Diplomatic Distance		-0.254	-0.261	-0.458	-0.263	-0.252
		(0.221)	(0.221)	(0.529)	(0.221)	(0.221)
Other gravity controls	–	✓	✓	✓	✓	✓
Source × Year FE	✓	✓	✓	✓	✓	✓
Pair FE	✓	✓	✓	✓	✓	✓
SE clustered by pair	✓	✓	✓	✓	✓	✓
Wild-cluster p (destination)	0.005	0.034	0.020	0.087	0.131	0.064
Source countries	224	224	224	49	224	224
Destination countries	21	21	21	21	21	21
Observations	16,866	16,866	16,866	5,374	16,866	16,866
R^2	0.362	0.363	0.363	0.344	0.363	0.363

Note: Dependent variable: asinh(bilateral FDI inflow) from source j to destination i, World Bank Harmonized Bilateral FDI Database, 2014–2020; see Appendix B.3 for data construction. The estimation sample is restricted to the 21 non-U.S. SWF home destinations active in PitchBook during 2014–2021. Because bilateral flows include negatives and zeros, the inverse hyperbolic sine is used. Columns (1)–(3): high-dilution exposure (share of above-median-dilution pre-2018 SWF co-investment deals targeting the U.S.) with progressive controls; column (3) adds U.S. trade share × Post and standardized pre-FIRRMA FDI level × Post, where pre-FIRRMA FDI level is the destination's 2014–2017 mean asinh bilateral inflow, standardized across destinations. Columns (4)–(6) include those balance controls: column (4) restricts source countries to the 49 nations where SWFs made high-dilution non-U.S. deals, and columns (5)–(6) use count-based and value-weighted exposure with full controls. All columns include source×year and pair FE. Gravity controls: FTA, log(GDP), GDP growth, asinh(bilateral trade), diplomatic distance. Standard errors are clustered by country pair. The wild-cluster row reports restricted wild-cluster bootstrap p-values clustered by SWF destination country. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

pair-clustered inference but not under destination clustering.[43] After FIRRMA, the coefficients are positive, with the largest estimate in 2020.

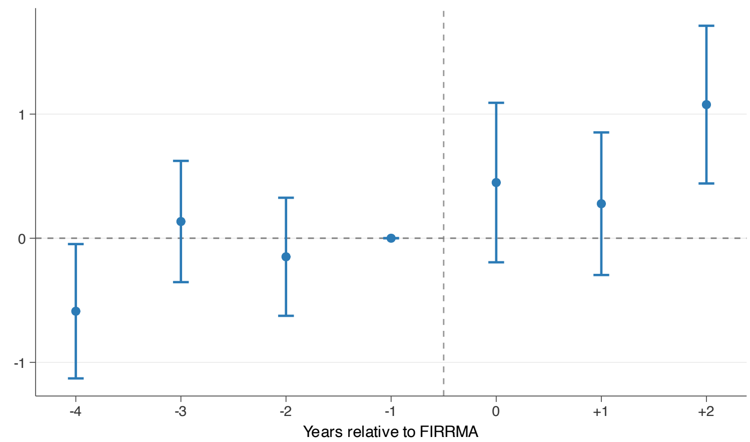

Figure 6: FIRRMA Event Study: FDI Inflows to SWF Home Countries

Notes: Annual event-study coefficients from the balance-control specification corresponding to column (3) of Table 6. The specification includes high-dilution exposure interacted with event time, U.S. trade share × Post, standardized pre-FIRRMA FDI level × Post, gravity controls, source×year FE, and pair FE. Sample: 21 non-U.S. SWF home destinations, 2014–2020. The omitted year is 2017. Error bars are 95% confidence intervals with standard errors clustered by country pair.

Margin decomposition. Because bilateral FDI flows can be negative, zero, or positive, the asinh specification mixes entry, continuation, and flow-size changes. Table 7 separates these margins following Haug, Nguyen and Owen (2023). The evidence points to stronger continuation among existing bilateral relationships rather than new relationships from previously inactive source countries. The entry margin does not increase for source–destination pairs with no positive pre-FIRRMA inflow (column 1). In contrast, existing relationships strengthen on two margins. Conditional on positive current flows, the intensive-margin coefficient is positive, 0.486, and marginally significant. Among pairs that were already active before FIRRMA, per unit of high-dilution exposure lowers the annual probability of falling to zero or negative flow by 22.1 percentage points. This pattern is consistent with the model's network-externality mechanism: SWF-linked activity appears to lower frictions for ongoing bilateral investment.

The decomposition also gives a cleaner economic scale. At observed exposure shares, the conditional log-flow coefficient in column (2) implies increases of about 4.9 percent for Singapore (exposure 0.098), 7.2 percent for China (exposure 0.143), and 8.4 percent for Australia (exposure

[43]The joint pre-period test is borderline with pair-clustered standard errors (p = 0.061), but far from rejection with destination-clustered standard errors (p = 0.292).

0.167), among source–destination pairs with positive current FDI flows. The exit-margin coefficient in column (3) implies that high-dilution exposure lowers the annual probability that an already-active pair falls to zero or negative inflows by 2.2 percentage points for Singapore, 3.2 percentage points for China, and 3.7 percentage points for Australia.

Table 7: FIRRMA Spillover: Entry / Intensive / Exit Decomposition

	(1) Entry	(2) Intensive	(3) Exit
SWF US Exposure (high-dilution) × Post	-0.017	0.486*	-0.221***
	(0.023)	(0.267)	(0.054)
SWF US Exposure (count) × Post	-0.043***	0.289*	-0.115***
	(0.014)	(0.164)	(0.037)
SWF US Exposure (value) × Post	-0.041***	0.212	-0.117***
	(0.012)	(0.129)	(0.029)
Outcome	$\mathbf{1}\{\text{flow} > 0\}$	$\log(\text{flow})$	$\mathbf{1}\{\text{flow} \leq 0\}$
Sample	Never-active pre	in_flow > 0	Ever-active pre
Gravity controls	✓	✓	✓
Balance controls	✓	✓	✓
Source × Year FE	✓	✓	✓
Pair FE	✓	✓	✓
SE clustered by pair	✓	✓	✓
Observations	6,556	6,587	10,310

Note: Each cell is a separate regression estimated on the 21 non-U.S. SWF home destination sample. Pre-period is 2014–2017; the FIRRMA shock occurs in 2018. A pair is classified as *never-active pre* if its bilateral inflow was non-positive in every pre-period year, and *ever-active pre* otherwise. Column (1) estimates the *entry* margin: a linear probability model for $\mathbf{1}\{\text{in\_flow} > 0\}$ on the never-active-pre cohort, asking whether HD-exposed pairs are more likely to begin receiving positive bilateral inflows post-FIRRMA. Column (2) estimates the *intensive* margin: an OLS regression of log(flow) on all pair-year observations with positive current flow, regardless of pre-period cohort, asking whether bilateral flow levels rise conditional on remaining positive. Column (3) estimates the *exit* margin: a linear probability model for $\mathbf{1}\{\text{in\_flow} \leq 0\}$ on the ever-active-pre cohort, asking whether already-active pairs are less likely to drop to zero or negative flow. Thus, the entry and exit cohorts are disjoint, but the intensive-margin sample overlaps with both because it conditions on current positive flow. All specifications include the full set of gravity controls (FTA membership, log destination GDP, GDP growth, asinh bilateral trade, diplomatic distance), the two balance controls from Table 6, source×year and pair fixed effects, and standard errors clustered by country pair. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Robustness. Appendix Table 29 reports the same specifications on the broader non-U.S. destination universe used in the previous version of the design. It compares the FDI inflows into the 9 positive-exposure SWF destinations with inflows into all 163 zero-exposure destinations. The magnitudes of the coefficients fall by half but remain significant. This is consistent with the SWF home countries having more concentrated FDI relationships with non-U.S. sources than the broader destination universe. I present the 21-destination sample as my main specification on conceptual grounds: the geofinance mechanism predicts crowd-in toward SWF home countries specifically, not toward non-SWF destinations. Appendix Table 30 then reports timing and oil-exporter checks using the balance-control specification. Placebo shocks in 2015 and 2016, estimated entirely before FIRRMA, yield small and statistically insignificant coefficients. Dropping oil-exporter SWF countries keeps the estimate positive and statistically significant. Leave-one-positive-exposure-SWF-country-out estimates remain positive throughout (Appendix Figure 16), including when China is omitted.

From FIRRMA to BIS. FIRRMA identifies crowd-in in bilateral FDI, but its exposure varies only at the destination level. The BIS design provides pair-level variation by tracing staggered SWF bank injections across lender countries and treatment years. It therefore offers complementary evidence on whether banking claims also rise after SWF-backed relationships form.

5.2 Banking Claims after SWF Equity Injections

This subsection tests whether SWF bank equity injections are followed by higher cross-border banking claims. I use bilateral claims from the BIS Locational Banking Statistics in a stacked-cohort difference-in-differences design centered on SWF equity injections into banks. Claims from the lender country to the SWF home country rise after treatment, build over the first two post-treatment years, and remain elevated through year 5. A cross-channel exercise asks whether SWF investments in non-financial firms generate similar banking spillovers.

Data and treatment. The BIS Locational Banking Statistics report bilateral cross-border claim stocks of banks resident in a reporting country on counterparties in each borrower country. These stocks measure outstanding banking claims rather than new loan originations, so I interpret the estimates as changes in bilateral banking exposure. The data cover claims beyond syndicated lending, including direct loans, debt securities, and deposits, and span 1995–2024 for 31 reporting countries and 218 borrower countries. Appendix B.4 describes the data construction.

I identify SWF bank equity injections from the Global SWF (GSWF) and Sovereign Wealth Fund Institute (SWFI) transaction databases, retaining direct bank equity injections and excluding asset-management mandates, insurance investments, and fund-of-fund commitments. The raw treatment universe contains 107 deals across 55 lender–SWF-country pairs, totaling $127 billion from 13 SWF countries. After merging to the LBS panel and applying data-availability restrictions, 12 treated pairs remain in the stacked-DiD design.[44] These treated pairs are concentrated among developed-country lenders and SWF-country borrowers.

[44] Of the 55 treatment pairs, 48 have disclosed deal sizes; of those 48, 12 have lender countries that report to the BIS LBS over the treatment window and enter the matched stacked-DiD design.

Specification and design. The treatment variable is *SWF investment intensity*: the total SWF equity injected, in USD billions, into banks in lender country l by SWFs from borrower country b. For example, disclosed Singaporean SWF injections into U.S. banks total \$19.8 billion across three events, generating an intensity of 19.8 for the USA→Singapore pair. This is a pair-level aggregate treatment: if the same lender–borrower pair receives multiple disclosed injections, I sum them into a single total intensity and assign that pair to a cohort at its first injection year.[45] The estimand is therefore the aggregate change in lender-country banking claims associated with the overall SWF–bank relationship in the pair, not the effect of only the first observed transaction. Within each cohort, I match up to 20 control pairs from the same lender country on pre-treatment log-loan levels and growth rates using Mahalanobis distance.[46] I then stack the cohorts and estimate

$$\log(Y_{lbt}) = \beta \cdot (\text{Intensity}_{lb} \times \text{Post}_t) + \alpha_{g(lb)t} + \alpha_{g(lb),lb} + \varepsilon_{lbt}, \tag{25}$$

where Y_{lbt} is a positive bilateral claim stock, Intensity_{lb} is the pair-specific treatment intensity, and Post_t equals one for periods at or after the cohort's treatment date. Cohort×period fixed effects absorb time-varying shocks common within a cohort. Cohort×pair fixed effects absorb time-invariant bilateral heterogeneity. β estimates whether more intensely treated lender–borrower pairs experience larger post-treatment increases in bilateral claims.

Investment intensity is endogenous: SWFs may target lender countries with stronger pre-existing bilateral ties or better growth prospects. The key assumption is therefore parallel trends within each stacked cohort. Conditional on cohort×period and cohort×pair fixed effects, treated pairs would have followed the same path as their matched controls absent SWF bank injections. Two features make this assumption more plausible. First, Mahalanobis matching aligns controls with treated pairs on pre-treatment log-loan levels and growth rates. Second, treatment is staggered across twelve lender–borrower pairs over 2007–2018, which reduces the scope for any single common shock to explain the full pattern. Although the matched stacks include many control pairs, inference is still based on 12 treated cohorts, so I report wild-cluster-bootstrap p-values in the main table as a small-sample diagnostic.

Main results and magnitude. Table 8 reports the annual main specification using Q4 stocks. Cross-border banking claims from the lender country to the SWF home country rise after SWF bank equity injections. In column (1), total claims rise by 0.058 log points per \$1 billion of SWF

[45] This biases the implied per-dollar effect downward in early years for pairs with later additional events.

[46] The matching reduces standardized differences on the pre-treatment log-loan series from 1.14 standard deviations between treated pairs and the full eligible candidate pool to 0.11 standard deviations between treated pairs and the matched controls, and auxiliary outcomes not used in the matching are similarly balanced (Appendix Table 31).

equity. Column (2) shows a similar estimate at 0.057 for loans and placed deposits, and column (3) shows a similar estimate at 0.053 for non-bank loans. The similarity across columns suggests that the increase is concentrated in loan-and-deposit claims to SWF-country borrowers rather than in debt securities or interbank positions. Column (4) shows a positive and significant effect on winsorized standardized loan flows, although that margin is no longer precisely estimated under wild-cluster-bootstrap inference. The loans estimate implies roughly 5.8% higher bilateral cross-border loans per additional $1 billion of SWF equity. This corresponds to about $345 million at the pre-treatment treated-pair median and about $600 million at the treated-pair mean. The estimate is a stock interpretation: it describes the average post-treatment outstanding claims, not new loan originations or cumulative annual flows.[47]

Dynamic evidence. Figure 7 reports year-by-year event-study coefficients for total claims, loans and deposits, and non-bank loans. Across the three stock outcomes, the pre-treatment coefficients are centered near zero, which supports the parallel-trends assumption and argues against a simple selection story in which SWF-country claims were already rising before the equity injections. After treatment, the coefficients turn positive, build over the first two post-treatment years, and remain elevated through year 5. The similar timing across total claims and the loan-based components suggests that the post-treatment increase is not confined to a narrow claim category; it reflects a broader rise in lender-country banking exposure to SWF home countries.

Cross-channel evidence. Table 9 asks whether non-financial SWF investments generate similar banking spillovers. I re-estimate the stacked-cohort design around unlisted non-financial SWF investments, excluding pairs with SWF bank equity injections. The estimates are positive but much smaller than the direct bank-channel effects; only non-bank loans are statistically significant. I therefore read this as suggestive evidence. If the crowd-in mechanism were a broad information-friction reduction, one might expect banking spillovers of similar scale when SWF capital enters firms rather than banks. However, banking claims respond much more strongly when SWF capital enters banks directly, consistent with a relationship-friction channel within banking links rather than a general information channel.

Robustness. These checks address temporal aggregation, placebo outcomes, borrower concentration, and residence-based measurement. Appendix Table 32 re-estimates the design at quarterly

[47]The scale calculation uses, for each treated lender–borrower pair, the pair's mean pre-treatment loans and placed deposits in the annual stacked sample. Across the 12 treated pairs, the median is $5.9 billion and the mean is $10.3 billion.

Table 8: Cross-Border Banking Claims Spillover

	(1) Total Claims	(2) Loans & Placed Dep.	(3) Non-Bank Loans	(4) Wins. Std. Flow Loans
SWF Invest ($bn) × Post	0.0581**	0.0566**	0.0530***	0.0252**
	(0.0195)	(0.0193)	(0.0151)	(0.0113)
Outcome transform	log	log	log	wins. flow/pre-stock
Observations	3,259	3,258	3,250	3,259
Cohort × Year FE	✓	✓	✓	✓
Cohort × Pair FE	✓	✓	✓	✓
Cluster	Cohort	Cohort	Cohort	Cohort

Note: Matched stacked-cohort DID using BIS Locational Banking Statistics (residence-based). Treatment intensity: total SWF equity injection in $bn into banks domiciled in the reporting country. When a lender-country × SWF-country pair receives multiple disclosed injections, those amounts are summed into a single pair-level intensity and the cohort is anchored at the pair's first injection date. Columns (1)–(3): log annual cross-border claim stocks (including loans, deposits and debt securities), using positive Q4 end-of-year stocks; column (2) restricts to loans & deposits; column (3) further restricts to non-bank counterparties. Column (4): reported annual LBS loan flow, summed from quarterly flows, divided by pre-treatment mean loan stock and winsorized at the 1st and 99th percentiles. Controls are Mahalanobis-matched on pre-treatment log-loan level and slope. Annual main specification uses a ±6-year window around treatment; quarterly dynamics are reported in the appendix. SE clustered by cohort, defined by the treated lender-country × SWF-country pair and its first injection year. Wild-cluster-bootstrap-t p-values clustered by cohort are 0.007, 0.009, 0.014, and 0.216 for columns (1)–(4), respectively. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

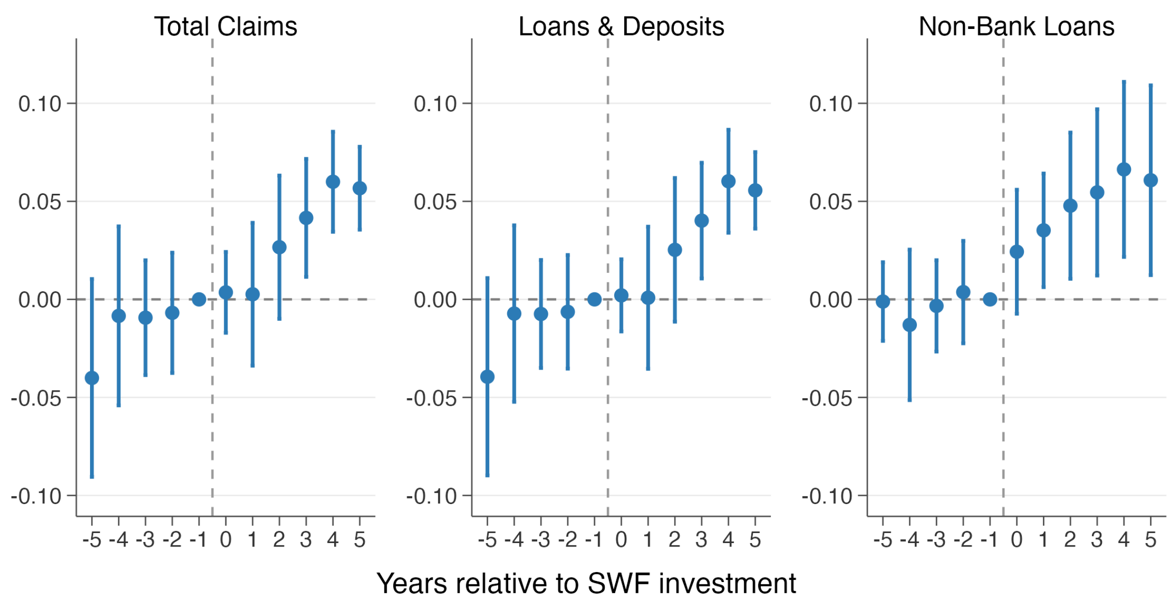

Figure 7: Cross-Border Claims: Annual Intensity-Weighted Event Study

Notes: Year-by-year stacked-cohort event study for BIS LBS cross-border claims. Panels report log total claims, log loans and deposits, and log non-bank loans. Treatment intensity: SWF equity injection ($bn). Reference year: $t-1$. Controls are Mahalanobis-matched using pre-treatment log-loan level and slope. Main annual design uses Q4 end-of-year stocks in a ±6-year window around treatment with cohort×year and cohort×pair FEs; SE clustered by cohort. Quarterly event-study dynamics are reported in the appendix figure.

Table 9: Cross-Channel Banking Claims Response from SWF Firm Investments

	(1) Total Claims	(2) Loans & Deposits	(3) Non-Bank Loans
Treatment intensity × Post	0.0128 (0.0091)	0.0133 (0.0093)	0.0135^{**} (0.0064)
Observations	23,138	22,983	21,526
Cohort × Year FE	✓	✓	✓
Cohort × Pair FE	✓	✓	✓
Cluster	Cohort	Cohort	Cohort

Note: Stacked-cohort DiD built around firm-channel treatment events, defined as the first SWF investment in an unlisted non-financial firm in each lender–borrower pair over 2002–2022 (GSWF/SWFI source; finance and listed/public-market deals excluded). The sample is restricted to pairs with no SWF bank equity injection in the same country pair, so the firm-channel effect is estimated independently of the bank equity injections mechanism. Treatment intensity is total SWF firm investment in the pair (USD bn); it enters interacted with Post. Outcomes are log annual BIS LBS claims measured using positive Q4 end-of-year stocks: total claims, loans and deposits, and loans to non-bank counterparties. Treated pairs are matched within lender country to up to 20 controls on pre-treatment log loans and log-loan growth rates. Cohort×year and cohort×pair fixed effects; SE clustered by cohort. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

frequency with cohort×quarter fixed effects. The coefficient remains close to the annual baseline, so the result is not an artifact of using Q4 stocks. Appendix Table 33, column (5), reports a liability-side placebo based on BIS liability positions; the estimate is small and insignificant and therefore inconsistent with a symmetric expansion of bilateral balance sheets.

Figure 17 drops each treated lender–borrower cohort in turn and re-estimates the intensity-weighted specification for loans. The coefficient remains positive throughout, so no single cohort drives the result.

Appendix Table 33, columns (6)–(7), compares the LBS estimates with BIS Consolidated Banking Statistics (CBS), which assign claims by parent-bank nationality rather than office residence. The CBS estimates on total claims (0.043) and on international claims (0.0599) are close to the LBS total-claims baseline (0.058), which suggests that the result is not solely explained by the residence-based reporting convention.

Taken together, Subsections 5.1 and 5.2 provide two complementary tests of Proposition 3. FIRRMA provides the cleaner crowd-in test in bilateral FDI. The BIS design provides complementary evidence that banking-system claims also rise after SWF bank equity injections, although the aggregate claims data cannot separate lending by recapitalized banks from lending by other banks. In both settings, the results point to a relationship-friction interpretation: crowd-in operates mainly through existing bilateral links.

6 Conclusion

Sovereign capital does more than finance firms. By exploiting financing-cost advantages, sovereign investors can direct recipients' cross-border activity toward their home countries. The resulting activity builds bilateral links that can lower frictions for later investors. Across four settings, private-market co-investments, crisis-era bank recapitalizations, FDI flows, and cross-border banking claims, the evidence points to the same pattern. Within SWF-backed co-investment rounds, a one-standard-deviation increase in financing frictions raises directed expansion by about half the non-SWF baseline rate, and SWF-backed activity is followed by 5–8 percent increases in FDI inflows and banking claims to SWF home countries. Together, these results connect a micro contracting margin to the geography of cross-border capital flows.

The framework developed here, geofinance, extends beyond sovereign wealth funds. It applies whenever a financier with a funding-cost advantage is willing to trade financial return for directed activity. Development finance institutions, state-owned enterprises, and strategic corporate investors are natural settings in which the same logic may apply.

The evidence leaves three margins for future work. First, the reduced-form estimates do not separately identify the financier's strategic-objective weight, the bilateral-friction function, and implementation costs. Estimating those objects would allow counterfactual analysis of how geopolitical shocks, such as wars or sanctions, redirect sovereign capital and the capital flows that follow. Second, SWF transactions are large and infrequent, so each empirical setting has limited treatment variation; richer adjacent settings could sharpen the magnitudes. Third, this paper focuses on financing frictions, but strategic capital may also operate through information, governance, or political channels.

Appendix Overview

A Model Derivations and Extensions

A.1 Benchmark regime and proofs of the main results

The propositions in Section 2 use an interior benchmark for contracted recipients: funding, directed deployment, outside-firm responses, and the contract rate are interior. The aggregate-capacity constraint may bind, with multiplier λ. The home-market value $A_i(r_i)$ is additively separable and independent of k_{is}, r_{is}, and Z, so it drops out of the derivative calculations below. This subsection derives the formulas; Appendix A.2 gives the KKT system and corner cases.

The Lagrangian is

$$\begin{aligned}\mathcal{L} = &\sum_{i\in C}\left[(r_{is}-\rho)x_i - \tfrac{c}{2}x_i^2\right] + \theta U(Z)\\ &+ \sum_{i\in C} y_i\left[\Pi_i(x_i, r_{is}, k_{is}, Z) - v_i(Z)\right]\\ &+ \lambda\left[\bar{X} - \sum_{i\in C} x_i\right]\\ &+ \Psi_Z\left[\sum_{i\in C} k_{is} + \sum_{j\notin C} k^*_{js}(Z) - Z\right].\end{aligned} \tag{26}$$

The inequality multipliers y_i and λ are nonnegative. The equality multiplier Ψ_Z is unrestricted in sign.

Contract rate r_{is}. From Equation (5), $\partial\Pi_i/\partial r_{is} = -x_i$. Stationarity gives

$$\frac{\partial\mathcal{L}}{\partial r_{is}} = x_i - y_i x_i = 0. \tag{27}$$

For any contracted recipient with $x_i > 0$ and an interior contract rate,

$$y_i = 1. \tag{28}$$

The participation constraint therefore binds, and the contract rate is pinned down by the reservation value.

Financing volume x_i (Proposition 2). Using $\partial\Pi_i/\partial x_i = r_i - r_{is}$,

$$\frac{\partial\mathcal{L}}{\partial x_i} = (r_{is}-\rho) - cx_i + y_i(r_i - r_{is}) - \lambda = 0. \tag{29}$$

Substituting $y_i = 1$ yields

$$x_i^* = \frac{r_i - \rho - \lambda}{c}, \tag{30}$$

so the contract rate r_{is} drops out of the marginal funding condition.

External action k_{is} (Proposition 1). From the operating payoff in Equation (5),

$$\frac{\partial \Pi_i}{\partial k_{is}} = G_i'(k_{is}) - \mu(Z) - r_i. \tag{31}$$

The financier's first-order condition is

$$\frac{\partial \mathcal{L}}{\partial k_{is}} = y_i \big[G_i'(k_{is}) - \mu(Z) - r_i \big] + \Psi_Z = 0. \tag{32}$$

Using $y_i = 1$ and rearranging,

$$G_i'(k_{is}) = \mu(Z) + r_i - \Psi_Z. \tag{33}$$

Comparing this expression with Equation (9) gives

$$\tau_{is}^* = \Psi_Z. \tag{34}$$

Aggregate network stock Z (Proposition 3). Differentiating $\mathcal{L}$ with respect to Z gives

$$\frac{\partial \mathcal{L}}{\partial Z} = \theta U'(Z) + \sum_{i \in C} y_i \left(\frac{\partial \Pi_i}{\partial Z} - \frac{\partial v_i}{\partial Z} \right) + \Psi_Z \left[\sum_{j \notin C} \frac{\partial k_{js}^*(Z)}{\partial Z} - 1 \right] = 0. \tag{35}$$

Maintaining that Z enters firm value only through the cost term $-\mu(Z)k$, and applying the envelope theorem to the outside option,

$$\frac{\partial \Pi_i}{\partial Z} - \frac{\partial v_i}{\partial Z} = -\mu'(Z)\big(k_{is} - k_{is}^{\text{out}}\big). \tag{36}$$

Total differentiation of the outside-firm condition

$$G_j'(k_{js}^*) = \mu(Z) + r_j$$

yields

$$G_j''(k_{js}^*) \frac{\partial k_{js}^*}{\partial Z} = \mu'(Z) \qquad \Longrightarrow \qquad \frac{\partial k_{js}^*}{\partial Z} = \frac{\mu'(Z)}{G_j''(k_{js}^*)}. \tag{37}$$

Substituting these expressions into the Z first-order condition and using $y_i = 1$ gives

$$\Psi_Z = \left[1 - \sum_{j \notin C} \frac{\mu'(Z)}{G_j''(k_{js}^*)}\right]^{-1} \left[\theta U'(Z) - \mu'(Z) \sum_{i \in C} \left(k_{is} - k_{is}^{\text{out}}\right)\right]. \tag{38}$$

This is the expression used in Proposition 3. □

Benchmark validity conditions. The closed-form expressions above characterize the benchmark candidate when the following conditions hold: (i) contracted recipients have $x_i^* > 0$, equivalently $r_i > \rho + \lambda$; (ii) directed deployment is interior, with $G_i''(k_{is}) < 0$, so $\chi_i = 0$ and the solution to the k_{is} first-order condition satisfies $k_{is} > 0$; (iii) outside firms are on an interior differentiable response margin with $G_j''(k_{js}^*) < 0$; (iv) the aggregate feedback satisfies $0 \le \sum_{j \notin C} \partial k_{js}^*(Z)/\partial Z < 1$; and (v) the contract rate is interior.

A.2 KKT conditions and corner regimes

To describe corner regimes, add the non-negativity constraints $x_i \ge 0$ and $k_{is} \ge 0$, with multipliers $v_i \ge 0$ and $\chi_i \ge 0$. Relative to Equation (26), define

$$\widetilde{\mathcal{L}} = \mathcal{L} + \sum_{i \in C} v_i x_i + \sum_{i \in C} \chi_i k_{is}.$$

KKT system. Primal feasibility is given by the constraints in Equation (13), together with $x_i \ge 0$ and $k_{is} \ge 0$. The inequality multipliers satisfy

$$y_i \ge 0, \qquad \lambda \ge 0, \qquad v_i \ge 0, \qquad \chi_i \ge 0, \qquad \Psi_Z \in \mathbb{R}.$$

Complementary slackness is

$$\begin{aligned} y_i \left[\Pi_i(x_i, r_{is}, k_{is}, Z) - v_i(Z)\right] &= 0, \\ \lambda \left(\bar{X} - \sum_{i \in C} x_i\right) &= 0, \\ v_i x_i &= 0, \qquad \chi_i k_{is} = 0. \end{aligned}$$

The stationarity conditions for the firm-specific controls are

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial r_{is}} = x_i - y_i x_i = 0,$$
$$\frac{\partial \widetilde{\mathcal{L}}}{\partial x_i} = (r_{is} - \rho) - c x_i + y_i (r_i - r_{is}) - \lambda + \nu_i = 0,$$
$$\frac{\partial \widetilde{\mathcal{L}}}{\partial k_{is}} = y_i \left[G_i'(k_{is}) - \mu(Z) - r_i \right] + \Psi_Z + \chi_i = 0.$$

The Z stationarity condition is unchanged except that y_i need not equal one:

$$\theta U'(Z) + \sum_{i \in C} y_i \left(\frac{\partial \Pi_i}{\partial Z} - \frac{\partial v_i}{\partial Z} \right) + \Psi_Z \left[\sum_{j \notin C} \frac{\partial k_{js}^*(Z)}{\partial Z} - 1 \right] = 0. \tag{39}$$

If outside firms are at an interior solution, their response is characterized by

$$\frac{\partial k_{js}^*}{\partial Z} = \frac{\mu'(Z)}{G_j''(k_{js}^*)}.$$

[48] Under the maintained concavity and regularity assumptions, this system provides the local characterization used in the benchmark regime and in the corner cases below.

Contracted-recipient benchmark regime. If $x_i > 0$ and $k_{is} > 0$, complementary slackness implies $\nu_i = \chi_i = 0$. Provided the contract rate is interior, the contract-rate condition then pins down $y_i = 1$, and the benchmark formulas above follow immediately. The aggregate-capacity constraint may still bind. This is the regime used for the propositions in the main text.

Firms outside the contracted set. If a candidate contract is not activated, the firm is excluded from C and is treated as an outside firm in the main text. No strategic capital is deployed, no directed action is contracted, and the contract rate is economically irrelevant. The propositions focus on contracted recipients because the empirical predictions concern firms and banks that receive strategic capital.

No induced action or corner deployment. Positive funding need not imply distorted real activity. If $\Psi_Z = 0$ and both the contract allocation and commercial benchmark are interior, the financier

[48] If an outside firm is at a boundary, the derivative is replaced by the relevant local one-sided derivative, or by zero when the boundary locally fixes k_{js}^*, as implied by that firm's own KKT conditions. The main benchmark assumes interior outside responses.

leaves the firm at $k_{is} = k_{is}^{\text{out}}$. If $\Psi_Z < 0$, the financier would prefer less external deployment than the commercial benchmark.

With no tied-financing constraint, $x_i > 0$ is compatible with the corner $k_{is} = 0$. Such a contract is funding without positive directed strategic-market activity, so it lies outside the directed-action benchmark.

Binding aggregate capacity. If $\lambda > 0$, aggregate funding capacity binds. A positive λ enters each contracted recipient's funding condition negatively, compressing funding all else equal:

$$x_i^* = \frac{r_i - \rho - \lambda}{c}.$$

The contract-rate cancellation survives because r_{is} still drops out of the funding first-order condition when $y_i = 1$.

A.3 First-best benchmark

The baseline model characterizes the strategic financier's contracting problem, not a welfare optimum. This subsection defines a first-best planner that directly chooses deployment and makes clear which externalities the strategic financier does and does not internalize.

Setup. A first-best planner chooses external deployment $k_{\ell s}$ for every firm $\ell \in \mathcal{I}$ and values the aggregate network stock at $W(Z)$, with $W' > 0$ and $W'' \leq 0$, separately from the friction-reduction channel $\mu(Z)$. As in the main model, ordinary home-market activity is pre-optimized into $A_\ell(r_\ell)$, which is constant in the external-deployment problem. The planner inherits the same financing wedge as the baseline model, so the comparison isolates the internalization of network externalities. Unlike the strategic financier, the planner problem does not separate contracted recipients from outside firms. The planner's external-deployment problem is

$$\max_{\{k_{\ell s}\}_{\ell \in \mathcal{I}}} \sum_{\ell \in \mathcal{I}} \Big[A_\ell(r_\ell) + G_\ell(k_{\ell s}) - \mu(Z) k_{\ell s} - r_\ell k_{\ell s} \Big] + W(Z), \tag{40}$$

where $Z = \sum_{\ell \in \mathcal{I}} k_{\ell s}$, subject to $k_{\ell s} \geq 0$ for all $\ell \in \mathcal{I}$.

First-best optimality. Let

$$Z^{FB} = \sum_{\ell \in \mathcal{I}} k_{\ell s}^{FB}.$$

Consider an interior planner allocation satisfying the relevant second-order conditions, with $k_{\ell s}^{FB} > 0$ and $G_\ell''(k_{\ell s}^{FB}) < 0$ for each firm ℓ. Because the planner directly controls every firm's deployment, changing $k_{\ell s}$ changes Z one-for-one. The first-order condition internalizes how a higher Z lowers the bilateral-friction cost on aggregate strategic-market deployment:

$$G_\ell'(k_{\ell s}^{FB}) - \mu(Z^{FB}) - r_\ell - \mu'(Z^{FB})Z^{FB} + W'(Z^{FB}) = 0. \tag{41}$$

Using the policy-wedge notation

$$G_\ell'(k_{\ell s}) = \mu(Z) + r_\ell - \tau_{\ell s},$$

the first-best wedge is

$$\tau_{\ell s}^{FB} = -\mu'(Z^{FB})\, Z^{FB} + W'(Z^{FB}). \tag{42}$$

At a given aggregate stock, the social wedge is common across firms. The planner applies it to all firms, while the strategic financier contracts only with firms in C. Since $\mu'(Z) < 0$, the friction-reduction component is positive when $Z^{FB} > 0$. A positive wedge lowers the effective marginal cost of strategic-market deployment relative to the private benchmark.

Comparison with the strategic financier. The planner and the strategic financier differ in which payoffs enter the marginal choice. The strategic financier values the whole network stock through $\theta U(Z)$. It does not, however, maximize total firm surplus, so the friction-reduction term enters only through the participation constraints of contracted recipients. Ignoring outside-firm feedback for exposition, the strategic wedge is common across contracted recipients and, for any $i \in C$, is

$$\tau_{is}^* = \theta U'(Z) - \mu'(Z) \sum_{n \in C} \left(k_{ns} - k_{ns}^{\text{out}}\right), \qquad \tau_{is}^{FB} = W'(Z) - \mu'(Z)Z.$$

The sum over $n \in C$ appears because a marginal increase in Z changes the participation value of all contracted recipients, not only recipient i. These objects need not be ordered. The planner internalizes the social value of lower bilateral frictions for all firms; the strategic financier internalizes its own network payoff and the compensation needed to contract with C. Which wedge is larger depends on $\theta U'(Z)$ versus $W'(Z)$, the contracted set, outside-firm feedback, and the allocation being compared.[49] The first-best benchmark is therefore a diagnostic of which externalities the

[49] If a planner allocation is at a corner, the comparison is read through the corresponding KKT inequality rather than the interior first-order condition. A weak ranking can be obtained under restrictive conditions. If rate interiority holds, outside-firm feedback is shut down $\left(\sum_{j \notin C} \partial k_{js}^*(Z)/\partial Z = 0\right)$, $k_{is}^{\text{out}} \geq 0$, and $\theta U'(Z) \leq W'(Z)$, then $\sum_{i \in C}(k_{is} - k_{is}^{\text{out}}) \leq$

financier does and does not internalize, not a monotone bound.

A.4 Dynamic incentive compatibility and relational contracts

The baseline model assumes that the strategic financier can enforce the required external action k_{is} through legally binding covenants. Let

$$\tau_{is}^{\text{static}} \equiv \tau_{is}^{*} = \Psi_Z$$

denote the wedge under perfect covenant enforcement. This subsection keeps the contracted-recipient benchmark regime, but replaces legal enforcement with relational enforcement through the threat of termination. In the binding dynamic incentive compatibility (DIC) regime, the financier may leave rents to sustain compliance, so the static rent-extraction result $y_i = 1$ need not apply.

Environment. The relationship repeats indefinitely with common discount factor $\beta \in (0, 1)$. I study a stationary extension of the static contract: the same $\{x_i, r_{is}, k_{is}\}$ is offered each period, and Z is not separately accumulated over time. A deviating firm takes the cheap financing but chooses k_{is}^{out}. The deviation is detected after one period, and future access to the contract ends.

The dynamic incentive compatibility constraint. Let

$$B_i(k, Z) = G_i(k) - \mu(Z)k - r_i k$$

denote the part of operating payoff that depends on strategic-market deployment after home-market activity has been pre-optimized. The one-period deviation gain from ignoring the directed-action requirement is

$$D_i(k_{is}, Z) \equiv B_i(k_{is}^{\text{out}}, Z) - B_i(k_{is}, Z).$$

Since k_{is}^{out} is the private optimum at fixed Z, $D_i(k_{is}, Z) \geq 0$, with equality only when the mandated action coincides with the private optimum. This is the operating payoff the firm recovers by ignoring the directed-action requirement. Financing terms cancel in D_i because the firm receives the same financing rate whether it complies or deviates. Let

$$R_i = (r_i - r_{is})x_i$$

$\sum_{i \in C} k_{is} \leq Z$, so $\tau_{is}^{FB} \geq \tau_{is}^{*}$ at the common Z. Once these restrictions are relaxed, no global ranking follows.

be the per-period financing rent. Stationary participation requires

$$R_i - D_i(k_{is}, Z) \geq 0, \qquad [y_i] \tag{43}$$

with multiplier $y_i \geq 0$. Compliance gives present value $(R_i - D_i(k_{is}, Z))/(1-\beta)$ relative to the outside option. A one-shot deviation gives R_i today and ends future surplus. The dynamic incentive compatibility constraint is therefore

$$\frac{R_i - D_i(k_{is}, Z)}{1-\beta} \geq R_i,$$

or equivalently

$$\beta R_i = \beta(r_i - r_{is})x_i \geq D_i(k_{is}, Z). \qquad [\xi_i] \tag{44}$$

Let $\xi_i \geq 0$ denote the DIC multiplier. When the DIC binds with $D_i(k_{is}, Z) > 0$, the financier leaves rents to preserve compliance: $R_i = D_i/\beta > D_i$. Participation is then slack, so complementary slackness gives $y_i = 0$. Static rent extraction would set $R_i = D_i$ and leave no continuation surplus, so costly directed action requires extra rent under relational enforcement. The relevant nontrivial relational regime is therefore the binding-DIC, slack-participation regime analyzed below.

If no admissible contract satisfies the DIC, induced action cannot be sustained and the financier must either relax the required action toward k_{is}^{out} or exit the relationship. The binding-DIC analysis assumes the required rent can be delivered through an interior rate choice; otherwise, the financier must relax the action or exit.

Interior wedge under binding DIC. Adding the stationary participation term $y_i[R_i - D_i(k_{is}, Z)]$ and the DIC term $\xi_i[\beta R_i - D_i(k_{is}, Z)]$ to the Lagrangian gives the rate condition

$$x_i - y_i x_i - \beta\xi_i x_i = 0, \tag{45}$$

so for a contracted recipient with $x_i > 0$,

$$y_i + \beta\xi_i = 1. \tag{46}$$

The x_i first-order condition is

$$(r_{is} - \rho) - cx_i + y_i(r_i - r_{is}) - \lambda + \beta\xi_i(r_i - r_{is}) = 0. \tag{47}$$

Using $y_i + \beta\xi_i = 1$, the contract rate again cancels and the separation result is preserved:

$$x_i^* = \frac{r_i - \rho - \lambda}{c}. \tag{48}$$

Under the maintained interior-deployment assumption, so $\chi_i = 0$, the k_{is} condition is

$$y_i\left[G_i'(k_{is}) - \mu(Z) - r_i\right] + \Psi_Z - \xi_i \frac{\partial D_i(k_{is}, Z)}{\partial k_{is}} = 0. \tag{49}$$

Because

$$\frac{\partial D_i(k_{is}, Z)}{\partial k_{is}} = -\left[G_i'(k_{is}) - \mu(Z) - r_i\right] = \tau_{is},$$

this condition implies

$$\tau_{is}^{\text{dyn}} = \frac{\Psi_Z}{y_i + \xi_i}.$$

In the binding-DIC regime with positive deviation gain, participation is slack, so $y_i = 0$ and $\xi_i = 1/\beta$. Therefore the dynamic wedge is

$$\tau_{is}^{\text{dyn}} = \beta\Psi_Z = \beta\,\tau_{is}^{\text{static}}. \tag{50}$$

Thus relational enforcement scales the static wedge by β, holding the contracted set and static network multiplier Ψ_Z fixed. The extension does not re-solve the network fixed point under dynamic enforcement.

Proposition 4 (Local attenuation under relational contracts)**.** *In this reduced-form relational-contract extension with common discount factor β, binding DIC, and slack participation, compliance sustained by the threat of relationship termination rather than by legal covenants attenuates the implementability wedge according to*

$$\tau_{is}^{dyn} = \beta\,\tau_{is}^{static} = \beta\Psi_Z.$$

The separation of financing volume x_i^* from the contract rate r_{is} is preserved.

Empirical support for the relational-contract assumption. The relational-contract interpretation requires persistent ties. The data show patterns consistent with such persistence in both settings. In PitchBook, companies whose first round included an SWF are about 50 times more likely to receive SWF capital in a later round than companies whose first round did not (15.5% vs. 0.3%). In the crisis-era bank sample, 13 of 88 bank–SWF-country pairs received an add-on injection after the initial deal, which directly supports persistent bilateral ties. The fact that 14 of 69 banks received SWF equity from more than one sovereign country points to repeated sovereign involvement in distressed banks, though not necessarily persistence of the same bilateral relationship. These facts are consistent with, but do not by themselves identify, the relational-contract mechanism.

Connecting the extensions. When the benchmark wedge is non-negative, the dynamic extension attenuates it:

$$\tau_{is}^{\text{dyn}} \leq \tau_{is}^{\text{static}}$$

because $\beta \in (0, 1)$. The first-best planner provides a contrasting benchmark because it directly controls all firms and internalizes the full network effect. Table 10 summarizes these regimes. The table is not an additional empirical test; it clarifies which assumptions strengthen or weaken the model's directed-action prediction.

Table 10: Implementability Wedge Across Regimes

Regime	Wedge	Key determinant
First-best planner	$\tau_{\ell s}^{FB} = -\mu'(Z^{FB})Z^{FB} + W'(Z^{FB})$	Direct command over all firms
Static strategic financier	$\tau_{is}^{\text{static}} = \Psi_Z$	Covenant enforcement
Dynamic relational contract (binding DIC)	$\tau_{is}^{\text{dyn}} = \beta\Psi_Z$	Firm patience β
Purely commercial finance	$\tau_{\ell s} = 0$	No strategic objective or directed-action requirement

Note: The first-best wedge is evaluated at the first-best allocation. The first-best planner directly controls all firms; the strategic financier operates through contracts with a subset. The dynamic row reports the binding-DIC, slack-participation regime in an infinitely repeated relationship with discount factor β. The static and dynamic strategic-financier rows use the same static contracted set and static-equilibrium network multiplier Ψ_Z; the dynamic row applies the reduced-form β attenuation rather than re-solving a separate dynamic network fixed point.

B Data Construction

B.1 PitchBook Co-Investment Panel

This subsection documents the PitchBook panel behind Section 3. The construction choices are part of the empirical design: the sample keeps deals with both SWF and non-SWF investors, drops same-country comparisons that would mechanically favor the SWF, and uses ID-based expansion matches to avoid fuzzy-name false positives.

Source tables. I link five PitchBook tables: *Deal* (2.5M transactions with deal date, size, post-money valuation, revenue, type, and status), *Investor* (400K profiles with type, headquarters country, and AUM), *DealInvestorRelation* (3.0M investor–deal participation records), *CompanyInvestorRelation* (2.4M acquirer–target linkages), and *Company* (1.0M firms with headquarters country, sector, founding year, and financials). All monetary values are in USD millions.

SWF classification. I start from PitchBook's "Sovereign Wealth Fund" investor type, which includes 98 investors across 51 countries. I then exclude 28 entities that are not SWFs, such as state pension funds, regional development agencies, and charitable foundations, and add 7 government entities that function as de facto SWFs, including the Future Fund of Australia, the Hong Kong Monetary Authority, and the Monetary Authority of Singapore. The resulting classification is then filtered by the co-investment and valuation-discount regression sample; 21 SWF home countries contribute SWF investor–deal observations to the main regression sample.[50]

Co-investment identification. A deal enters the co-investment sample if it includes at least one SWF and at least one deep-pocket non-SWF investor, defined as PE/buyout, VC, asset manager, hedge fund, investment bank, university, family office, growth/expansion, government investor or mutual fund. Of 2.5M completed deals with valid dates, 6,789 satisfy this filter.

Panel filters.

1. *Investor join*: DealInvestorRelation and CompanyInvestorRelation are linked to Investor by the PitchBook investor identifier. Rows whose investor IDs do not appear in Investor are dropped rather than imputed, which is conservative but avoids assigning SWF status or investor home country using uncertain identifiers.

2. *Target-country assignment*: I use the company-level headquarters country when available. If it is missing, I fall back only to validated site-location tokens: exact country names are

[50] The 44 SWF investors that enter the main valuation-discount regression sample are, using PitchBook investor names: Australia Future Fund; Future Fund; QIC; Mumtalakat; Brunei Investment Agency; CIC Capital Corporation; Central Huijin Investment; China Investment Corporation; China-Africa Development Fund; Silk Road Fund Company; Bpifrance; Caisse des Depots Group; Hong Kong Monetary Authority; Ireland Strategic Investment Fund; CDP Equity; Kuwait Investment Authority; Khazanah Nasional; New Zealand Superannuation Fund; Government Pension Fund Norway; Oman Investment Authority; Oman's State General Reserve Fund; Qatar Holding; Qatar Investment Authority; Russian Direct Investment Fund; Saudi Arabia's Public Investment Fund; GIC (Singapore); Government of Singapore; Monetary Authority of Singapore; Temasek Holdings; Public Investment Corporation; Korea Investment Corporation; National Development Fund of Taiwan; Abu Dhabi Developmental Holding Company (ADQ); Aabar Investments; Abu Dhabi Growth Fund; Abu Dhabi Investment Authority; Abu Dhabi Investment Council; Abu Dhabi Investment Office; Abu Dhabi Investment Company (Invest AD); Investment Corporation of Dubai; Mubadala Development Company; Mubadala Investment Company; Oman Investment Fund and Tawazun Strategic Development Fund.

accepted, and exact US state names or postal abbreviations are mapped to the United States. Unvalidated strings remain missing.

3. *Same-country drop*: Remove investor–deal pairs where the target headquarters country equals the investor's home country, avoiding the mechanical confound that firms do more business at home.

4. *Same-as-SWF drop*: Remove non-SWF investors from the same country as the deal's SWF, since these cannot contribute clean cross-country within-deal variation.

5. *Re-require both arms*: After each filter, verify that every deal still contains at least one SWF and one non-SWF row; drop deals that lose an arm.

6. *Contamination filter*: For multi-round companies, drop later-round rows where the *same investor* (by ID) appeared in an earlier round, to avoid double-counting repeat relationships. Later-round rows with new investors are retained.

7. *Deduplication*: Remove exact duplicate investor–deal rows inherited from duplicates in the raw DealInvestorRelation table.

The final panel contains 10,101 investor–deal pairs across 2,476 deals and 2,417 companies, with 2,810 SWF rows and 7,291 non-SWF rows. After the validated geography fallback, only 49 rows across 9 companies still have unresolved target country. Most of the attrition from the 6,789 candidate co-investment deals to the final 2,476 deals comes from enforcing clean cross-country comparisons, re-requiring both investor arms after those exclusions, and dropping contaminated later rounds with repeat investors. The regression samples are smaller because the financing proxies impose additional non-missingness restrictions: the main valuation-discount sample contains 3,713 investor–deal pairs across 882 deals, the common valuation-discount-plus-dilution sample contains 3,660 pairs across 870 deals, and the revenue subsample used in the P/Revenue horse race contains 1,829 pairs across 399 deals.

Outcome construction. I identify expansion events through two ID-based routes. In route 1, the panel company appears as an active investor, acquirer, new investor, or add-on sponsor in PitchBook's DealInvestorRelation table for expansion-type deals (M&A, JV, VC, buyout, or corporate). In route 2, the panel company appears as an acquirer in CompanyInvestorRelation. I combine the two routes and deduplicate them on company, target country, and date, which yields 9,896 unique expansion events. The company-to-investor bridge uses exact normalized name matches only. I do not use fuzzy matching, alias expansion, or manual predecessor-name stitching, so any resulting measurement error should attenuate the estimates by missing true expansions rather than inflate them through false matches.

For each investor–deal pair, three outcomes are constructed:

- *New entry* (Y_{ext}): equals 1 if the target expanded into the investor's home country within 5 years after the deal, or through the end of the observed expansion window for later deals, and had no prior expansion activity in that country before the deal. Pairs with pre-deal activity are coded as zero (continuation, not new entry).

- *Count* ($\text{asinh}(\text{Count})$): inverse hyperbolic sine of the number of expansion events in the investor's home country over the same observed post-deal window.

- *Value* ($\text{asinh}(\text{Value})$): inverse hyperbolic sine of the sum of recorded deal sizes (USD M) over the same observed post-deal window. About 46% of expansion events have recorded deal sizes; missing sizes are imputed as zero so that the value regression uses the same sample as the count regression, biasing value coefficients toward zero.

Valuation discount. For each deal d in industry sector s at year t:

$$\text{ValDiscount}_d = -\Big(\log(\text{PostVal}_d) - \overline{\log(\text{PostVal})}_{s(d),t(d)}\Big),$$

where the benchmark $\overline{\log(\text{PostVal})}_{s,t}$ is computed from the full PitchBook deal universe, not just the co-investment panel. The sign flip ensures that higher values correspond to lower valuations relative to peers and therefore more expensive outside finance, that is, higher r_i. I winsorize the variable at the 1st and 99th percentiles and standardize it to unit SD. I construct dilution (= dealsize/postvaluation) analogously, winsorize it at the 99th percentile, and standardize it.

Pricing premium (P/Revenue). The separation test in Table 2 uses $\log(\text{PostVal}/\text{Revenue})$ as an alternative pricing measure. Revenue in PitchBook is deal-specific rather than taken from the static Company.dta snapshot. Some pre-revenue firms, mainly in biotech, mining exploration, and early-stage technology deals, report zero or negative revenue. I exclude those deals from the log(P/R) calculation and propagate the resulting missing values through that regression sample. This does not affect the main valuation-discount specification.

Pre-deal geographic orientation. For each investor–deal pair, I compute the target company's pre-deal expansion share toward the investor's home country: the number of expansion events in that country during the three years before the deal divided by the total number of expansion events across all countries in the same window. This variable controls for pre-existing geographic orientation and enters progressively interacted with the valuation discount.

Table 11 summarizes the final investor–deal panel used in the main PitchBook regressions. It connects the construction steps above to the paper's within-deal design by showing the outcome rates, SWF and non-SWF row counts, and the financing-friction measures that define the main interaction.

B.2 DealScan Syndicated Lending Panel

This subsection documents the construction of the bank–country–quarter syndicated lending panel from Refinitiv DealScan.

Raw data. The source file is the DealScan general-syndicate extract, which contains lender–tranche-level records for syndicated loan facilities worldwide. Each row represents one lender's participation in one tranche. I retain the following fields: lender parent name and operating country, borrower country, deal and tranche identifiers, deal active date, tranche type, tranche amount (USD converted), lender share (USD), number of lenders, primary role, loan purpose fields, tranche maturity dates, tenor, average life, all-in spread drawn (bps), margin (bps), and fee variables (upfront, commitment, and annual fees in bps).

Lender parent assignment. DealScan assigns each lender to a parent entity. I apply one override: any row whose lender name contains "Merrill Lynch" (case-insensitive) is reassigned to the parent "Merrill Lynch," regardless of DealScan's current parent mapping. This preserves Merrill Lynch as a distinct treated unit in the pre-2008 period rather than merging Merrill-branded subsidiaries into post-acquisition parents such as BofA Securities.

Exposure formula. For each lender–tranche row, I compute committed dollar exposure as follows. Rows where the lender holds a purely advisory or administrative role (adviser, agent, custodian, paying agent, etc.) are first excluded from the exposure calculation and tracked separately as service-role counts. For the remaining lending rows:

1. If the lender's reported DealScan share is non-missing and positive, I convert that percentage

Table 11: PitchBook Co-Investment Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (investor–deal level)						
New entry (5yr post-deal)	3,713	0.05	0.23	0.00	0.00	0.00
Expansion count (5yr)	3,713	0.39	2.35	0.00	0.00	0.00
asinh(expansion count)	3,713	0.15	0.53	0.00	0.00	0.00
asinh(expansion value $M)	3,713	0.29	1.20	0.00	0.00	0.00
SWF investor	3,713	0.27	0.45	0.00	0.00	1.00
Valuation discount (std.)	3,713	0.00	1.00	-0.71	-0.09	0.63
Dilution (std.)	3,660	-0.02	0.99	-0.86	-0.48	1.46
Deal size ($M)	3,660	1009.44	2697.76	75.00	210.00	653.60
Pre-acq. geographic share	3,713	0.02	0.11	0.00	0.00	0.00
Baseline (country deal share)	3,713	0.15	0.19	0.01	0.03	0.44
Panel B: Panel Dimensions						
Investor–deal pairs (regression obs.)	3,713					
Deals	882					
Target companies	850					
SWF investor–deal pairs	1,017 (27.4%)					
Unique SWF investors	44					
SWF home countries	21					
Non-SWF investor–deal pairs	2,696					
SWF Investment Breadth (per SWF investor):						
Deals per SWF	23 (median 5, range 1–264)					
Target countries per SWF	5.8 (median 3)					
Sectors per SWF	3.4 (median 3)					
Panel C: Broader Clean Co-Investment Panel						
Investor–deal pairs	10,101					
Deals	2,476					
Target companies	2,417					
SWF investor–deal pairs	2,810					
Non-SWF investor–deal pairs	7,291					
SWF investors	48					
SWF home countries with any row	24					

Notes: Panel A reports investor–deal level variables used in the main regressions: new entry and expansion-count/value outcomes within 5 years of the deal, or through the end of the observed PitchBook expansion window for later deals, the SWF-investor indicator, the standardized valuation discount and dilution measures, deal size in USD millions, and the two pre-deal geographic-orientation controls. Panel B reports dimensions for the regression sample with non-missing valuation discount. Panel C reports the broader cleaned co-investment panel before imposing the valuation-discount restriction, which is the source for the 10,101 investor–deal pairs, 2,476 deals, and 2,417 target companies cited in the text. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

into USD by multiplying LenderShare/100 by the tranche amount. In the raw lending rows, approximately 28% have reported shares.

2. If the lender share is missing, I count the number of lead arrangers in the tranche (Admin Agent, Arranger, Bookrunner, Lead Arranger, Mandated Lead Arranger, Syndication Agent, or similar). This lead count is assigned to all rows in the tranche, including non-lead participants.

3. If the lender is a lead and the lender share is missing, I allocate the tranche amount equally across all lead arrangers in that tranche.

4. If the lender is not a lead and leads exist in the tranche (lead count > 0), the lender receives zero exposure. This follows the standard DealScan convention (Sufi, 2007; Ivashina and Scharfstein, 2010): participant shares are poorly reported, and equal-splitting across all lenders would overstate lead arranger exposure.

5. If no leads are identified in the tranche (all lenders are participants), I split the tranche amount equally across all lenders.

This is the main paper's *commitment* measure. It is conservative by construction: when participant shares are unreported, exposure is assigned to the reported lead side of the syndicate rather than spread mechanically across all lenders. In the restricted 18-bank baseline rows used for the main panel, approximately 29% of lending rows receive exposure through step (1), 47% through step (3), and the remaining 24% receive zero through step (4). Deal counts, which count distinct tranches regardless of dollar exposure, are unaffected by the exposure allocation formula and therefore serve as the more robust intensive-margin outcome. Steps (2) and (5) are intermediate routing rules that determine whether a row ultimately falls into the equal-split or zero-exposure categories.

Sum-to-one robustness. The baseline commitment rule is not designed to recover exact within-tranche ownership. As an appendix robustness check, I therefore construct a second panel that first collapses each tranche to unique parent-bank participants and then forces the attributed shares to sum to one within each tranche. When all parent-level shares are observed, I normalize the reported shares to sum to one. Otherwise, I impute lead and participant shares from fully observed loans with the same syndicate structure (number of lead arrangers and number of participants), following the logic of Chodorow-Reich (2014). If a syndicate structure is not observed with complete shares, I fall back to the baseline lead-equal-split rule. Re-estimating the main continuous DiD on this alternative panel leaves the results essentially unchanged relative to Table 5. Deal counts and new entry are unaffected by the exposure-allocation formula, and the loan-volume estimate remains positive under the sum-to-one construction. I therefore keep the commitment-based panel as the baseline and use the sum-to-one construction only as a measurement robustness check.

Aggregation to bank–country–quarter. Exposure and deal counts are summed within each (lender parent, borrower country, year-quarter) cell to produce the *book panel*. A *balanced panel* is then constructed by cross-joining all observed (bank, country) pairs with all observed quarters, filling missing cells with zero exposure and zero deal count.

Summary statistics. Table 12 documents the bank–country–quarter panel used in the crisis design. It reports the outcome variables, the ex ante SWF-status indicator, the bank-level write-down ratio, and the pair-level controls used in the fully controlled specification. The table connects the appendix construction to Table 5: the balanced panel contains many zero cells, so the paper reports new entry separately and estimates log count and log volume on the positive-lending subsamples.

B.3 Bilateral FDI and Gravity Controls

The bilateral FDI data used in Sections 5.1 and related spillover tests are drawn from the World Bank Harmonized Bilateral FDI Database (Steenbergen et al., 2022). This subsection documents the cleaning steps applied to the raw database before estimation.

Source. The WBG-HBFDI database harmonizes four underlying sources: OECD Bilateral FDI Statistics (BMD4 and historical BMD3 series, 2005–2019), the IMF Coordinated Direct Investment Survey (CDIS, 2009–2019), UNCTAD Bilateral FDI Statistics (2001–2012), and China's official FDI bulletins (2003–2019). The database covers 247 reporting economies and 247 partner economies over 2001–2021, with four variables per pair-year: inward/outward flows and inward/outward stocks, all in USD millions. I use inward flows as the outcome variable: for a reporting economy i (destination) and partner economy j (source), the inward-flow series $\text{FDI}^{\text{in}}_{ji,t}$ represents FDI flowing from j into i in year t.

Source deduplication. The harmonized database is constructed through a sequential augmentation process, where each source fills gaps left by higher-priority sources (Table 6 of Steenbergen et al. 2022). The priority hierarchy for inward flows is: (1) OECD-BMD4 from the host (reporting) economy; (2) OECD-BMD4 mirror from the partner economy; (3) China source; (4) UNCTAD;

Table 12: DealScan Bank–Country–Quarter Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (bank–country–quarter level)						
Total cross-border exposure ($M)	24,206	378.5	3357.3	0.0	0.0	29.1
Deal count	24,206	2.12	9.60	0.00	0.00	1.00
log(total exposure), positive cells	6,862	5.325	1.740	4.111	5.175	6.399
log(deal count), positive cells	7,356	1.181	1.068	0.000	1.099	1.792
New entry (LPM)	24,206	0.005	0.074	0.000	0.000	0.000
Lending share (within-bank quarter)	24,206	0.010	0.058	0.000	0.000	0.001
Write-down ratio (bank)	24,206	0.127	0.154	0.031	0.069	0.190
SWF status (country)	24,206	0.130	0.336	0.000	0.000	0.000
Post ($\mathbf{1}[t \geq 2007Q3]$)	24,206	0.429	0.495	0.000	0.000	1.000
WD × SWFStatus	24,206	0.018	0.076	0.000	0.000	0.000
WD × Pre-share	24,206	0.0015	0.0125	0.0000	0.0000	0.0002
WD × FTA	24,178	0.029	0.084	0.000	0.000	0.000
WD × Common language	23,968	0.025	0.089	0.000	0.000	0.000
WD × Political distance	22,406	0.226	0.388	0.038	0.106	0.215
Panel B: Panel Dimensions						
Bank–country–quarter observations	24,206					
Banks	18 (5 treated, 13 control)					
Borrower countries	155 total (14 SWF-status)					
Quarters in window	14 ($[-8, +5]$ around 2007Q3)					

Notes: Panel A reports bank–country–quarter level variables used in the main continuous DiD (Table 5). Outcomes are total cross-border exposure, deal count, log transformations on positive cells, and the new-entry LPM. The bank-level write-down ratio and country-level SWF-status indicator define the treatment interaction WD × SWFStatus, and the additional pair-level controls (Pre-share, FTA, common language, political distance) enter the fully-saturated specification interacted with Post. Panel B reports panel dimensions: 18 baseline banks (5 treated, 13 control), 155 borrower countries in total (14 SWF-status), and the $[-8, +5]$ quarter window around 2007Q3, that is, eight pre-treatment quarters and six quarters from 2007Q3 through 2008Q4. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(5) IMF-CDIS. Despite this sequential design, the 2022 vintage of the database contains approximately 39,500 duplicate pair-year observations, concentrated entirely in 2018–2019 where OECD and CDIS coverage overlaps. Of these, roughly 7,500 pairs have conflicting values from different sources (e.g., OECD-BMD4 reports +532 USD million while CDIS reports −108,582 USD million for the same pair-year). I deduplicate by retaining the highest-priority source for each pair-year, breaking ties by preferring the reporter economy ("r") over the partner economy ("p") as the data origin. This reduces the raw database from 588,058 to 548,487 observations.

Estimation window and gravity controls. The cleaned WBG-HBFDI database covers 2001–2021, but the FIRRMA regressions use the 2014–2020 panel after imposing the event window around the 2018 shock and merging the required controls. Bilateral gravity variables are drawn from CEPII Gravity V2022 (Head and Mayer, 2014): FTA membership, destination GDP (used in log current-USD form), bilateral trade flows (Comtrade), and GDP growth. Diplomatic distance is measured as ideal-point distance from the UN General Assembly voting database (Bailey, Strezhnev and Voeten, 2017). CEPII trade coverage is the binding control and is available through 2020, so the main panel ends in 2020. The 2014 GDP growth observation uses the 2013–2014 lag. Pairs missing gravity controls are dropped from the estimation sample.

Winsorization. Bilateral FDI flows contain extreme values driven by financial-center conduit transactions, such as Luxembourg→Ireland flows above \$100 billion in a single year. To limit the influence of those outliers without dropping observations, I winsorize inward flows at the 1st and 99th percentiles before applying the asinh transformation. I do this within each estimation window, including the main 2014–2020 panel and each placebo window.

Negative and zero flows. Bilateral FDI flows can be negative when disinvestment, profit repatriation, or intercompany loan repayment exceeds new investment in a given year. In the control-complete FIRRMA estimation sample (2014–2020, excluding US pairs), approximately 17% of observations have negative inward flows and 58% are exact zeros. The inverse hyperbolic sine transformation accommodates both: $\text{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$ is defined for all real x, approximately linear near zero, and approximately $\ln(2|x|)$ for large $|x|$.

Summary statistics. Table 13 documents the source×destination×year panel used in the FIRRMA design. The table links the raw FDI data to Table 6 by reporting the transformed FDI outcome, the exposure measures, the gravity controls, and the two balance controls after the 2014–2020 sample restriction. It also makes the estimation support explicit: the crowd-in estimates are not estimated on the full WBG-HBFDI universe, but on the subset with the controls needed for the source×year and pair-fixed-effect specification.

Table 13: FIRRMA Bilateral FDI Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (source × destination × year, 2014–2020)						
Bilateral FDI inflow ($M, winsorized)	16,866	109.2	685.5	0.0	0.0	2.2
asinh(bilateral FDI inflow)	16,866	0.625	3.294	0.000	0.000	1.539
Exposure: high-dilution (destination)	16,866	0.244	0.353	0.000	0.091	0.333
Exposure: count-based (destination)	16,866	0.339	0.379	0.000	0.238	0.700
Exposure: value-weighted (destination)	16,866	0.373	0.432	0.000	0.076	0.927
Post ($\mathbf{1}$[year ≥ 2018])	16,866	0.432	0.495	0.000	0.000	1.000
FTA membership	16,866	0.210	0.407	0.000	0.000	0.000
log(destination GDP)	16,866	20.391	1.415	19.581	20.027	21.250
Destination GDP growth	16,866	0.009	0.100	-0.043	0.014	0.067
asinh(bilateral trade)	16,866	0.528	0.982	0.001	0.038	0.538
Diplomatic distance (UN voting)	16,866	0.850	0.753	0.130	0.667	1.504
US trade share (destination)	16,866	0.093	0.065	0.050	0.079	0.128
Pre-FIRRMA FDI level (std., destination)	16,866	-0.081	0.883	-0.709	-0.206	0.247
Panel B: Panel Dimensions						
Source–destination–year observations	16,866					
Source–destination pairs	2,969					
Source countries	224					
Destination countries	21					
Destinations with positive HD exposure	9					
Years (FIRRMA shock 2018)	2014–2020 (4 pre + 3 post)					

Notes: Panel A reports source×destination×year variables used in the FIRRMA crowd-in regression (Table 6). The outcome is asinh of the winsorized bilateral FDI inflow. Three exposure measures are constructed at the destination level from pre-2018 SWF deal characteristics: high-dilution (above-median dilution), count-based, and value-weighted. Gravity controls follow CEPII Gravity V2022 plus UN-voting diplomatic distance. The balance controls are U.S. trade share and standardized pre-FIRRMA FDI level; the latter is the destination's 2014–2017 mean asinh bilateral inflow, standardized across destinations. Panel B reports panel dimensions: source–destination–year observations, source/destination country counts, and the count of destinations with positive high-dilution exposure. The 2014–2020 window provides four pre-FIRRMA years and three post years. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 BIS Locational and Consolidated Banking Statistics

The main spillover test in Table 8 uses BIS Locational Banking Statistics (LBS), which report claims by office location or residence. The robustness check in Table 33 uses BIS Consolidated Banking Statistics (CBS), which report claims by the nationality of the parent bank.

SWF investment treatment universe. SWF bank equity injections are identified from the Global SWF (GSWF) and Sovereign Wealth Fund Institute (SWFI) transaction databases, supplemented and cross-checked against the IFSWF member-fund deal disclosures, S&P Capital IQ, Mergermarket, and PrivCo. Across all sectors, the cleaned and deduplicated unlisted private-market SWF deal universe covers about \$1.28 trillion of cross-border investment over 2002–2022. Within that broader universe, I retain direct bank equity injections, dropping asset-management mandates, insurance investments, and fund-of-fund commitments. The resulting bank-treatment master contains 107 bank-injection deals totaling \$127 billion across 55 lender–SWF-country pairs and 13 SWF source countries. The BIS spillover design treats these as pair-level aggregate relationships: if a lender-country $\times$ SWF-country pair receives multiple disclosed injections, I sum them into a single total intensity and anchor the cohort at the pair's first injection date. After BIS reporting, disclosure, timing, and matching restrictions, 12 treated lender–borrower pairs enter the stacked DiD used in Table 8. The firm-channel treatment in Table 9 further restricts this universe to unlisted non-financial deals over 2002–2022, totaling \$1.13 trillion in Appendix Table 14.

Data cleaning. The BIS introduced the instrument breakdown (loans vs. debt securities) in 1995, so I restrict the sample to 1995–2024. Negative stock positions (fewer than 0.1% of observations), which arise from exchange-rate valuation adjustments and short positions in debt securities (Bank for International Settlements, 2019), are floored at zero because they are reporting and valuation artifacts rather than economically meaningful negative lending stocks. The main stock specifications use log claims on strictly positive Q4 stock positions; in the matched annual regression sample this restriction drops one loans-and-deposits observation and nine non-bank-loan observations. Annual aggregation uses the Q4 end-of-year stock and drops pair-years without a Q4 observation, following Avdjiev et al. (2020). For the flow column, I use reported LBS loan flows, sum quarterly observations to the pair-year level, and scale the annual flow by the pair's pre-treatment mean loan stock; I do not construct flows from changes in stocks. Matching uses the pre-treatment mean and linear trend of log(loans), standardized within the treated-control pool.

Summary statistics. Table 14 documents the matched stacked-cohort panel used in the BIS spillover regression. It connects the BIS data construction to Table 8 by reporting both the LBS outcomes and the SWF equity-injection treatment intensity after coverage, timing, and matching restrictions. The table also shows the cost of the design: the matched panel is cleaner than the raw treatment universe, but it covers only the country pairs for which BIS reporting and pre-treatment matching variables are available.

Table 14: BIS LBS Stacked-Cohort Panel: Summary Statistics

	N	Mean	SD	p25	p50	p75
Panel A: Regression Variables (cohort × pair × year, ±6-year window)						
Total cross-border claims ($M)	3,259	22976.2	48149.6	973.0	5896.0	20846.0
Loans & placed deposits ($M)	3,258	18392.3	37497.6	684.0	5094.4	17571.5
Non-bank loans ($M)	3,250	5442.6	11114.3	285.5	1922.5	5764.8
log(total claims)	3,259	8.325	2.275	6.880	8.682	9.945
log(loans & deposits)	3,258	8.093	2.332	6.528	8.536	9.774
log(non-bank loans)	3,250	7.141	2.072	5.654	7.561	8.660
SWF intensity ($bn)	3,259	0.267	1.730	0.000	0.000	0.000
SWF intensity × Post ($bn)	3,259	0.144	1.276	0.000	0.000	0.000
Post ($\mathbf{1}[\text{year} \geq \text{cohort treat}]$)	3,259	0.541	0.498	0.000	1.000	1.000
Treated pair indicator	3,259	0.048	0.213	0.000	0.000	0.000
Panel B: Panel Dimensions						
Cohort–pair–year observations	3,259					
Cohorts (lender–borrower × first-treatment year)	12					
Reporting countries (full annual LBS panel)	31					
Borrower countries (full annual LBS panel)	218					
Treated lender–borrower pairs	12					
Control pairs (Mahalanobis-matched)	172					
Event-time window	$[-6, 6]$ around cohort treatment					
Panel C: SWF Investment Treatment (treated pairs only)						
Bank-channel deals (raw)	107					
Bank-channel lender–SWF-country pairs (raw)	55					
SWF home countries with bank injections	13					
Total bank-channel SWF equity ($bn)	127					
SWF equity per pair, post-disclosure filter ($bn)	48	2.65	4.76	0.11	0.43	2.73
Pairs in stacked DiD (LBS-reporting + disclosed)	12					
SWF equity per stacked-DiD pair ($bn)	12	5.57	5.99	0.73	4.15	8.32
Non-financial unlisted SWF deals (2002–2022, cross-channel)	2,636					
Non-financial unlisted firm-channel pairs	378					
Total non-financial unlisted SWF investment ($bn)	1134					
Firm-investment intensity per pair ($bn)	378	3.00	9.02	0.10	0.46	1.87

Notes: Panel A reports cohort×pair×year variables used in the BIS spillover regression (Table 8). Outcomes are total cross-border claims, loans and placed deposits, and non-bank loans, both in raw USD millions and as log-transformed positive stocks. The treatment intensity is total SWF equity injected (USD billions) into banks domiciled in the lender country, by SWFs from the borrower country, interacted with the post-treatment indicator. When a pair receives multiple disclosed injections, those amounts are summed into a single pair-level total and switched on from the pair's first injection date. Panel A pools treated and matched-control rows; conditional-on-treatment summaries are reported in Panel C. Panel B reports panel dimensions. Panel C summarizes the SWF investment treatment universe: bank-channel deals and lender–SWF-country pairs (raw, before LBS coverage and timing restrictions) with the per-pair SWF equity-investment distribution, and the analogous non-financial firm-investment universe used as treatment in the cross-channel test (Table 9). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Additional PitchBook Results

This appendix collects robustness checks and supporting evidence for the PitchBook co-investment results in Section 3. The checks address the main threats to the within-deal design: sector composition, valuation premia and the pricing-versus-financing distinction, country concentration, random SWF-label assignment, the decomposition of absolute expansion rates, tail treatment, outcome-window choice, positive-outcome conditioning, mechanically overlapping same-country co-investors, valuation-discount construction, and firm-level counterfactual selection.

C.1 SWF Representation

Table 15 reports a composition check within the main co-investment sample. Higher valuation discount is associated with greater SWF representation among the retained institutional investor slots in a round. A one-standard-deviation increase in valuation discount raises the SWF share of investor slots by 2.4 percentage points (column 1) and the probability that a given investor row is an SWF by 3.5 percentage points (column 2). Dilution, measured as deal size over post-money valuation, gives similar estimates in columns (3)–(4).[51] This table should not be read as a direct test of the model's financing volume x_i. PitchBook reports investor-level ticket amounts too sparsely for that test. The main test in Section 3 asks whether the SWF–non-SWF directed-expansion differential loads on valuation discount.

Table 15: Financing Frictions and SWF Representation

	SWF Share (1)	is_SWF (2)	SWF Share (3)	is_SWF (4)
Val. Discount	0.0238***	0.0350***		
	(0.0055)	(0.0078)		
Dilution			0.0315***	0.0413***
			(0.0051)	(0.0072)
Year FE	Yes	Yes	Yes	Yes
SE	Company	Company	Company	Company
Observations	882	3,713	870	3,660

Note: Columns (1) and (3) are deal-level (one row per deal); the dependent variable is the SWF share of investor slots in the deal. Columns (2) and (4) are investor–deal level; the dependent variable is an indicator for whether the investor slot is held by an SWF. Columns (1)–(2) use valuation discount; columns (3)–(4) use dilution. All columns include year fixed effects. SE are clustered by company in all columns. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

[51] High-friction rounds tend to have smaller retained syndicates, so the SWF-share outcome can partly reflect fewer non-SWF comparison investors. This denominator concern is strongest for the valuation-discount share specification. For dilution, the association survives controlling for the log number of retained investor slots, and high-dilution rounds also contain more SWF investor slots. I therefore interpret the table as a syndicate-composition check rather than a direct test of financing volume.

C.2 Follow-on Financing

Table 16 reports three checks on follow-on financing. Panel A asks whether an SWF investor is more likely to return to the same company. Let i index target companies, j investors, and d financing rounds. Columns (1)–(2) estimate

$$\text{Return}_{ijd}^{5y} = \alpha_d + \beta\,\text{SWF}_{jd} + X'_{ijd}\gamma + \varepsilon_{ijd},$$

where Return_{ijd}^{5y} equals one if the exact investor appears in another financing round for the same company within five years. The deal fixed effect compares SWF and non-SWF investors in the same original round. Column (1) includes no controls. Column (2) adds investor j's own prior follow-on rate, computed from earlier portfolio-company relationships whose five-year return window had closed before round d. This is an investor-history control, not a measure of the target firm's prior fundraising. Column (2) also controls for same-home-country co-investors in the original round. The SWF coefficient is 3.35 percentage points in column (1) and 2.49 percentage points in column (2), so SWF investors are more likely than same-deal non-SWF co-investors to return.

Panels B and C ask whether expected follow-on financing helps explain the expansion gap. Following Gil and Marion (2013), who proxy continuation value with future market opportunities, I use market thickness. SWF-home thickness is the standardized log count of prior five-year SWF-backed deals by investors from the focal investor's home country in the target firm's sector. Generic thickness is the analogous count for non-SWF deep-pocket investors from the same home country and sector. These are investor-home-country-by-sector deal-flow measures, not target-country measures.[52]

Panel B applies this proxy to new entry. The baseline SWF × valuation-discount coefficient is 0.0371. Adding SWF-home and generic thickness leaves it nearly unchanged at 0.0372, while the two thickness terms are positive but imprecise. For first entry, the result remains tied to the financing-friction proxy rather than to market thickness. Panel C turns to post-deal expansion activity. For both asinh-transformed expansion count and value, the SWF × valuation-discount response is larger in thicker generic home-country sectors. Since generic and SWF-home thickness are strongly correlated within SWF investor rows, I read Panel C as evidence that later expansion is stronger where follow-on financing opportunities are more abundant.

[52] In unreported checks, target-country-by-sector thickness does not significantly predict future SWF financing and does not significantly amplify the new-entry or expansion responses.

Table 16: Follow-on Financing and Market Thickness

	Same-Investor Return		New Entry		Expansion Activity	
	Base (1)	+ Controls (2)	Base (3)	+ Thickness (4)	Count (5)	Value (6)
Panel A: Exact investor follow-on financing						
SWF investor	0.0335***	0.0249**				
	(0.0106)	(0.0107)				
Investor baseline return controls	–	✓	–	–	–	–
Other same-country investor controls	–	✓	–	–	–	–
Panel B: New entry and ex ante thickness						
SWF × Val. Discount			0.0371***	0.0372***		
			(0.0117)	(0.0112)		
SWF × SWF-home thickness				0.0137		
				(0.0253)		
SWF × generic thickness				0.0439		
				(0.0307)		
Panel C: Expansion activity and ex ante thickness						
SWF × VD × generic thickness					0.1669***	0.3431***
					(0.0486)	(0.1224)
SWF-home thickness interactions	–	–	–	–	✓	✓
Observations	3,713	3,713	3,713	3,713	3,713	3,713
Deals / firms	882 / 850	882 / 850	882 / 850	882 / 850	882 / 850	882 / 850
Deal FE	✓	✓	✓	✓	✓	✓
Investor-country × year FE	–	–	✓	✓	✓	✓
SE clustered by company	✓	✓	✓	✓	✓	✓

Note: Investor–deal level regressions. In Panel A, the dependent variable equals one if the exact investor appears in another PitchBook financing round for the same company within five years. Investor baseline return controls in column (2) are the investor's prior same-investor follow-on propensity, the log number of mature prior investor–company relationships used to estimate it, and an indicator for having no mature prior relationship history. Same-country investor controls are the log number of other investors from the focal investor's home country in the original deal and an indicator for whether any of those same-country investors is an SWF. In Panel B, the dependent variable is new entry into the investor's home country within five years, conditional on no prior activity there before the deal. In Panel C, the dependent variables are asinh-transformed post-deal expansion count and value, with zero outcomes retained. SWF-home thickness is the standardized log count of prior five-year SWF-backed deals from the focal investor's home country in the firm's sector. Generic thickness is the analogous measure for non-SWF deep-pocket investors from the same home country and sector. Thickness is measured by investor home country and sector, regardless of the prior target company's country. Columns (1)–(2) include deal fixed effects. Columns (3)–(6) include deal fixed effects and investor-home-country × deal-year fixed effects. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Sector Heterogeneity

Table 17 asks whether the PitchBook result is a sector artifact. It drops each primary industry sector in turn and re-estimates the main specification. The aggregate count and value estimates remain stable, so the expansion-count result is not a one-industry result. The new-entry margin is more fragile: it concentrates in consumer-facing deals, where entry plausibly requires local distribution, regulatory approval, or reputation building. That qualification narrows the external-validity claim but fits the model's scope condition that directed expansion should be strongest when local entry is costly and commercially actionable.

Table 17: Sector Heterogeneity: Leave-One-Sector-Out

	New Entry	asinh(Count)	asinh(Value)
Baseline (all sectors)	+0.037***	+0.103***	+0.213***
Business (B2B)	+0.042***	+0.108***	+0.217***
Consumer (B2C)	+0.015	+0.090***	+0.213***
Energy	+0.039***	+0.105***	+0.217***
Fin. Services	+0.040***	+0.105***	+0.222***
Healthcare	+0.040***	+0.107***	+0.211***
Info. Technology	+0.042***	+0.081***	+0.158***
Materials	+0.038***	+0.105***	+0.215***

Notes: Each row drops one PitchBook primary industry sector (Business (B2B), Consumer (B2C), Energy, Financial Services, Healthcare, Information Technology, or Materials) and re-estimates the baseline specification (SWF × Valuation Discount with deal FE + Investor-country × Deal-year FE; SE clustered by company). New Entry = **1**[no pre-deal activity in investor country AND post-deal activity]. asinh(Count) and asinh(Value) use all post-deal activity (not conditioned on pre=0). The new-entry effect concentrates in Consumer Products (B2C); the aggregate count and value estimates are robust to dropping any single sector. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4 SWF Valuation Premium

Table 18 shows SWF-backed deals have higher log post-money valuation over revenue. The premium is large with sector–year fixed effects and remains about 17–18% after adding round controls or using within-company identification. This result does not weaken the pricing-versus-directed-response interpretation. Instead, it shows that SWFs often pay favorable financial terms, while the main horse race in Section 3 shows that those terms do not predict which firms subsequently expand toward the SWF home country.

C.5 Leave-One-SWF-Country-Out

Figure 8 drops each SWF investor home country in turn. This check addresses the main external-validity concern in PitchBook: the identifying variation is concentrated in a small number of active

Table 18: SWF Valuation Premium: log(Post-Money Valuation / Revenue)

	Outcome: log(Post-money valuation/Revenue)		
	Sector × Year (1)	+Round Controls (2)	Company FE (3)
SWF present	0.4439***	0.1593***	0.1678***
	(0.0605)	(0.0589)	(0.0526)
log(Deal size)		0.1077***	
		(0.0041)	
Sector × Year FE	Yes	Yes	–
Company FE	–	–	Yes
Year FE	–	–	Yes
Round controls	–	Yes	–
SE clustered by company	Yes	Yes	Yes
Observations	113,403	112,207	52,291

Note: Deal-level regressions. Revenue is recorded at the time of the deal (from PitchBook Deal.dta). Column (1): sector×year FE only. Column (2): adds round controls (log deal size, round number, stage). N is smaller because controls require non-missing values. Column (3): company FE + year FE (within-company identification), restricted to companies with ≥2 deals. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

SWF countries. The coefficient remains positive under every omission, preserving the sign of the result, but the sensitivity to Singapore and the UAE qualifies how broadly the PitchBook estimate should be generalized.

C.6 Permutation Inference

Table 19 asks whether the PitchBook pattern could arise from accidental assignment of SWF labels within deals. The test randomly reassigns SWF labels within the design's identifying cells and recomputes both a model-free statistic and the regression coefficient. The observed statistics sit above the simulated null distribution, which is useful in a sparse-event setting where asymptotic clustered standard errors may be fragile.

C.7 Expansion Rate Decomposition

Figure 9 decomposes the redirection pattern behind Figure 4. It shows absolute new-entry rates separately for SWF and non-SWF investor home countries, rather than only the SWF share of expansion events. This qualification matters for interpretation: the positive interaction reflects a relative narrowing of the SWF–non-SWF gap among high-discount deals, not a uniform rise in SWF-country entry rates. Bar labels report the exact cell rates shown in the figure. Panel B shows that the same pattern is not present when deals are sorted by log(P/Revenue), which reinforces the distinction between financing need and financial premium.

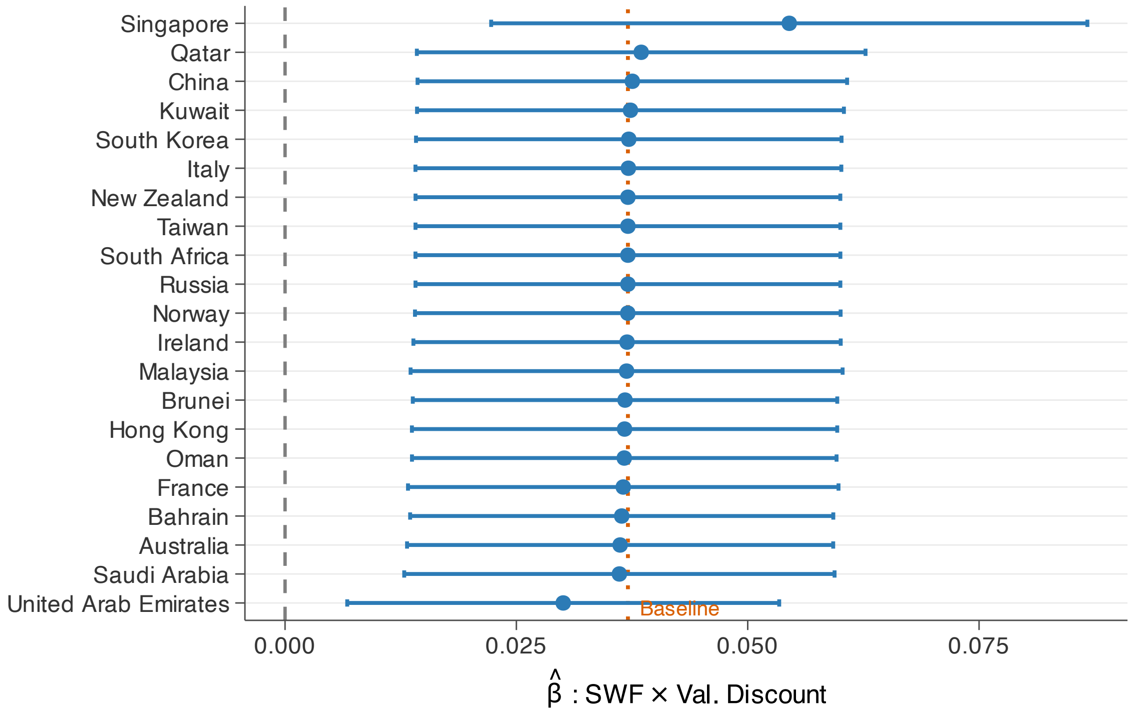

Figure 8: Leave-One-SWF-Country-Out: SWF × Val. Discount Coefficient

Notes: Each row drops one SWF investor home country and re-estimates the full-FE valuation-discount specification (deal FE + Investor-country × Deal-year FE). Orange dashed line: full-sample baseline. Error bars: 95% CIs; SE clustered by company. Singapore and the UAE together account for 73% of SWF investor–deal rows and 88% of SWF new-entry events in the valuation-discount sample. Dropping the UAE attenuates the estimate, while dropping Singapore amplifies it; both omissions leave the coefficient positive and statistically significant.

Table 19: Permutation Inference

	Observed	Null Mean [SD]	RI $p_{\text{two-sided}}$
Panel A: Model-free ratio statistic			
SWF vs. non-SWF ratio	0.959	-0.032 [0.017]	0.000
Panel B: Regression coefficient			
New Entry	0.0371***	-0.0001 [0.0073]	0.000
asinh(Count)	0.1028***	0.0005 [0.0135]	0.000
asinh(Value)	0.2131***	0.0010 [0.0371]	0.000
Permutations		1,000	

Note: Within-deal random reassignment of SWF investor labels (1,000 permutations). Panel A tests a model-free ratio statistic: $E[Y|\text{SWF} = 1]/E[\text{baseline}|\text{SWF} = 1] - E[Y|\text{SWF} = 0]/E[\text{baseline}|\text{SWF} = 0]$; RI p = share of permuted ratios with $|\text{ratio}^{\text{perm}}| \geq |\text{ratio}^{\text{obs}}|$. Panel B permutes the SWF indicator and re-estimates the main specification (deal FE + Investor-country × Deal-year FE, SE clustered by company); RI p = share of permuted SWF × Val. Discount coefficients with $|\hat{\beta}^{\text{perm}}| \geq |\hat{\beta}^{\text{obs}}|$. Stars on observed coefficients reflect asymptotic p-values from the main regression. Two-sided p-values reported. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

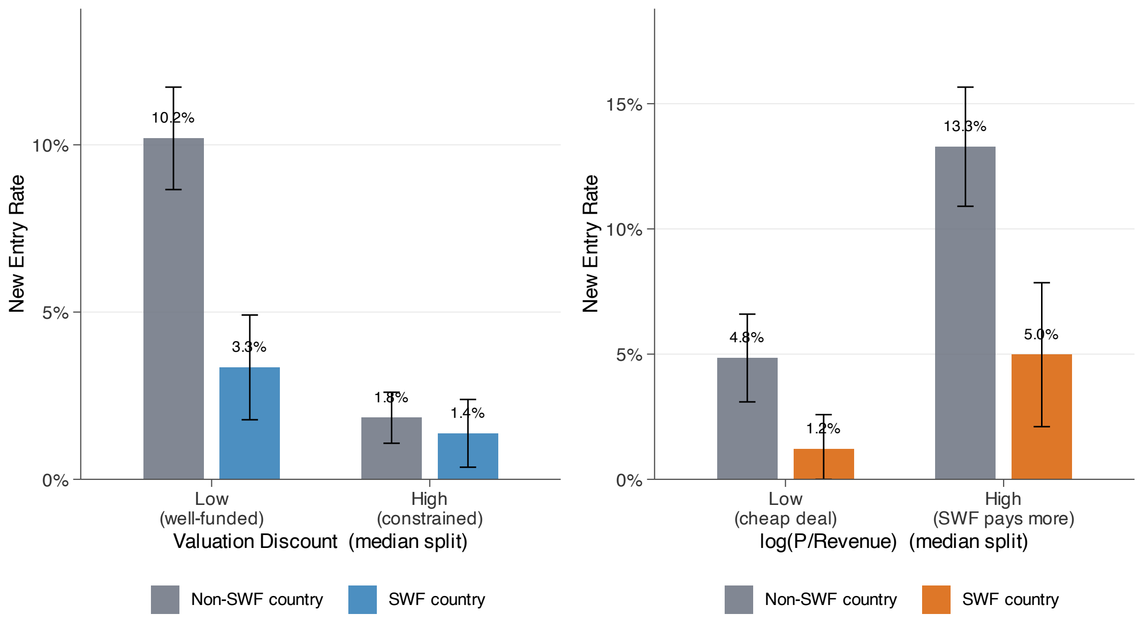

Figure 9: Expansion Rates by Investor Type and Pricing Variable

Notes: New-entry rate (share of investor–deal rows with $Y_{\text{ext}} = 1$) by median split of the indicated sorting variable. Grey bars: non-SWF co-investor home countries. Colored bars: SWF home countries. Panel A uses the full valuation-discount sample of 882 deals and sorts by valuation discount. Panel B uses the revenue-available subsample of 399 deals and sorts by log(P/Revenue). Error bars: 95% CIs. In both bins, absolute entry rates are lower for SWF than for non-SWF rows; the identifying pattern is that the SWF–non-SWF gap narrows sharply in the high-valuation-discount bin of Panel A, but not in Panel B. The exact rates are 10.2% and 3.3% below the valuation-discount median and 1.8% and 1.4% above it in Panel A (non-SWF and SWF, respectively). In Panel B, the corresponding rates are 4.8% and 1.2% below the log(P/Revenue) median and 13.3% and 5.0% above it. That narrowing in Panel A corresponds to the positive interaction $\hat{\beta}_2$ in the separation regression (Table 2).

C.8 Winsorization Sensitivity

Table 20 tests whether the valuation-discount result is driven by extreme valuation residuals. It re-estimates the main specification with no winsorization, alternative two-sided winsorization cutoffs, and two-sided trimming. The coefficient is stable across the non-winsorized and winsorized specifications, so the financing-friction result is not an artifact of a small number of extreme valuation-discount observations. Trimming the tails attenuates the estimate, but the coefficient remains positive.

Table 20: Winsorization Sensitivity

	Outcome: New Entry (5yr)			
	No winsorization	Winsorize p5/p95	Winsorize p1/p99 (main)	Trim p5/p95
	(1)	(2)	(3)	(4)
SWF × Val. Discount	0.0361***	0.0355***	0.0371***	0.0206*
	(0.0114)	(0.0118)	(0.0117)	(0.0114)
Deal FE	Yes	Yes	Yes	Yes
Investor-country × Deal-year FE	Yes	Yes	Yes	Yes
Observations	3,713	3,713	3,713	3,345

Note: Each column re-standardizes valuation discount under a different tail treatment and re-estimates the main specification (deal FE + Investor-country × Deal-year FE; SE clustered by company). Column (3) is the main-text specification, which winsorizes valuation discount at the 1st and 99th percentiles. Column (4) trims (drops) rather than winsorizes observations outside the 5th–95th percentile range, so N is smaller. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.9 Alternative Outcome Windows

Table 21 asks whether the new-entry result depends on the length of the post-deal window used to construct the dependent variable, while keeping the main regression sample fixed. The 3-year, 5-year and 7-year estimates are all positive and significant at 1%, with coefficients rising from 0.0319 to 0.0371 to 0.0410. The window exercise therefore suggests that the directed-expansion response is not sensitive to the post-deal window length.

C.10 Intensive-Margin Robustness: Log on Conditional-Positive Subsample

Table 22 separates the intensive margin from the entry margin by estimating $\log(Y)$ only on investor–deal pairs with strictly positive post-deal expansion. This is a deliberately conditional check: it drops the zero outcomes through which much of the main effect operates. Conditional on expansion, the count coefficients are negative across specifications and only marginally significant in the most saturated specification. The value coefficients are also negative, but the tightest specification is

Table 21: Alternative Outcome Windows

	New Entry (alternative post-windows)		
	3-year (1)	5-year (2)	7-year (3)
SWF × Val. Discount	0.0319***	0.0371***	0.0410***
	(0.0113)	(0.0117)	(0.0121)
Deal FE	Yes	Yes	Yes
Investor-country × Deal-year FE	Yes	Yes	Yes
SE clustered by company	Yes	Yes	Yes
Dep. var. mean	0.0485	0.0536	0.0568
Observations	3,713	3,713	3,713

Note: Each column constructs the new-entry outcome using a different post-deal window (3, 5, or 7 years) and re-estimates the main specification (deal FE + Investor-country × Deal-year FE; SE clustered by company) on the same regression sample as the main table. New entry equals one if the target expands into the investor's home country within the window and there was no prior activity in that country before the focal deal. Only the outcome window changes across columns; the underlying deal sample does not. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

imprecise. The conditional-positive estimates therefore suggest that the main asinh results are driven by entry and total activity, not by larger SWF-relative transaction count or value among pairs that already expand.

C.11 Same-Country Non-SWF Co-Investors

The baseline sample drops non-SWF investors headquartered in the same country as the deal's SWF because their outcome, expansion into the investor's home country, is mechanically identical to the SWF outcome. If an SWF-backed firm expands into the SWF home market, a same-country non-SWF co-investor inherits the same outcome without adding independent within-deal variation. Table 23 reports two diagnostics: an expanded sample that keeps those rows, and a stricter sample that drops the entire financing round whenever it contains a same-country non-SWF co-investor. The estimates remain positive across entry, expansion count, and expansion value, so the result is not driven by how these mechanically overlapping co-investors are handled.

C.12 Valuation Discount Robustness

The main-text valuation discount is the negated sector–year residual of log post-money valuation. Table 24 addresses two mechanical concerns about this proxy for financing need. First, post-money valuation includes the size of the financing round, so the measure could partly capture round size rather than the firm's outside financing cost r_i. Second, within a sector–year cell, lower absolute valuations may reflect firm scale rather than financing frictions. These checks matter for the design because the PitchBook test relies on valuation discount as the empirical counterpart to the model's

Table 22: Intensive-Margin Robustness: log(Count) and log(Value) on Positive Subsample

	log(Count) | Count > 0			log(Value) | Value > 0		
	(1)	(2)	(3)	(4)	(5)	(6)
SWF × Val. Discount	-0.1374	-0.2362	-0.4738*	-1.6907***	-2.0061***	-0.9029
	(0.2946)	(0.1830)	(0.2500)	(0.5073)	(0.4579)	(0.7085)
SWF	-0.5856**	-0.3651*		-1.3270*	-1.1412	
	(0.2626)	(0.1865)		(0.6805)	(0.6859)	
Pre-acq. share		0.5359***			0.6158***	
		(0.1216)			(0.1173)	
Pre-acq. share × Val. Disc.		0.1397			-0.3645**	
		(0.0912)			(0.1790)	
Deal FE	✓	✓	✓	✓	✓	✓
Investor-country × Deal-year FE	–	–	✓	–	–	✓
Pre-acq. controls	–	✓	–	–	✓	–
SE clustered by company	✓	✓	✓	✓	✓	✓
Control mean	0.7333	0.7333	0.7333	3.6067	3.6067	3.6067
Observations	363	363	363	251	251	251

Note: Investor–deal level regressions on the conditional-positive subsample (Cragg second part). Columns (1)–(3) use log(Count) on pairs with post-deal expansion count > 0 ($N = 363$). Columns (4)–(6) use log(Value) on pairs with post-deal expansion value > 0 ($N = 251$). Within each outcome, specifications are (a) deal FE only; (b) deal FE + pre-deal acquisition share and its interaction with valuation discount; (c) deal FE + Investor-country × Deal-year FE. Under specification (c), pre-deal controls are absorbed by the Investor-country × Deal-year FE. SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23: Same-Country Non-SWF Co-Investors: Stability Check

	Expanded Sample			Drop Same-Country Deals		
	New Entry (1)	asinh(Count) (2)	asinh(Value) (3)	New Entry (4)	asinh(Count) (5)	asinh(Value) (6)
SWF × Val. Discount	0.0332***	0.0915***	0.1906***	0.0434***	0.1217***	0.2665***
	(0.0100)	(0.0153)	(0.0459)	(0.0116)	(0.0204)	(0.0596)
SWF	-0.0291*	-0.0401**	-0.1477**	-0.0352	-0.0238	-0.1578
	(0.0149)	(0.0203)	(0.0621)	(0.0227)	(0.0463)	(0.1129)
Deal FE	Yes	Yes	Yes	Yes	Yes	Yes
Investor-country × Deal-year FE	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered by company	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,015	4,015	4,015	3,160	3,160	3,160

Notes: Investor–deal level panel. Dependent variables are new market entry, asinh(expansion count), and asinh(expansion value), measured within 5 years post-deal or through the end of the observed PitchBook expansion window for later deals. Columns (1)–(3) use the expanded sample that keeps non-SWF investors whose home country matches the deal's SWF home country. Columns (4)–(6) drop the entire financing round if it contains any such same-country non-SWF co-investor. Outcomes are constructed from the same PitchBook acquisition-matching procedure used in the main specification. All columns use the sector-year valuation-discount benchmark, include deal FE and Investor-country × Deal-year FE, and cluster SE by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

outside-option friction.

Panel A reconstructs the discount from pre-money valuation, which removes the round-size component, and horse-races it against standardized log revenue on the revenue-available common sample. The pre-money and post-money discounts are nearly identical on the broader common sample (correlation 0.97). With the revenue control included, the pre-money discount continues to predict new entry and expansion count, while the revenue interaction does not.

Panel B horse-races the post-money valuation discount against standardized log revenue on the revenue-available subsample. Valuation discount continues to predict directed expansion, while the revenue interaction does not. This makes the simple low-revenue or early-stage-firm interpretation less plausible.

Table 24: Valuation Discount Robustness: Round-Size and Firm-Scale Checks

	New Entry (1)	asinh(Count) (2)	asinh(Value) (3)
Panel A: Pre-money valuation discount and firm scale			
SWF × Val. Discount (pre-money)	0.1217***	0.1044*	0.0616
	(0.0428)	(0.0617)	(0.1888)
SWF × log(revenue), std.	0.0399	-0.0634	-0.2286
	(0.0420)	(0.0699)	(0.1762)
SWF	-0.0075	0.1282	0.0811
	(0.0469)	(0.0830)	(0.2363)
Observations	1,144	1,144	1,144
Panel B: Horse race against firm scale (revenue-available subsample)			
SWF × Val. Discount (post-money)	0.0785***	0.1358***	0.2451***
	(0.0241)	(0.0333)	(0.0941)
SWF × log(revenue), std.	0.0139	-0.0256	-0.0901
	(0.0179)	(0.0296)	(0.0794)
SWF	-0.0230	0.0535	0.0498
	(0.0327)	(0.0573)	(0.1614)
Observations	1,829	1,829	1,829
Deal FE	✓	✓	✓
Investor-country × Deal-year FE	✓	✓	✓
SE clustered by company	✓	✓	✓

Notes: Investor–deal level regressions. Panel A uses a valuation discount constructed from pre-money valuation (negated log pre-money residual relative to sector×year mean, winsorized at 1/99 percentiles, standardized to unit SD) on the revenue-available subset of investor–deal pairs with non-missing post- and pre-money valuations. Panel B uses the main-text post-money valuation discount on the subsample of investor–deal pairs with non-missing deal-level revenue. In both panels, SWF × log(revenue) is a standardized log-revenue horse-race interaction. All columns include deal FE + Investor-country × Deal-year FE; SE clustered by company. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.13 Matched Counterfactual: Would These Firms Have Expanded Anyway?

The within-deal specification in Section 3 compares SWF and non-SWF co-investors in the same financing round, which absorbs deal-level heterogeneity. It does not fully answer a firm-level counterfactual: among similar firms without SWF backing, how many would have expanded into the same SWF country anyway? Table 25 addresses that question with a weighted exact-cell design on cross-border SWF-backed (deal, SWF country) pairs.

The treatment unit is a (deal, SWF country) pair. Each cross-border SWF-backed deal contributes one treatment observation per distinct SWF country among its investors, so that a deal with two SWF co-investors from different countries generates two treatment observations with the same deal covariates but different target countries. Same-country SWF pairs, such as Bpifrance backing a French firm, are dropped. Controls are non-SWF-backed deals paired to candidate SWF countries within exact cells defined by candidate country, three-year deal-year bin, PitchBook primary sector, and a seven-region headquarters grouping. Within those exact cells, I estimate a propensity score using five standardized covariates: valuation discount, asinh deal size, asinh employee count, year founded, and the cumulative count of distinct prior-investor home countries observed before the deal year. The baseline estimand then uses overlap weights with exact-cell fixed effects and clustering by company and candidate country. The outcome is an indicator for expansion into the treated SWF country within five years, constructed using the same expansion types and active-investor statuses as the main analysis (Appendix B.1).

In the baseline (Panel A), 7.21% of SWF-backed (deal, SWF country) pairs expand into the SWF country within five years, compared with 3.23% of the weighted control pairs. The estimated effect is 3.73 percentage points (SE 1.17) across 277 exact cells. Comparable non-SWF firms do expand into the same SWF country, but at less than half the SWF-backed rate. This addresses the firm-level counterfactual that the within-deal design cannot fully answer on its own.

Panel B reports sensitivity checks. Dropping Singapore-targeting treated pairs, the largest cohort, lowers the treated expansion rate from 7.21% to 3.71%, reduces the point estimate to 1.93 percentage points, and makes the estimate statistically indistinguishable from zero. The unweighted exact-cell design, trimmed-propensity sample, ATT weights, and 3- and 7-year outcome windows all continue to produce positive effects. The Singapore sensitivity therefore qualifies both the breadth and the average magnitude of the PitchBook evidence, but the weighted exact-cell exercise still supports the claim that SWF-backed firms expand into SWF countries more often than comparable non-SWF-backed firms.

Table 25: Matched Counterfactual: SWF-Backed vs. Observationally Similar Non-SWF-Backed Firms

	N treated (1)	Treated rate (%) (2)	Control rate (%) (3)	Effect (pp) (4)	SE (pp) (5)
Panel A: Baseline weighted exact-cell design					
Overlap weights + exact-cell FE, 5-yr outcome window	524	7.21	3.23	3.73^{***}	1.17
SE clustered by company and candidate country; 277 exact cells					
Panel B: Sensitivity					
Drop Singapore target country	236	3.71	1.67	1.93	1.44
Unweighted exact-cell FE	524	7.63	0.85	6.31^{***}	1.85
Trim propensity to [0.05, 0.95]	140	12.69	9.31	4.71^{***}	1.25
ATT weights	524	7.63	3.85	3.56^{***}	1.07
Outcome window: 3 years	753	5.40	2.53	2.68^{***}	0.67
Outcome window: 7 years	369	7.27	4.03	2.99^{***}	1.11

Notes: Estimation is on the PitchBook deal universe, restricted to deals with a computable post-money valuation discount and to firms whose name matches an investor name in PitchBook (so expansion events can be detected on both sides). Treatment unit is a (deal, SWF country) pair: a deal with SWF co-investors from multiple distinct countries contributes one treatment observation per country. Same-country SWF pairs (e.g., Bpifrance backing a French firm) are dropped. Controls are non-SWF-backed deals paired to candidate SWF countries within exact cells defined by candidate country, three-year deal-year bins, the seven PitchBook primary sectors, and a seven-region HQ grouping (North America, Western Europe, Asia, Latin America, MENA, Oceania, Other). The baseline row uses overlap weights with exact-cell fixed effects. Propensity scores are estimated within exact cells using five standardized covariates: valuation discount, asinh deal size, asinh employee count, year founded, and the cumulative count of distinct prior-investor home countries observed before the deal year (computed at the company-year level, so within-year deals share a value). Outcome: indicator for expansion into the treated SWF country within the outcome window, using the same expansion types and active-investor statuses as the main analysis (Appendix B.1). All specifications restrict to fully observed outcome windows and report SE clustered by company and candidate country. Panel B reports sensitivity to dropping Singapore target-country pairs, removing the overlap weights, trimming the propensity score support, switching to ATT weights, and changing the outcome window to three or seven years. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Bank Results

This appendix reports robustness checks for the bank syndicated lending results in Section 4. The checks address the main design risks: whether write-down exposure proxies for pre-crisis bank characteristics, whether the result depends on restricting the sample to large cross-border syndicated-loan arrangers, whether zero-inclusive aggregate count and volume specifications give the same qualitative pattern, whether any single SWF destination drives the estimate, whether alternative SWF-treatment definitions change the estimand, and whether random destination labels can reproduce the result.

D.1 Pre-Crisis Balance: Write-Down Ratio

Figure 10 asks whether the bank write-down ratio is simply proxying for pre-crisis bank characteristics. It regresses WriteDownRatio on five standardized covariates computed from the bank–country–quarter panel over 2004Q3–2007Q2. Four covariates are pre-window levels, and the fifth is a within-window growth measure that compares 2004Q3–2005Q4 with 2006Q1–2007Q2. The balance check is reassuring overall: all five coefficients are statistically insignificant, including pre-crisis SWF-country lending share. The main specification additionally absorbs bank–country level differences through bank×country fixed effects and the WD × Pre-share × Post control. Taken together, the figure argues against the most direct continuation story that distressed banks were already systematically more tilted toward SWF-country lending before the crisis.

D.2 Expanded G-SIB Universe

Table 26 asks whether the bank result depends on the 18-bank baseline sample. It re-estimates the continuous DID on the broader G-SIB universe used for the bank-level robustness checks. The reason to include this universe is to stress-test sample selection: the baseline banks are large global syndicated-loan arrangers with substantial cross-border DealScan activity, while the broader G-SIB list adds systemically important banks that are not always close controls on this lending margin. For example, Nomura appears with 775 DealScan deals across 33 borrower countries and five SWF destinations, CIBC with 3,262 deals and three SWF destinations, and Wells Fargo with 2,124 deals; these banks are less comparable to baseline global arrangers such as BNP Paribas, HSBC, RBS, Deutsche Bank, and Barclays. The first-stage universe contains 30 banks, while the realized DealScan lending panel contains 29 because Wachovia does not appear as a separate parent after its 2008 acquisition by Wells Fargo.

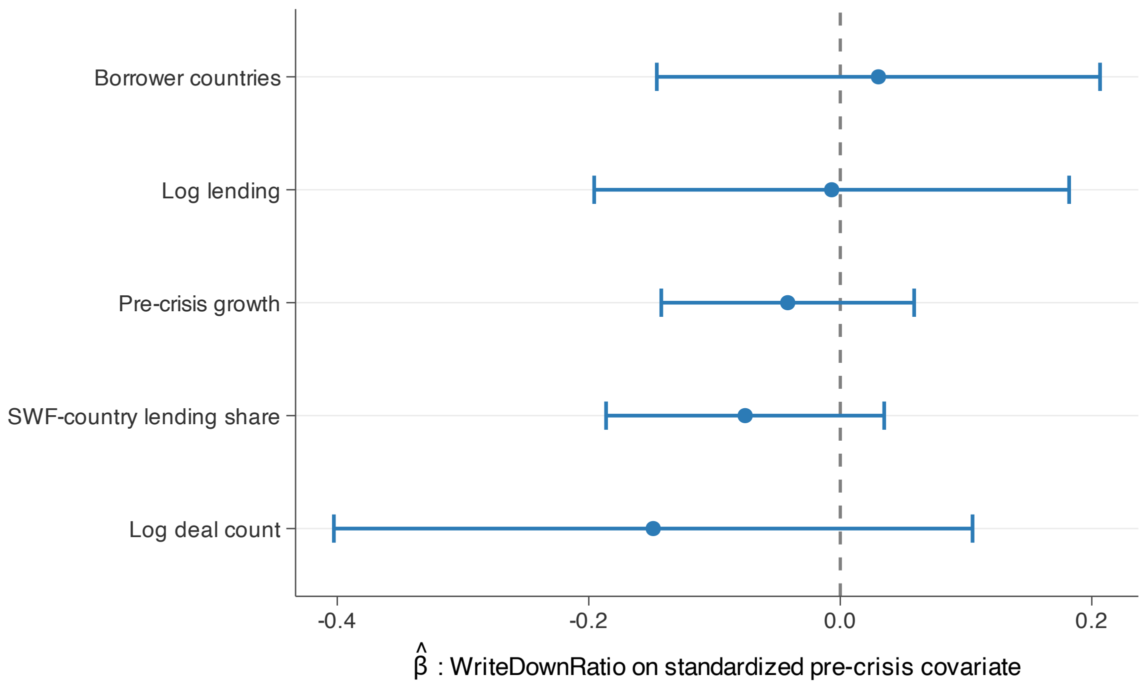

Figure 10: Bank Pre-Crisis Balance: WriteDownRatio on Standardized Covariates

Notes: Cross-sectional OLS of the bank-level WriteDownRatio on five standardized pre-crisis covariates (N=18 banks). Pre-crisis covariates aggregate the bank–country–quarter panel over 2004Q3–2007Q2: total cross-border lending (USD), deal count, number of borrower countries, share of total lending to SWF home countries, and a pre-crisis lending growth measure defined as the log of total lending in 2006Q1–2007Q2 relative to 2004Q3–2005Q4. Each covariate is standardized to mean zero and unit standard deviation. Error bars: 95% confidence intervals from OLS standard errors. Coefficients near zero indicate balance; any residual level differences are absorbed by the Bank×Country and WD × Pre-share × Post controls in the main specification.

The expanded-bank results are therefore best read as a demanding external-validity check, not as the preferred estimand. The lending-reallocation coefficients remain positive in the expanded sample. Under the full-controls specification, the new-entry coefficient is 0.026 and statistically significant, the log-count coefficient is 1.136 and statistically significant, and the log-volume coefficient is positive but imprecise. The table therefore supports the entry and deal-count results among a broader set of syndicated-loan arrangers, while showing why the more comparable 18-bank sample remains the baseline.

Table 26: 29-Bank Continuous DID Results

	New Entry			log(Count)			log(Volume)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
WD × SWFStatus × Post	0.037***	0.036***	0.026***	1.114***	1.009***	1.136***	1.301***	1.183***	0.715
	(0.007)	(0.007)	(0.009)	(0.308)	(0.330)	(0.363)	(0.455)	(0.421)	(0.534)
WD × Pre-share × Post	–	-0.027*	-0.020	–	-1.643**	-1.518*	–	-1.699	-2.221*
		(0.013)	(0.053)		(0.776)	(0.772)		(1.625)	(1.165)
Add. controls (FTA, language, pol. dist.)	–	–	✓	–	–	✓	–	–	✓
Bank×Country FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Bank×Quarter FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Country×Quarter FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
SE clustered by bank	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	37,296	37,296	34,192	10,761	10,761	9,880	9,901	9,901	9,102
R^2	0.260	0.260	0.269	0.809	0.809	0.815	0.795	0.795	0.804

Note: Same progressive specification as Table 5 re-estimated on the expanded G-SIB universe. The bank-level first-stage universe contains 30 banks (5 treated, 25 controls, including Wachovia). The realized DealScan lending panel contains 29 banks (5 treated, 24 controls) because Wachovia does not appear as a separate parent after its 2008 acquisition by Wells Fargo. Columns progress within each outcome group: (1)/(4)/(7) baseline; (2)/(5)/(8) adds WD × Pre-share × Post; (3)/(6)/(9) adds all controls. N is larger than the 18-bank baseline. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3 Aggregate Count and Volume Effects

Table 27 re-estimates the bank DID on aggregate count and volume outcomes, using the inverse hyperbolic sine transformation to retain zero-lending cells. Because zero-lending cells are retained, the coefficients combine entry and scale responses. The aggregate-count estimates are positive and statistically significant across specifications. The aggregate-volume estimates are also positive: they are significant in the first two specifications and remain positive, though less precise, after adding the full pair-level control bundle.

Figure 11 reports the corresponding binned event study for the aggregate count and volume outcomes using the full-controls specification. The pre-period bins are not statistically distinguishable from zero. After the crisis cutoff, aggregate count rises clearly, while aggregate volume also turns positive but is noisier.

Table 27: Aggregate Count and Volume Effects

	Aggregate Count			Aggregate Volume		
	(1)	(2)	(3)	(4)	(5)	(6)
WD × SWFStatus × Post	0.335**	0.312**	0.376**	0.933*	0.865*	0.787
	(0.135)	(0.133)	(0.157)	(0.451)	(0.444)	(0.530)
WD × Pre-share × Post	–	-1.788*	-1.912**	–	-5.108	-5.020
		(0.936)	(0.808)		(3.211)	(3.336)
Add. controls (FTA, language, pol. dist.)	–	–	✓	–	–	✓
Bank×Country FE	✓	✓	✓	✓	✓	✓
Bank×Quarter FE	✓	✓	✓	✓	✓	✓
Country×Quarter FE	✓	✓	✓	✓	✓	✓
SE clustered by bank	✓	✓	✓	✓	✓	✓
Observations	24,206	24,206	22,196	24,206	24,206	22,196
R^2	0.803	0.803	0.809	0.715	0.715	0.723

Note: Bank–country–quarter panel around 2007Q3, $[-8, +5]$ quarters. Columns (1)–(3) use the inverse hyperbolic sine of deal count; columns (4)–(6) use the inverse hyperbolic sine of allocated loan volume. Zero-lending cells are retained, so these specifications capture aggregate effects that combine entry and scale responses. Within each outcome, specifications are (1)/(4) baseline; (2)/(5) adds WD × Pre-share × Post; (3)/(6) adds full pair-level controls (Pre-share, FTA, common language, and UN-voting political distance), each interacted with WD and Post. Bank×Country, Bank×Quarter, and Country×Quarter FEs throughout. SE clustered by bank. The asinh coefficients should be read as index effects rather than percentage changes. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

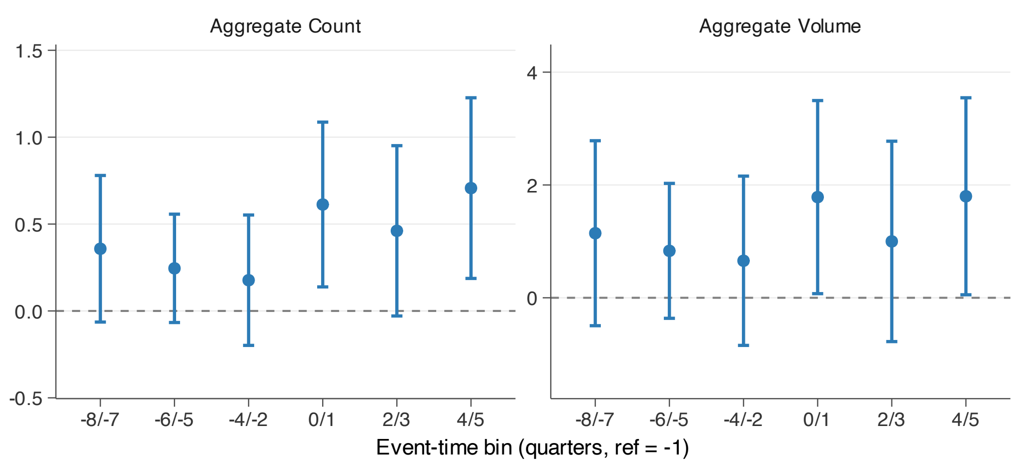

Figure 11: Aggregate Event Study: Deal Count and Loan Volume

Notes: The figure shows WriteDownRatio$_b$ × SWFStatus coefficients by binned event-time period relative to 2007Q3. Outcomes are inverse-hyperbolic-sine deal count and inverse-hyperbolic-sine allocated loan volume. Reference bin: $t = -1$. Error bars: 95% CIs; SE clustered by bank. Full-controls specification with bank×country, bank×quarter, and country×quarter fixed effects, plus pre-share, FTA, common-language, and political-distance controls interacted with WriteDownRatio and the post indicator.

D.4 Leave-One-SWF-Country-Out

Figure 12 drops each SWF destination country in turn from the full-controls log-count DID specification. This addresses the bank-sample concentration concern: with a finite set of SWF destinations, the estimated reallocation could in principle be carried by one destination market. The coefficient remains positive in every omission and remains statistically significant at the 5% level except when Australia is omitted. The exercise therefore rules out a sign reversal or a result driven mechanically by one country.

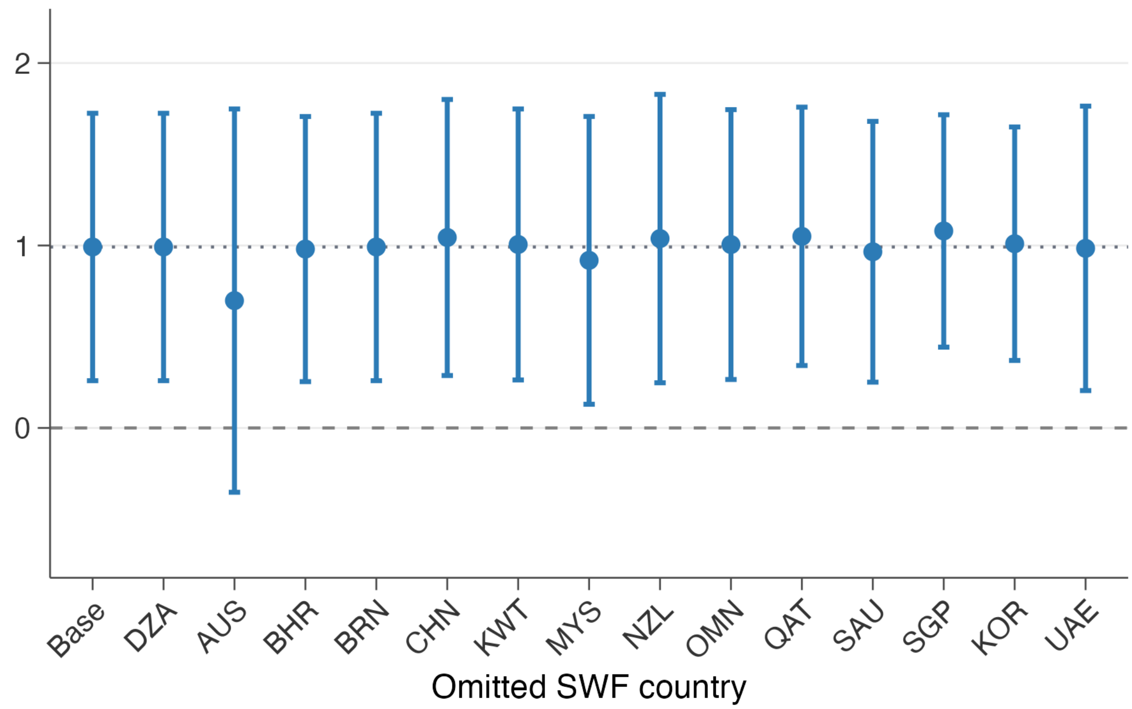

Figure 12: Leave-One-SWF-Country-Out: Deal Count

Notes: Each point is the WD coefficient from the log(deal count) specification, restricted to cells with positive deal count, omitting one SWF country from the destination set. Dashed line: full-sample baseline. Error bars: 95% CIs; SE clustered by bank. Full-controls specification with three-way FEs.

D.5 Alternative SWF Treatment Definitions

The baseline bank specification uses an ex ante destination-level treatment. SWFStatus$_c$ equals one for countries that had active cross-border sovereign investors before the crisis: the 7 countries whose funds later injected equity into treated banks plus 8 additional pre-2007 SWF countries in the defined set.[53] This definition is chosen before observing the realized crisis-era bank–SWF matches. It asks whether banks with greater demand for sovereign capital redirected lending toward sovereign-capital destinations, not only toward the specific countries that ultimately provided their equity.

Table 28 replaces the baseline SWF-status set with the 21 SWF home countries that contribute observations to the PitchBook valuation-discount regression sample. This check uses the same full-controls specification as the main bank table. The coefficients remain positive across all three outcomes. New entry is positive and marginally significant, 0.0153; the log-count coefficient is 0.6947 and significant at the 5% level; and the log-volume coefficient is 1.4065 and significant at the 5% level. The PitchBook-based definition therefore preserves the sign of the bank result, although it shifts the strongest response toward the intensive margins. I treat this as a destination-set diagnostic rather than a preferred specification, because the baseline bank definition is chosen ex ante for the pre-crisis syndicated-lending setting.

Table 28: Alternative SWF Treatment Definition: PitchBook SWF Home Countries

	New Entry (1)	log(Count) (2)	log(Volume) (3)
Panel A: PitchBook SWF home countries (21 countries)			
WD × SWF (PitchBook) × Post	0.0153* (0.0083)	0.6947** (0.3175)	1.4065** (0.5524)
Bank × Country FE	✓	✓	✓
Bank × Quarter FE	✓	✓	✓
Country × Quarter FE	✓	✓	✓
SE clustered by bank	✓	✓	✓
Observations	22,196	6,707	6,262

Note: Bank–country–quarter panel around 2007Q3, $[-8, +5]$ quarters. The table replaces the baseline ex ante SWF-status destination set with the 21 SWF home countries that contribute observations to the PitchBook valuation-discount regression sample. All columns use the full-controls specification with bank×country, bank×quarter, and country×quarter fixed effects; SE clustered by bank. Columns include WD × Post interactions with pre-lending share, FTA, common language, and UN-voting political distance. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E FIRRMA Spillover Robustness

This appendix reports robustness checks for the FIRRMA crowd-in results in Section 5.1. The checks address the main limits of the design. Appendix E.1 expands the destination universe beyond the 21 SWF home countries used in the main specification. Appendix E.2 reports pre-treatment

[53]The defined set contains 15 countries in total. Libya has no lending observations in the realized panel, leaving 14 effective SWF-status destinations in the estimation sample.

balance on destination characteristics, Appendix E.3 documents non-U.S. high-dilution SWF deal volume around FIRRMA, and Appendix E.4 displays the cross-country exposure distribution. Appendix E.5 tests placebo shock dates and drops oil-exporting SWF countries, and Appendix E.6 reports leave-one-exposed-SWF-country-out estimates.

E.1 Full-Destination Main Specification

Table 29 repeats the FIRRMA specification on the broader non-U.S. destination universe rather than restricting destinations to the 21 SWF home countries used in the main text. This is a demanding comparison because the additional destinations have zero SWF exposure by construction and need not be on the same investment margin as SWF home countries. The estimates remain positive and statistically significant across the high-dilution, count-based, and value-weighted exposure measures. The full-control high-dilution coefficient is smaller than in the 21-country design, 0.311 rather than 0.713, which is consistent with dilution from adding many zero-exposure destinations outside the main support. I therefore read this table as a robustness check on sign and statistical stability, not as the preferred magnitude.

E.2 Pre-Treatment Balance: Exposure

Figure 13 asks whether the high-dilution exposure measure is proxying for destination-level fundamentals within the 21-country SWF destination sample. It regresses exposure on six pre-2018 destination covariates: GDP, GDP growth, U.S. trade share, the level and slope of pre-2018 FDI inflows, and the number of source partners. This balance check is necessarily at the destination level because the treatment varies at that level; bilateral covariates are absorbed by source×year and pair fixed effects in the main regression. The figure shows two imbalances. Exposed SWF destinations have greater pre-2018 U.S. trade exposure, which is expected because the treatment is built from U.S.-targeted SWF deals. They also have lower pre-2018 bilateral FDI inflow levels. The other covariates, including destination GDP, GDP growth, pre-2018 FDI growth, and the number of source partners, are not statistically distinguishable across exposure levels. I therefore read the balance test as a reason to include U.S.-trade-share and pre-FDI-level post interactions in the main table, and to read the estimates together with the dynamic and placebo checks.

Table 29: FIRRMA Exposure and Third-Party FDI Inflows: Full Destination Sample

	High-Dilution Exposure				Count	Value
				Target Sources		
	(1)	(2)	(3)	(4)	(5)	(6)
SWF US Exposure × Post	0.304***	0.328***	0.311***	0.726**		
	(0.097)	(0.097)	(0.098)	(0.299)		
SWF US Exposure (count) × Post					0.228***	
					(0.086)	
SWF US Exposure (value) × Post						0.247***
						(0.083)
US Trade Share × Post			0.178	0.081	0.201*	0.202*
			(0.116)	(0.240)	(0.116)	(0.116)
Pre-FIRRMA FDI Level × Post			-0.088***	-0.157***	-0.092***	-0.091***
			(0.019)	(0.039)	(0.019)	(0.019)
asinh(Bilateral Trade)		0.105	0.127	0.104	0.130	0.132
		(0.155)	(0.155)	(0.253)	(0.155)	(0.155)
Diplomatic Distance		0.007	0.015	0.084	0.015	0.016
		(0.059)	(0.059)	(0.117)	(0.059)	(0.059)
Other gravity controls	–	✓	✓	✓	✓	✓
Source × Year FE	✓	✓	✓	✓	✓	✓
Pair FE	✓	✓	✓	✓	✓	✓
SE clustered by pair	✓	✓	✓	✓	✓	✓
Source countries	224	224	224	49	224	224
Destination countries	172	172	172	172	172	172
Observations	112,541	112,541	112,541	39,407	112,541	112,541
R^2	0.296	0.296	0.296	0.292	0.296	0.296

Note: Same specifications as Table 6, estimated on the broader non-U.S. destination universe (172 destinations) as a robustness check. Dependent variable: asinh(bilateral FDI inflow), 2014–2020. Columns (3)–(6) include U.S. trade share × Post and standardized pre-FIRRMA FDI level × Post. All columns include source×year and pair FE. Standard errors are clustered by country pair. Significance levels: $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

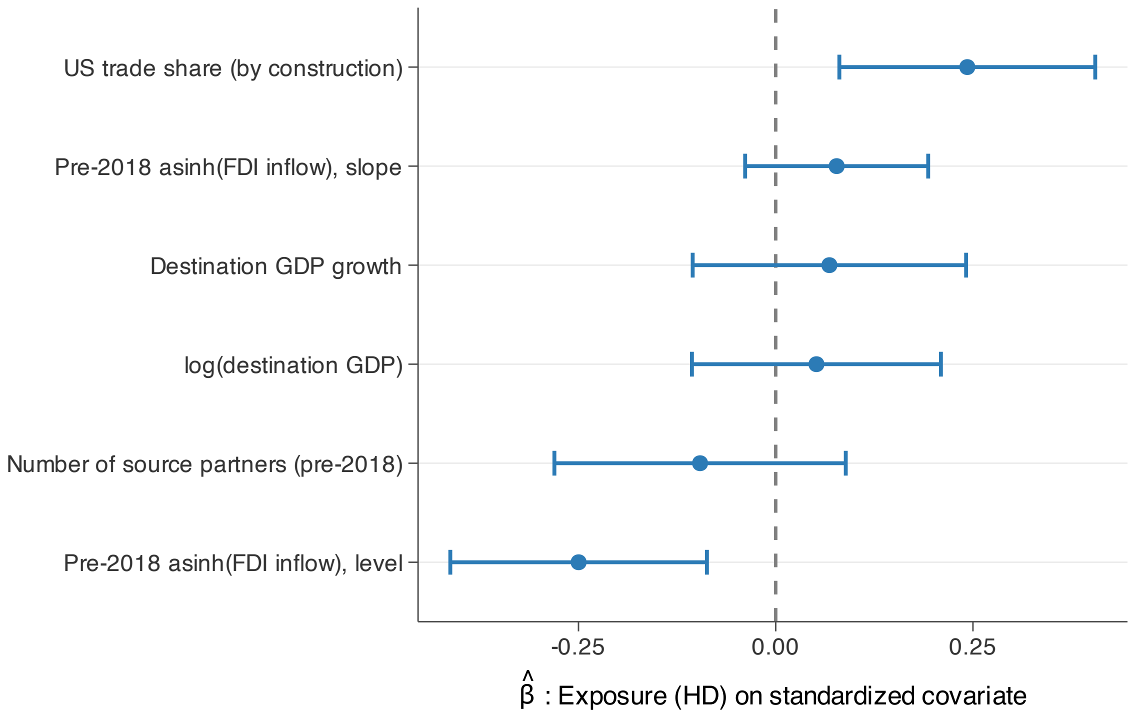

Figure 13: FIRRMA Exposure Balance: Pre-2018 Destination Covariates

Notes: Cross-sectional OLS of destination-level high-dilution exposure on six standardized pre-2018 destination covariates. Sample: 21 non-U.S. SWF home destinations in the main FIRRMA panel (Table 6); pre-2018 means computed over 2014–2017. Each covariate is standardized to mean zero and unit standard deviation. Error bars: 95% confidence intervals. The U.S. trade-share coefficient is positive, consistent with the fact that exposure is built from pre-2018 SWF deals targeting the United States. The negative coefficient on pre-2018 FDI inflow levels means that more-exposed destinations start from lower bilateral inflows on average; pair fixed effects absorb fixed bilateral levels in the main regression, while the event-study and placebo tests address differential timing.

E.3 Non-U.S. high-dilution deal volume

Figure 14 reports annual non-U.S. high-dilution SWF deal counts around FIRRMA. Panel A scales deal counts by the number of SWF home countries in each exposure group; Panel B reports raw annual counts. Non-U.S. high-dilution deal volume rises after 2018 in both groups, including among SWF countries with positive pre-FIRRMA U.S. high-dilution exposure. I read the figure as descriptive evidence that the relevant non-U.S. deal margin expanded after FIRRMA.

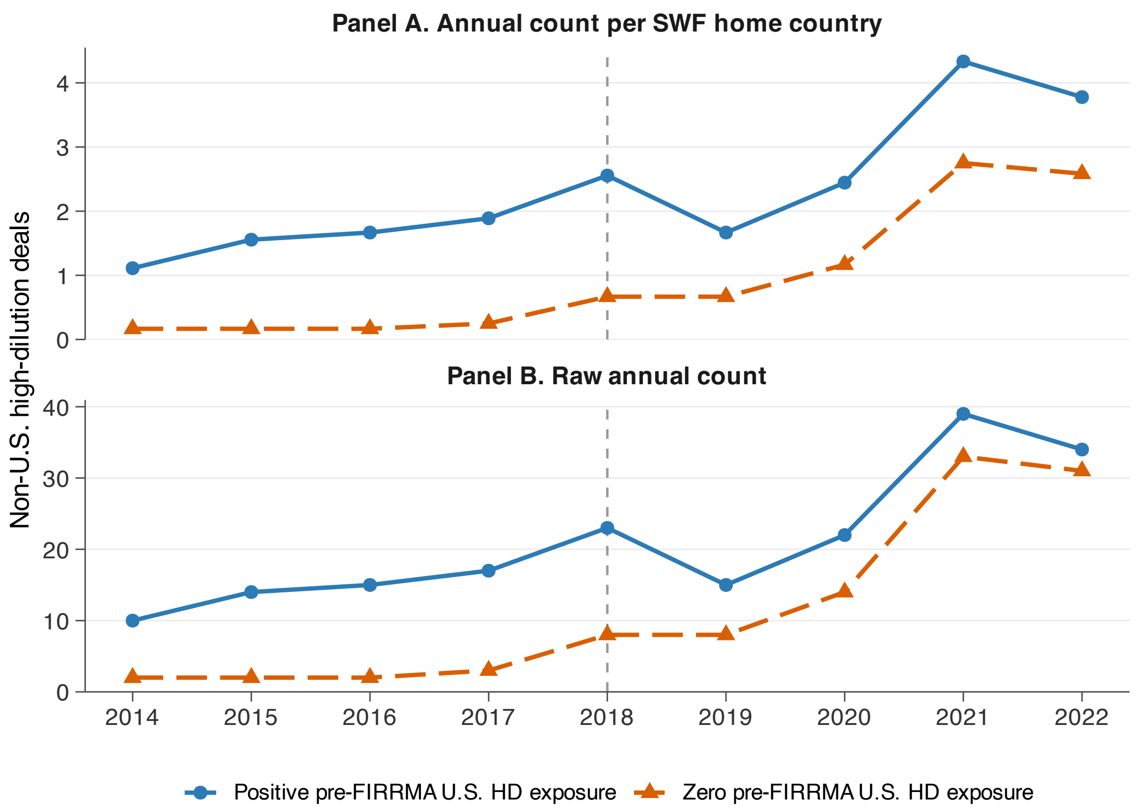

Figure 14: Non-U.S. High-Dilution SWF Deal Volume around FIRRMA

Notes: Annual PitchBook counts of SWF-backed high-dilution deals targeting non-U.S. companies, 2014–2022. High-dilution deals are those with dilution above the median among SWF co-investment deals. Positive-exposure countries are SWF home countries with at least one U.S.-targeted high-dilution deal in 2014–2017; zero-exposure countries have none. Panel A divides annual counts by the number of SWF home countries in the exposure group. Panel B reports raw annual counts. The dashed vertical line marks 2018, the FIRRMA enactment year. The figure is descriptive and does not condition on controls or fixed effects.

E.4 Exposure by SWF Country

Figure 15 plots the cross-country distribution of high-dilution FIRRMA exposure for the 21 SWF home destinations. Nine countries have positive exposure and twelve have zero. Within the positive group, three countries reach 67–100% mechanically: Ireland and New Zealand sit at 100% on a single pre-2018 SWF deal each, and South Korea at 67% on four deals. The substantive variation comes from France (33% on 19 deals), Australia (17% on 10), China (14% on 12), and Qatar (12% on 31). Singapore has the largest pre-2018 sample at 202 deals, but only a 9% share, because most Singaporean SWF deals before FIRRMA targeted non-US firms. The zero-exposure comparison group includes UAE (n=32) and Kuwait (n=19) alongside a small tail with no pre-2018 SWF deals (Brunei, Hong Kong, Italy, Taiwan). This dispersion across positive-exposure countries motivates the leave-one-out exercise in Appendix Figure 16.

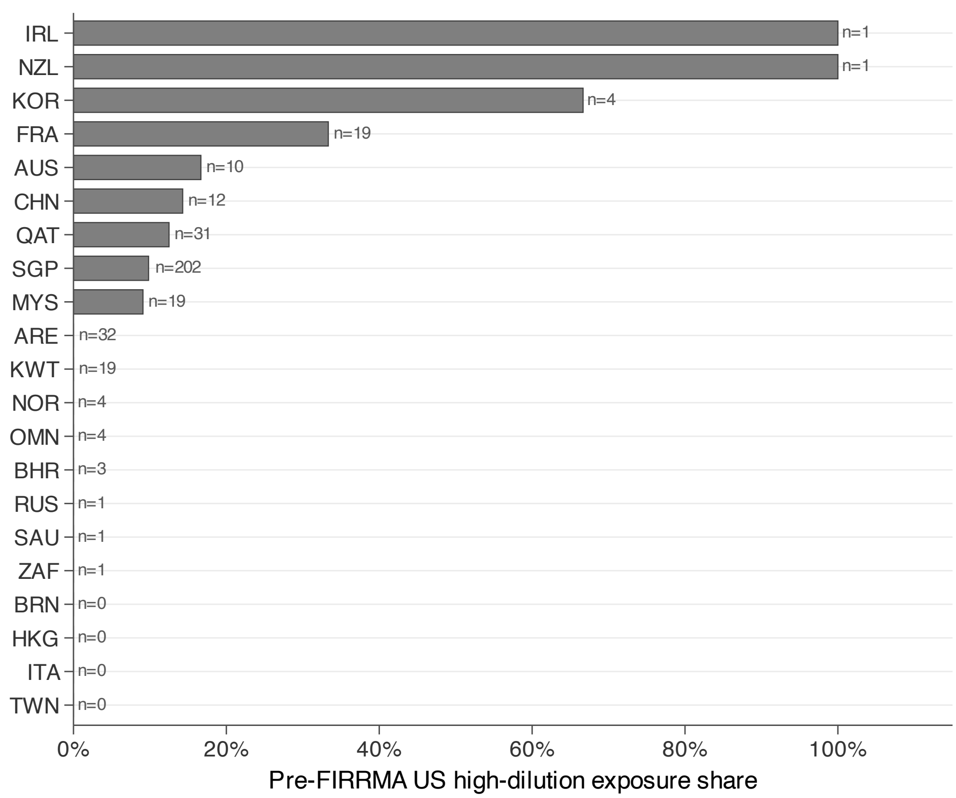

Figure 15: FIRRMA High-Dilution Exposure by SWF Country

Notes: Bars report each SWF home country's pre-FIRRMA high-dilution exposure, defined as the share of 2014–2017 SWF co-investment deals with observed dilution that both targeted the United States and involved above-median dilution. Countries are sorted by exposure. Labels report the number of pre-2018 SWF deals used to construct each country's exposure share. The figure makes clear that some unit-exposure countries are based on very thin pre-FIRRMA deal counts.

E.5 Placebo Shocks and Oil-Exporter Drop

Table 30 tests two confounds in the FIRRMA design. The placebo columns move the shock date earlier while keeping the panel entirely pre-FIRRMA and holding fixed the realized 2014–2017 exposure measure, asking whether that realized exposure predicts similar pre-2018 FDI shifts when the shock date is moved earlier. The placebo estimates are much smaller than the baseline and statistically insignificant: 0.129 for a 2015 placebo shock and 0.203 for a 2016 placebo shock, compared with the baseline estimate of 0.713. The oil-exporter exclusion asks whether the result is driven by concurrent oil-related shocks in Gulf and other commodity-linked SWF countries, including Norway. Dropping oil-exporter SWF countries leaves a positive and statistically significant coefficient of 0.615. The table therefore separates the FIRRMA interpretation from two obvious alternatives: earlier timing patterns in the realized exposure and oil-linked SWF shocks.

Table 30: FIRRMA Spillover: Robustness

	Baseline (1)	Placebo 2015 (2)	Placebo 2016 (3)	No Oil Exporters (4)
SWF US Exposure (high-dilution) × Post	0.713***	0.129	0.203	0.615**
	(0.228)	(0.261)	(0.286)	(0.247)
asinh(Bilateral Trade)	-0.729**	-0.359	-0.625**	-0.976***
	(0.309)	(0.258)	(0.294)	(0.331)
Diplomatic Distance	-0.261	-0.104	-0.253	-0.345
	(0.221)	(0.176)	(0.205)	(0.229)
Gravity controls (GDP, FTA)	✓	✓	✓	✓
Balance controls	✓	✓	✓	✓
Source × Year FE	✓	✓	✓	✓
Pair FE	✓	✓	✓	✓
SE clustered by pair	✓	✓	✓	✓
Observations	16,866	16,486	14,209	13,585
R^2	0.363	0.375	0.384	0.394

Note: Column (1): baseline with full gravity and balance controls (high-dilution exposure). Columns (2)–(3): placebo shocks at 2015 and 2016, respectively, using the same realized 2014–2017 exposure measure as the baseline but restricting the estimation window to pre-FIRRMA years ending in 2017. Column (4): oil-exporter SWF countries dropped (UAE, Saudi Arabia, Qatar, Kuwait, Norway, Oman, Bahrain, Gabon, Brunei). Dependent variable: asinh(bilateral FDI inflow). Balance controls are U.S. trade share × Post and standardized pre-FIRRMA FDI level × Post. All columns include source×year and pair FEs. SE clustered by country pair, as in the main FIRRMA specification; because treatment varies at the destination-exposure level, inference should be read together with the placebo-timing and leave-one-country-out checks. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.6 Leave-One-SWF-Country-Out

Figure 16 drops each positively exposed high-dilution SWF destination country in turn. This check addresses the finite-country concern in the macro FDI design while focusing on the countries that identify the high-dilution exposure coefficient. The coefficient remains positive across all omissions, so the result is not driven by a single exposed destination. The movement across omissions is still informative: countries whose omission shifts the estimate most are the destinations carrying more of the identifying variation, so the figure should be read as a sensitivity diagnostic rather than as a claim that every SWF country contributes equally.

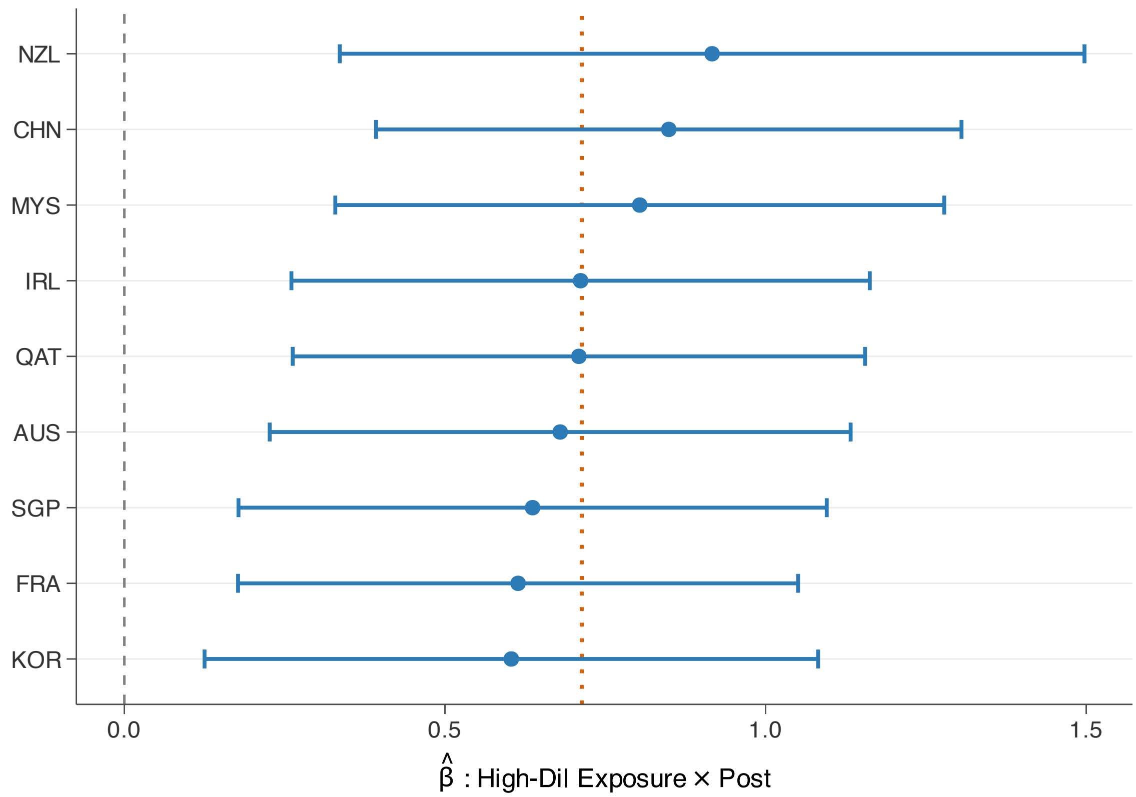

Figure 16: Leave-One-SWF-Country-Out: FIRRMA Spillover

Notes: Each row drops one positively exposed high-dilution SWF destination country and re-estimates the balance-control high-dilution specification from Table 6, column (3). Orange dashed line: full-sample column (3) baseline. Error bars: 95% CIs; SE clustered by country pair.

F BIS Spillover Robustness

This appendix reports robustness checks for the BIS cross-border banking-claims spillover results in Section 5.2. Appendix F.1 reports matching balance, Appendix F.2 tests whether one treated lender–borrower cohort drives the result, and Appendix F.3 checks whether the result survives quarterly timing. Appendix F.4 reports a liability-side placebo and compares the LBS residence-based estimates with CBS nationality-based claims.

F.1 Matching Balance

Table 31 reports the balance achieved by the cohort-by-cohort Mahalanobis matching used in the main BIS specification. The matching uses pre-treatment levels and slopes of log loans and deposits, the outcome closest to the main banking-claims channel. Standardized differences fall sharply after matching, including for auxiliary outcomes that were not used in the match. The stacked-cohort estimates therefore compare treated pairs with controls that had similar pre-treatment banking relationships, rather than with the full pool of unmatched country pairs.

Table 31: BIS Mahalanobis Matching Balance

	Treated	Pre-Match Pool		Matched Controls (k=20)	
	Mean	Mean	|Std. Diff.|	Mean	|Std. Diff.|
Pre-mean log(loans & deposits)*	8.228	5.013	1.136	7.975	0.114
Pre-slope log(loans & deposits)*	0.268	0.150	0.265	0.221	0.187
Pre-mean log(total claims)	8.433	5.228	1.111	8.218	0.095
Pre-mean log(non-bank loans)	6.881	4.407	1.008	6.964	0.045
Number of pairs (across cohorts)	12	765		172	

Notes: Cohort-aggregated balance for the matched stacked-cohort design used in Table 8. Pre-treatment statistics are computed within the three pre-treatment years before each cohort's first-treatment date. "Treated" pools the 12 treated lender–borrower pairs; "Pre-Match Pool" pools the full set of eligible non-treated pairs in the same lender country (765 unique pairs across cohorts); "Matched Controls (k=20)" pools the cohort-specific Mahalanobis-matched controls (172 unique pairs across cohorts). Because controls can be reused across cohorts, the number of unique matched-control pairs is smaller than the mechanical upper bound of 240 cohort-control slots (12 cohorts times 20 controls), implying 68 control-pair reuses across cohorts. The matching routine uses up to 20 controls per cohort, and in the realized matched sample all 12 cohorts use 20 controls. Absolute standardized differences are computed as $|\bar{x}_{\text{treated}} - \bar{x}_{\text{control}}|/\sqrt{(s^2_{\text{treated}} + s^2_{\text{control}})/2}$. The two starred variables are used in the Mahalanobis distance; total claims and non-bank loans are auxiliary checks not used in the match. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.2 Leave-One-Cohort-Out

Figure 17 drops each treated lender–borrower cohort in turn, including that cohort's matched controls. This is the direct concentration check for the BIS design: it asks whether the pooled coefficient is coming from a single treated pair. The loans coefficient remains positive in every omission. The estimates range from 0.0475 to 0.0768, compared with the full-sample estimate of 0.0566.

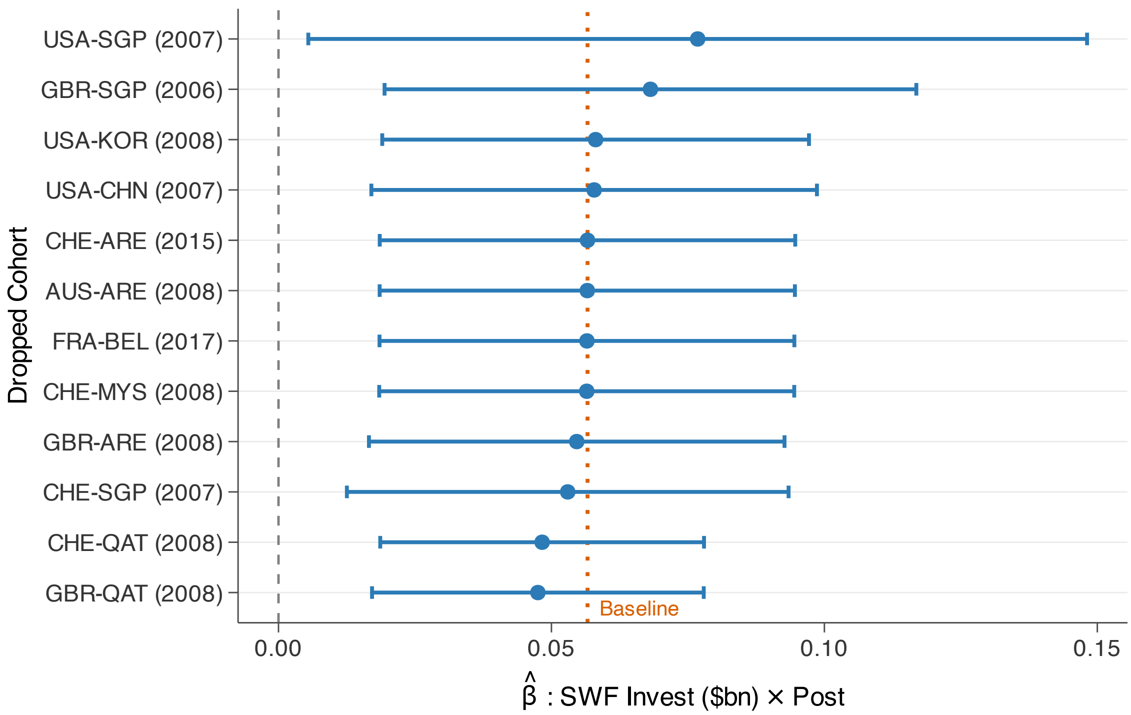

Figure 17: Leave-One-Cohort-Out: LBS Cross-Border Loans Spillover

Notes: Each row drops one treated lender-country × SWF-country cohort, including its matched controls, and re-estimates the intensity-weighted DiD from Table 8 (column 2: log loans & deposits). Orange dashed line: full-sample baseline ($\hat{\beta}$ = 0.0566). Error bars: 95% CIs; SE clustered by cohort. All estimates remain positive; the LOO range is [0.0475, 0.0768]. Eleven of twelve estimates are significant at the 5% level, and the remaining estimate is significant at the 10% level.

F.3 Quarterly Specification

Table 32 re-estimates the BIS design at quarterly frequency. The annual specification is the baseline because BIS stocks are cleaner at year end and match the main stacked-cohort window. The quarterly version is a timing check: similar estimates would mean the crowd-in pattern is not created by annual aggregation. Treatment timing uses the exact deal quarter when it is observed; in the raw post-filter treatment universe, 33 of 48 lender–borrower pairs have dated quarters and 15 are imputed to Q1 of the treatment year. The quarterly analog using all 12 matched cohorts gives a log-loans coefficient of 0.0515, close to the annual baseline of 0.0566. One matched cohort uses a Q1-imputed treatment quarter; dropping that cohort gives the same estimate at displayed precision, so treatment-quarter imputation is not driving the result.

Table 32: Cross-Border Banking Claims Spillover: Quarterly Specification

	(1) Total Claims	(2) Loans & Placed Dep.	(3) Non-Bank Loans
SWF Invest ($bn) × Post	0.0531***	0.0515**	0.0557***
	(0.0165)	(0.0167)	(0.0127)
Outcome transform	log	log	log
Observations	12,276	12,270	12,217
Matched cohorts	12	12	12
Cohort × Quarter FE	✓	✓	✓
Cohort × Pair FE	✓	✓	✓
Cluster	Cohort	Cohort	Cohort

Note: Quarterly analog of the annual specification in Table 8. Same matched stacked-cohort design using BIS Locational Banking Statistics (residence-based). Treatment intensity: total SWF equity injection in $bn. The treatment quarter equals the observed deal quarter when available and Q1 of the treatment year otherwise. Pre-matching window: 12 quarters. Control pairs are Mahalanobis-matched on pre-treatment log-loan levels and slopes. SE clustered by cohort. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.4 Liability Placebo and Nationality Robustness

Table 33 compares the main LBS residence-based estimates with CBS nationality-based claims. This check matters because LBS assigns claims to the location of the reporting banking office, not to the nationality of the banking group. For example, a loan booked by the London branch of a U.S. bank is a U.K. claim in LBS, not a U.S. claim. CBS instead follows the nationality of the parent banking group, so it provides a complementary check that the result is not created by the residence-based booking convention or by activity routed through international banking centers. Column (5) first reports a liability-side placebo based on BIS liabilities; the estimate is small and statistically insignificant, which is inconsistent with a symmetric expansion of both sides of the bilateral balance sheet. The CBS estimates are the same sign and of similar order: the total-claims estimate is smaller than the LBS total-claims baseline (0.0426 versus 0.0581), while the CBS international-claims estimate is slightly larger (0.0599). The comparison therefore suggests that the macro spillover result is not an artifact of the LBS reporting convention, while still leaving some scope for differences between residence and nationality measurement.

Table 33: Cross-Border Banking Claims Spillover: Liability Placebo and Nationality Robustness

	LBS (Residence)					CBS (Nationality)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Total Claims	Loans & Placed Dep.	Non-Bank Loans	Wins. Std. Flow Loans	Liabilities (placebo)	Total Claims	International Claims
SWF Invest (\$bn) × Post	0.0581**	0.0566**	0.0530***	0.0252**	0.0225	0.0426*	0.0599***
	(0.0195)	(0.0193)	(0.0151)	(0.0113)	(0.0197)	(0.0210)	(0.0181)
Outcome transform	log	log	log	wins. flow/pre-stock	log	log	log
Observations	3,259	3,258	3,250	3,259	3,255	4,325	4,055
Matched cohorts	12	12	12	12	12	17	17
Cohort × Year FE	✓	✓	✓	✓	✓	✓	✓
Cohort × Pair FE	✓	✓	✓	✓	✓	✓	✓
Cluster	Cohort	Cohort	Cohort	Cohort	Cohort	Cohort	Cohort

Note: Same matched stacked-cohort design as Table 8 but at annual frequency (±6 years), using Q4 end-of-year annual stocks. Control pairs are Mahalanobis-matched on pre-treatment log-loan levels and slopes. Columns (1)–(3) are BIS Locational Banking Statistics (residence-based) log stock outcomes: total claims, loans and placed deposits, and non-bank loans. Column (4) is the reported annual LBS loan flow, summed from quarterly flows, divided by pre-treatment mean loan stock and winsorized at the 1st and 99th percentiles. Column (5) is the log BIS liability-side position, used as a placebo outcome. Columns (6)–(7) are BIS Consolidated Banking Statistics (nationality-based, immediate counterparty) log stock outcomes: total claims and international claims. CBS total claims include local claims in local currency; CBS international claims exclude these, making column (7) the closest CBS analog to the LBS cross-border measure. The CBS estimates are the same sign and of similar order as the LBS baseline: CBS total claims are smaller, while CBS international claims are slightly larger. The CBS sample includes 17 cohorts (vs. 12 for LBS) because CBS covers additional reporting countries. SE clustered by cohort. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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